COUNTRY REVIEWS

OECD Economic Surveys

Denmark

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OECD ECONOMIC SURVEYS

1999-2000

Denmark

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BASIC STATISTICS OF DENMARK

THE LAND

Area (sq. km):	43 094
Agricultural area (sq. km) (1997)	26 880

Inhabitants in major cities, 1.1.1998 (thousand):

Copenhagen	488
Århus	216
Odense	145
Ålborg	119

THE PEOPLE

Population 1999, thousands	5 319
Number of inhabitants per sq. km	123
Net natural increase (average 1991-1997, thousands)	6.5
Net natural increase per 1 000 inhabitants, 1997	1.5

Total employment, 1999, thousands	2 679
By sector:	
Agriculture	105
Industry	462
Construction	161
Market services	1 001
Community, social and personal services	950

PRODUCTION

Gross domestic product, 1999 (Kr billion):	1 215.8
GDP per head (1999 US$)	32 748

Gross fixed capital formation in 1999 (Kr billion):	241.3
Per cent of GDP	19.8
Per head (US$)	6 500

THE GOVERNMENT

Public consumption in 1999 (percentage of GDP)	25.7
General government current revenue in 1998 (percentage of GDP)	56.6
Public gross fixed capital investment in 1998 (percentage of GDP)	1.7

Composition of Parliament (number of seats):

Social Democrates	64
Liberals	43
Conservatives	17
People's Socialists	13
Danish People's Party	12
Centre Democrats	8
Social Liberals	7
Left Alliance	5
Christian Democrats	4
Progressive Party	4
Independent	2
Total	179

Last general elections: 11.03.1998

Next general election: 11.03.2002 (at the latest)

FOREIGN TRADE

Exports of goods and services as percentage of GDP, 1999	36.9
Main exports in 1999, percentage of total marchandise exports:	
Agricultural products	10.7
of which: products of animal origin	8.6
Manufactured products	76.0
of which: machinery and instruments	26.9
Other manufatured products	49.1
Other products	13.3

Imports of goods and services as percentage of GDP, 1999	32.7
Main imports in 1999, percentage of total marchandise imports:	
Intermediate goods for agriculture	2.4
Intermediate goods for other sectors	42.6
Fuels and lubricants	3.5
Capital goods	13.2
Transport equipment	6.3
Consumer goods	29.9

THE CURRENCY

Monetary unit: Krone

Currency units per US$, average of daily figures:

Year 1999	6.980
June 2000	7.851

N*ote*: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Denmark by the Economic and Development Review Committee on 3 May 2000.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 15 June 2000.

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The previous Survey of Denmark was issued in January 1999.

Assessment and recommendations

A "soft landing" has been achieved…

The pace of activity in Denmark slowed in 1999. This followed a long period of very strong growth in domestic demand and the adoption of policy measures aimed at cooling the economy. Overall GDP growth was 1.6 per cent, with domestic spending remaining soft for most of the year, reflecting weak private consumption, falling fixed investment and a sharp adjustment in stocks. However, exports grew by 7 per cent, well ahead of market growth, benefiting in part from a lower effective exchange rate. Imports grew only slowly, so that the current account returned to a surplus of 1 per cent of GDP. Higher export growth offset weakness in domestic manufacturing orders and helped maintain industrial production growth. Capacity utilisation has fallen from its peak in early 1998 but started to rise again last spring, and business confidence has risen steadily from its trough at the end of 1998.

… although prospects for inflation remained worrisome until recently

With output having moved above levels considered sustainable as the recovery matured, wage and price developments remained a concern in 1999. CPI inflation kept rising until very recently to over 3 per cent and Denmark breached the EMU inflation criterion in early 2000, although this is expected to be only temporary. In contrast, domestic market-determined inflation (abstracting from oil prices, imports, taxes and housing), which peaked in the first half of 1999 at 3.1 per cent, fell back to 2.7 per cent by the fourth quarter. Hourly earnings growth has moderated only slightly to 4 per cent. Employment growth has slowed, even though registered unemployment has continued to fall to around 5¼ per cent of the labour force. Rather than signalling a further rise in labour market pressures, the persistent decline in measured unemployment may be sustainable, since it has coincided with continuing efforts to tighten the unem-

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ployment benefits regime. The recently concluded accord between the social partners suggests moderate wage growth can be expected in 2000, easing inflation risks. An important feature of the agreement was an increase in holidays, which is generally expected to reduce average working hours over the next three years, even if the costs thereof may be partially offset by increased working-time flexibility. The pact covers a four-year period, providing greater certainty to employers and a stable backdrop to bargaining at the enterprise level.

The economy is likely to remain close to its potential level

By the end of 1999, excess demand was more or less eliminated, and the danger of overheating had receded. Looking ahead, the economy is projected to expand at rates close to potential growth, at 2¼ per cent in 2000 and 2½ per cent in 2001. Exports may continue to benefit from increasingly buoyant activity in the rest of Europe, and domestic spending is projected to accelerate gradually as export-led growth feeds into higher employment and inflation eases, allowing real disposable incomes to recover. With no significant output gap expected in the short term, the risks are quite finely balanced between a renewal of inflationary pressures and a pick-up in unemployment. If wage settlements ultimately agreed at the plant level turn out higher than currently expected, possibly reflecting the recent strengthening in demand, this could be sufficient to set the economy back on a path of rising inflation, especially if labour supply continues to shrink. On the other hand, if international developments lead to a larger increase in short-term interest rates and especially if these feed through into higher long-term mortgage rates, the economy might be pushed into a renewed slowdown. Managing to keep the economy on a reasonably steady growth path will be a challenge to policy makers, given the limited instruments available.

The monetary authorities have continued to set official interest rates solely with a view to fixing the exchange rate

Monetary conditions play a major role in determining the short-term outlook for Denmark, but they are effectively determined by the economic situation in Europe as a whole. Expansionary monetary conditions, and the inability of fiscal policy to tighten enough to counteract them effectively, led to the overheating in 1997 and 1998. The appreciation of the effective exchange rate in 1998 helped to take some steam

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out of the economy. The economy has been out of step with the rest of Europe during the 1990s, producing sometimes inappropriate monetary conditions. However, the "soft landing" achieved in 1999, may help align the business cycle with that of the euro area in the future, reducing the risk of untimely monetary impulses. Denmark has a long-standing and highly credible policy of maintaining a fixed exchange rate. It is also committed to operating monetary policy in the framework of the Exchange Rate Mechanism, which ties the Danish currency to the euro and minimises risk premia over the longer term. Under these conditions, Denmark may be better off adopting the euro and thereby gaining the benefits of sharing a single currency with most of its main trading partners. However, the decision rests on the outcome of a referendum scheduled for September.

Fiscal measures have helped stabilise activity, but continued high surpluses may be difficult to sustain

With the effective exchange rate and interest rates largely beyond its ability to influence, fiscal policy is Denmark's main tool for macroeconomic stabilisation in the short run. A budget surplus of 3 per cent of GDP was achieved in 1999, after several years of consolidation. The so-called "Whitsun" package, adopted in mid-1998 and being phased in up to 2002, contained measures oriented towards dampening activity in the short term and improving savings incentives and resource allocation over the medium term. With no major policy changes in the budget for 2000, the surplus will be maintained, at least on a cyclically-adjusted basis, as long as the agreements with local governments are adhered to. However, as some other OECD countries have found, maintaining a surplus can be difficult in the face of calls for tax cuts or additional spending, and such pressures may emerge in the negotiations over the budget for 2001. Nonetheless, these pressures should be resisted.

Public debt reduction should continue and the damaging effects of high taxes be reduced

Public finance faces several medium-term challenges. The first is to continue to reduce the level of outstanding debt by at least maintaining the budget surplus. This would provide more room to manoeuvre if and when there is an economic shock, especially one that might affect Denmark to a greater extent than its EU partners. Maintaining the surplus would also help to prepare for the spending pressures of an ageing population that will begin to arise over the next

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few years. Second, the top marginal tax rate remains rather high and is paid by a growing number of taxpayers. More generally, high tax rates may discourage human capital investment and entrepreneurship and make short working hours and "do it yourself" activities more attractive. Efforts to reduce the distortions in the tax system have been made through reforms since 1994, although some further improvements in the efficiency of the tax system could be made. However, tax policy options may be constrained over the medium term by the need to respond to pressures on the overall tax base arising, inter alia, from the removal of restrictions on cross-border trade in 2004. In any case, the magnitude of the welfare state makes it difficult to reduce the overall tax burden without compromising public finances. With public expenditure at almost 55 per cent of GDP in 1999, and public consumption at more than 25 per cent of GDP, gaining better control over expenditure and enhancing the value for money in the public sector are key elements.

The threshold for the top income tax rate is too low, and other tax distortions should be addressed

Tax reforms have improved private financial savings and lowered statutory tax rates. But because an increasing number of taxpayers find themselves in the top tax bracket and local government taxes have increased, the average effective marginal tax rate on labour income has declined only slightly. More than a third of full-time workers currently face a marginal rate of 62 per cent, and the proportion is likely to rise as a result of the 1999 tax reform. Since wages are relatively compressed, raising the threshold for the top tax rate could be achieved for relatively little revenue foregone while the impact in terms of strengthening work incentives could be significant. Furthermore, the large differences between the tax treatment of positive and negative net capital income, private share portfolios and pension saving schemes – which widened further with the latest tax reforms – tend to distort households' portfolio choices. Some of these problems could be resolved by lowering the tax rate on positive net capital income. Although tax expenditures related to corporate taxation have been reduced in scope, depreciation rates higher than economic depreciation still reduce the effective tax rate and produce a bias in favour of capital-intensive industries. Broadening the corporate tax

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base by reducing depreciation allowances would provide room for reduction in statutory rates for all firms.

Expenditure control needs strengthening, especially at lower levels of government

Making room for even modest cuts in marginal tax rates without undermining the budget surplus will require restraint in and better control over spending. Real consumption expenditure by county and municipal authorities has ratcheted up in recent years, having been repeatedly higher than initially provided for in the annual agreements with central government or in the budgets finally adopted, although the outcome for 1999 was in line with planned levels. The municipal budgets for 2000 reflect the annual agreement, so that one source of past slippage has been avoided this year. However, sustained progress in this area will require a clearer division of responsibilities between central and local governments. The increased tendency in recent years for central government to engage more directly in setting objectives for locally-provided services seems to be partly in conflict with the very decentralised governance structure of the Danish public sector. In addition, the large variation in unit costs of providing services across municipalities suggests scope for rationalisation of the present system through a combination of better co-operation between authorities, voluntary mergers or outsourcing. Although progress in altering existing arrangements has proved very difficult, it is important that rationalisation efforts be pursued.

Greater use of outsourcing could lead to improved services at lower cost

More effort could be made to incorporate techniques for improving the allocation of resources within the public sector as well as the efficiency of their use. The general government share of total employment, at 30 per cent, is higher in Denmark than in almost all other OECD countries, and the wage bill is a higher proportion of public consumption than elsewhere. Greater recourse to outsourcing, through tendering for the provision of publicly-funded goods and services, should therefore be considered. As in other countries, this could be achieved without compromising government objectives for service delivery and quality as long as supply contracts are carefully designed. Tendering would also stimulate supplier innovation, ultimately providing better services at lower cost. Experience so far has been encouraging.

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User charges would reduce over-use of public services and provide clearer signals about demand

User charges for public services are not applied consistently and are mostly insignificant, with little effect on incentives. Experience from Denmark and elsewhere has shown that even relatively modest charges, if well-designed, can provide clearer signals about demand for public services and encourage more efficient use of resources. The free provision of many services by government has long been a feature of Danish society, and the utilisation of user charges for social services has actually diminished over the past decade. But it is well established that "free" services tend to be over-used, while the taxpayer foots the bill. A more consistent and widespread application of user charges could play an important role in achieving tighter control over public expenditures. An initial step would be to calculate and publish the full costs of providing each service so that the degree of subsidy is made transparent.

Structural unemployment has been reduced significantly, but some additional reforms could still be made…

Denmark has achieved a considerable reduction in unemployment since 1994, and, while some of this improvement reflects the pickup in activity, structural policy reforms have played an important role. These reforms have concentrated on improving active labour market policies and tightening the eligibility criteria. However, it is doubtful whether these policies can do much more to reduce the structural unemployment rate, and they may even be put under some strain if unemployment edges back up. Further strengthening the eligibility criteria and stronger reliance on economic incentives should be considered. For example, calculating the unemployment benefit based on annual taxable income (rather than the last twelve weeks' wages) could be one way to make the benefit digressive. Gradually reducing the high statutory replacement rate applying to most of the unemployed in line with the duration of the unemployment spell would encourage job search. Another initiative would be to remove the extended benefit rights that currently apply to the unemployed aged 55 to 59. These privileges, together with their link to early retirement benefit entitlement, prevent active labour market policies from working as effectively as they might for this age group.

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Raising labour supply will require tighter rules on social transfers

Over the medium term, the government wants to increase labour supply to enhance productive capacity and help position the economy to deal with an ageing population. However, the objective looks unlikely to be attained on present policies. Average hours worked, already low by OECD standards, are falling further because of the high marginal tax rates and recently-agreed additions to standard holiday time. In contrast, Denmark has one of the highest participation rates in the OECD, even though almost everyone of working age outside the workforce is receiving a public benefit. But there are two particular weak spots in labour force participation – the withdrawal of older workers and the excess time spent in education. In general, reaching the government's objectives may require a careful reassessment of the rules governing transfer schemes so that they do more to discourage take-up and promote a return to the labour market. The impact of the recent reform of the early retirement benefit scheme should be monitored closely, and the changes should be strengthened if outcomes fall short of expectations. Within the announced framework for reform of the early retirement (disability) pension scheme, it is important to ensure that the benefit structure underpins rather than compromises the effort to improve job opportunities for people with less than full working capacity. In particular, the replacement rate for the least disabled group should not be increased. The sickness benefit could also be made less generous, by introducing a waiting period before benefits become available.

… and the excessively long time spent in education still needs to be reduced

Progress has been made on improving performance and managing costs in the formal education system. Basic skills acquisition has been enhanced, and the dropout rate from upper secondary education has been reduced. The agreements reached in the 1999 wage bargaining round should lead to increased teaching hours and establish a link between teachers' pay and performance. Nevertheless, more efforts are needed to reduce the excessively long time spent in education and the corresponding high average age of entry into the work force. Reducing the length of time over which students can receive state support would help. Initiatives to strengthen cost awareness and improve the incentive structure in the adult education system were

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agreed in 1999. In particular, the relevant public expenditures will henceforth remain unchanged in nominal terms, so that public financing will diminish over time, leaving employers and employees – the direct beneficiaries – to bear a greater share of the cost. But the list of courses for which public subsidies are made available should also be pruned to eliminate narrow, firm-specific programmes.

Rent controls should be removed

The housing market, which was reviewed in depth in the previous Survey, remains subject to a wide range of interventions and policies that have resulted in the misallocation of resources. Rent controls have discouraged maintenance and new investments in private housing, perversely adding to the stock of buildings qualifying for urban renewal subsidies. In a further negative twist, urban renewal is proving more costly on average than constructing new social housing. A two-pronged approach is in order: removing rent controls should be a high priority, and urban renewal policies need to incorporate stronger incentives for cost containment. Rent regulation in the social housing area also needs to be modified to eliminate unwarranted rent differentials within and across vintages. Furthermore, the social housing sector would benefit from granting increased property rights to tenants, provided social problems are not exacerbated, in order to encourage tenant responsibility. In addition, the housing associations are accumulating significant wealth, and it is unclear to whom it belongs or what should be done with it.

Competition should be strengthened further in many sectors

The vigour of competition in product markets has improved but remains weak in many sectors. The Competition Act has been strengthened, but merger control and limits on fines are two dimensions where it still falls short. The more active role being taken by the Competition Council in advocating intensified competition, even in matters where it has no direct authority, is welcome. Nevertheless, more could be done to ensure vibrant competition. It is important to remove remaining exemptions from the Act, such as co-operative agreements in distribution, resale price maintenance for Danish publications, and pharmaceuticals. And practices that are exempt under the "necessary or direct consequence of public regulation" provision should be more tightly controlled. Most

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utilities are still heavily regulated. Reform is underway in the electricity sector, but further initiatives to reduce vertical integration are still needed. This industry should be carefully monitored to ensure effective competition.

A more rigorous approach is needed to environmental policies

Environmental issues are a priority concern in Denmark. In some areas – notably greenhouse gas emissions and waste recycling – ambitious environmental targets have been adopted and a wide-ranging set of policies intended to achieve them has been developed. Most individual policies are carefully designed, partly to meet political, fiscal and distributional objectives. But it seems that some of the targets have been adopted without a full appreciation of the resulting costs or benefits, and the resulting structure sometimes appears rather complex. Progress in using economic instruments may be undermined by overlapping directives, special treatment or complicated exemptions that subvert the principal advantage of economic instruments – namely, ensuring that environmental externalities are internalised equally in all activities. Such special treatment both raises costs and reduces the likelihood of reaching the targets. More rigorous and systematic valuation of costs and benefits, where possible and even if subject to uncertainty, would provide a basis for more transparent and consistent comparison with the many non-quantifiable costs and benefits.

Water charges should treat users equally, and agricultural discharges should be taxed or traded under a permits scheme

Water is an example where different sectors currently face differing incentives to reduce polluting discharges. The targets for municipal water treatment and industrial discharges have been met, but at high cost, whereas discharges from agriculture have declined much less. The environmental component of current water charges and taxes could be used to equalise abatement costs, but major rebates are given to industrial users, and a completely separate system handles agricultural discharges. Although fines are imposed for "excess" fertiliser use (compared with a norm defined with respect to agricultural technology but with no reference to environmental costs), no incentives are given for reductions beyond the norm. Some farmers have even switched to more fertiliser-intensive types of wheat production to take advantage of special concessions in the Aquatic Action Plan II. A better approach would be to tax net nitrogen discharges from farms or to establish a tradable

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permits scheme. An offsetting and transitional generalised tax credit could be granted if there were concerns about the potential income effects on some farmers.

Current high costs of reducing greenhouse gas emissions could be cut by rationalising tax and permit-trading schemes

Denmark has a number of ambitious energy-related targets and appears willing to pay a considerable price to achieve them. Its laudable policy of "leading by example", through which it hopes to induce other countries to adopt equally ambitious targets, results in relatively high costs, particularly in the case of CO_2 and other energy-related emissions. These costs should be made as transparent as possible so that the relative merits of various options to meet Kyoto commitments might be clearer. The existence of a widely-varying set of tax rates, with lower rates for heavy energy users, means that overall costs – per tonne of Danish CO_2 emissions abated – are higher than would appear at first sight to be necessary. An alternative policy would imply higher taxation on energy-intensive industry (and lower taxation on households), with the likely result that some plants (though perhaps fewer than industry lobbyists might suggest) would close, and that some of their production – and CO_2 emissions – would migrate to countries with less stringent targets. The Danish authorities argue that such "carbon leakage" defeats the object of their programme, which is to have an impact on global emissions, not just those located in Denmark. If Danish credibility and influence as a leader in this field depends on other countries taking the same view on leakage, these added costs may be worthwhile. But in the context of the Kyoto Protocol, where all industrialised countries – including many to whom Danish emissions might "leak" – also have targets to meet, such an approach would certainly represent costly protection of particular industries. The Danish authorities recognise this and have stated that they will over time modify the system of differentiated CO_2 tax rates as the Kyoto Protocol comes into force. Consideration should be given to phasing the differentiation out entirely.

More cost-effective policies may be achieved through greater integration

The authorities also recognise that the wind turbine programme has been expensive, although views differ on the value of the environmental benefits that have been obtained. In this and a number of other areas policy is already moving towards much more cost-effective ways of

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achieving environmental goals through increasing use of domestic cap-and-trade schemes and emission taxation. It should be possible to increase the efficiency gains by integrating some of these, notably the green certificate system for wind turbines, the CO_2 trading system for electricity generators, and the CO_2 tax. This would allow the current sector-by-sector targets to be abandoned and the elimination of sector- and use-specific tax rates, as well as ensuring that the wind turbine programme is not used to protect and provide rents to an "infant" industry that, with a major presence in world markets, has now grown up. Danish concerns over leakage makes full integration of the various measures difficult.

Transport policies could be more consistent in encouraging emission reductions

Emissions are also an issue for transport. Transport policies may not currently provide the right incentives for reducing emissions. First, subsidies to public transport may be justified for social reasons (although procedures for allocating them are far from transparent), but they are often assumed to be good for the environment as well. In fact, the environmental advantages of public transport depend critically on passenger load factors. Economic efficiency and environmental effectiveness would be improved by ceasing to deliver subsidies through fuel tax exemptions and by ensuring that allocation of subsidies is based as much as possible on systematic and consistent cost-benefit analysis. Second, the current taxation of private cars and fuels is skewed. Most revenue is raised from a tax that discourages car ownership but where the relation to the emissions thereby generated is weak. It might appear desirable to switch taxation to fuel, but insofar as such a move results in a shift in fuel purchases across the border to Germany, its efficiency in terms of reducing overall emissions would be lowered, and tax revenues would also be lost. As a consequence, costs to the nation would be high compared to environmental benefits.

Economic instruments should be allowed to function in the area of waste management

Waste management is an area where Danish environmental policy appears to rely on a strong system of economic instruments in the form of differentiated waste taxes and charges. But government suggestions that cost-benefit analysis has been carried out are belied by the reliance, in practice, on national targets for the proportions of different

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types of waste to be recycled, incinerated or disposed of in landfill sites. Such a "waste hierarchy" approach is not necessarily cost-effective; hence further development of economic instruments should be considered. The system of taxes on individual containers and packaging products makes sense, provided they are set to reflect environmental costs, although there is inevitably a trade-off with administrative complexity. Ongoing monitoring or periodic review would help to establish the appropriate balance. However, the present regulations on re-use of glass bottles and the ban on metal cans as drink containers are believed to favour domestic producers, and the EU Commission is contesting the regulations before the European Court. It would be best if they were revised to meet environmental objectives without harming trade and competition.

Summing up, short-term goals have been achieved, and the outlook is good, but more reforms are needed

Denmark managed the rare feat of a "soft-landing" in 1999, after a long and finally unsustainable period of strong growth. Now, real GDP is set to rise by around 2¼ per cent this year, and 2½ per cent the next, leaving output close to its potential level. Fiscal policy has had to play a major role in offsetting the expansionary effects of monetary policy, which turned out to be stronger than policy makers had expected. Closer convergence with the business cycle in the Euro area would alleviate the macroeconomic policy challenges in the future, as would greater control over public spending. Although Denmark has achieved more progress on improving the budget balance than most OECD countries, there is no room for complacency: tax rates are too high, and ageing will soon add to spending pressures. Increased outsourcing and more reliance on user charges could help to restrain the growth of public consumption. Structural reforms undertaken over the past decade have led to higher trend growth rates, but more remains to be done. While Denmark has succeeded in reducing structural unemployment, it has made less progress on meeting its labour-supply objectives. Increased labour supply will be essential to assure long-term fiscal sustainability. There is still too little effective competition in many product markets, despite the best efforts of the Competition Authority, and the rental housing market continues to function rather poorly. There has been a laudable increase in the use of

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economic instruments in the area of environmental policy. But it should be possible to achieve environmental improvements more cost-effectively if greater use of cost-benefit analysis were made and policies were rationalised in some areas. In sum, while Denmark has made good progress on many fronts, further efforts to improve policies would help to secure a more prosperous future.

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I. Macroeconomic developments and policy challenges

The sustained recovery that got underway in Denmark in 1993 tapered off towards the end of 1998, and 1999 was a year of relatively slow growth, resulting in a "soft landing". This was sufficient to have eliminated the small positive output gap that had emerged, and output remains close to its potential level. This presents challenges for policy, with the economy quite finely poised between the risks of renewed overheating or slower growth and higher unemployment over 2000 and 2001. This chapter reviews the recent developments and the macroeconomic context and policies. The short-term outlook is then presented and the risks and challenges now facing Denmark are reviewed.

Recent developments

The recovery had a number of key features that set the scene for the more recent developments (Figure 1):

- Activity was kick-started in 1993, after a long and deep recession, by a brief fiscal expansion, together with a change in the regulation of mortgage lending. Labour market and tax reforms have helped to sustain the recovery.
- The economy expanded by 5½ per cent in 1994, and then growth settled down to more moderate annual growth rates of between 2½ and 3 per cent. Most of the output gap was eliminated early in the recovery.
- Growth was domestically-generated, with strong private and public consumption and an investment boom, drawing in much higher imports. Exporters steadily lost market share, while the current account surplus shrank and eventually turned to deficit in 1998.
- Registered unemployment was sharply reduced, from more than 12 per cent of the labour force in early 1994 to around 6½ per cent by mid-1998. Survey-based measures show a shallower decline, suggesting that some of the improvement represented a tightening of benefit rules and

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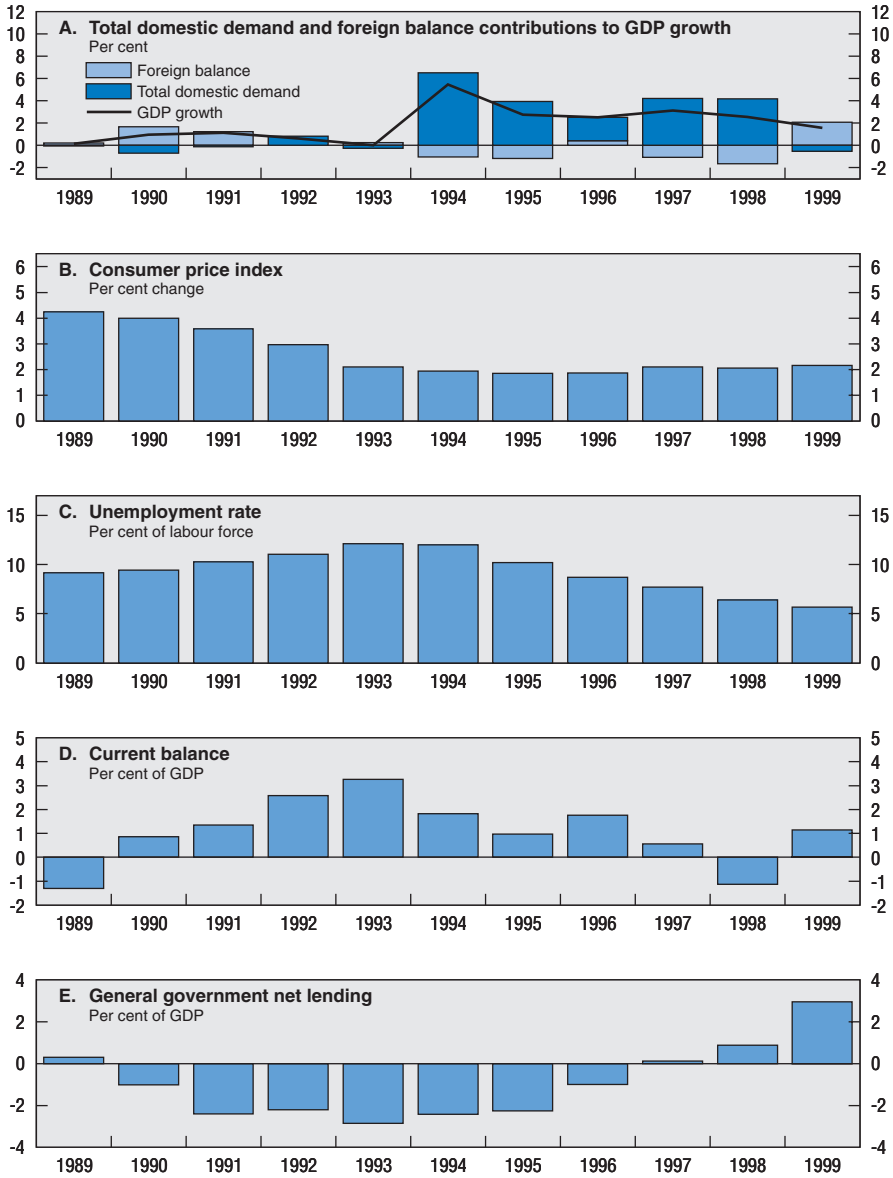
Figure 1. **Macroeconomic performance**



Source: OECD.

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implying a smaller increase in labour-market pressures. Nevertheless, some signs of labour shortages started to emerge.

– Short-term interest rates fell dramatically and by the end of 1996 they were 10 percentage points lower than their temporary peak in late 1992, only edging very slowly upwards towards the end of 1997. Long-term mortgage-credit bond yields fell by around 3 percentage points between the end of 1994 and mid-1998. Together with easier mortgage lending conditions and fairly strong growth in disposable incomes, this gave a boost to real house prices, which soared, albeit after a long decline.

The main concern of policy makers during the second half of 1998 and 1999 was the process of unwinding the excess demand that had built up, both by dampening spending and by boosting potential output, especially through increases in labour supply. Fiscal policy and ongoing labour-market reform were the key instruments employed, and shifts in monetary conditions and effective exchange rates also contributed to cooling the economy. Nevertheless, worries remained about whether a soft landing had in fact been achieved until late in 1999. Interpretation of economic data was complicated by the effects of the 1998 general labour market conflict which boosted measured growth rates in 1999, and led to a recorded increase in actual hours worked, although the settlement itself reduced their normal levels. There have been also some puzzling and conflicting indicators – the persistent rise in housing prices is one, and another is the continuing reduction in registered unemployment despite slower growth in employment, resulting in a slight fall in labour supply.

Fiscal policy was restrictive

Denmark had already achieved a significant consolidation of public finances over the course of the recovery, reaching a surplus on the government balance in 1997 (Figure 2). Against this backdrop, the government's budget for 1999 was designed to dampen the economy in the short run and included several initiatives intended to deliver benefits primarily over the medium term, particularly for labour supply. These included a third phase of labour-market reform, changes in the early retirement system, improved rehabilitation and better job opportunities for the disabled, and improved educational efforts (see Chapter III). The budget bill agreed in December 1998, together with the agreements with local authorities provided for a general government balance of 2.5 per cent of GDP (including the boost arising from the reclassification of compulsory pension contributions, approximately DKr 7 billion or ½ per cent of GDP, as public-sector revenue). The 1999 budget also included the first-year effects of the reduced tax deductibility of mortgage interest payments (from the "Whitsun" package[1]). This measure alone was estimated to dampen activity in 1999 by ½ percentage point.

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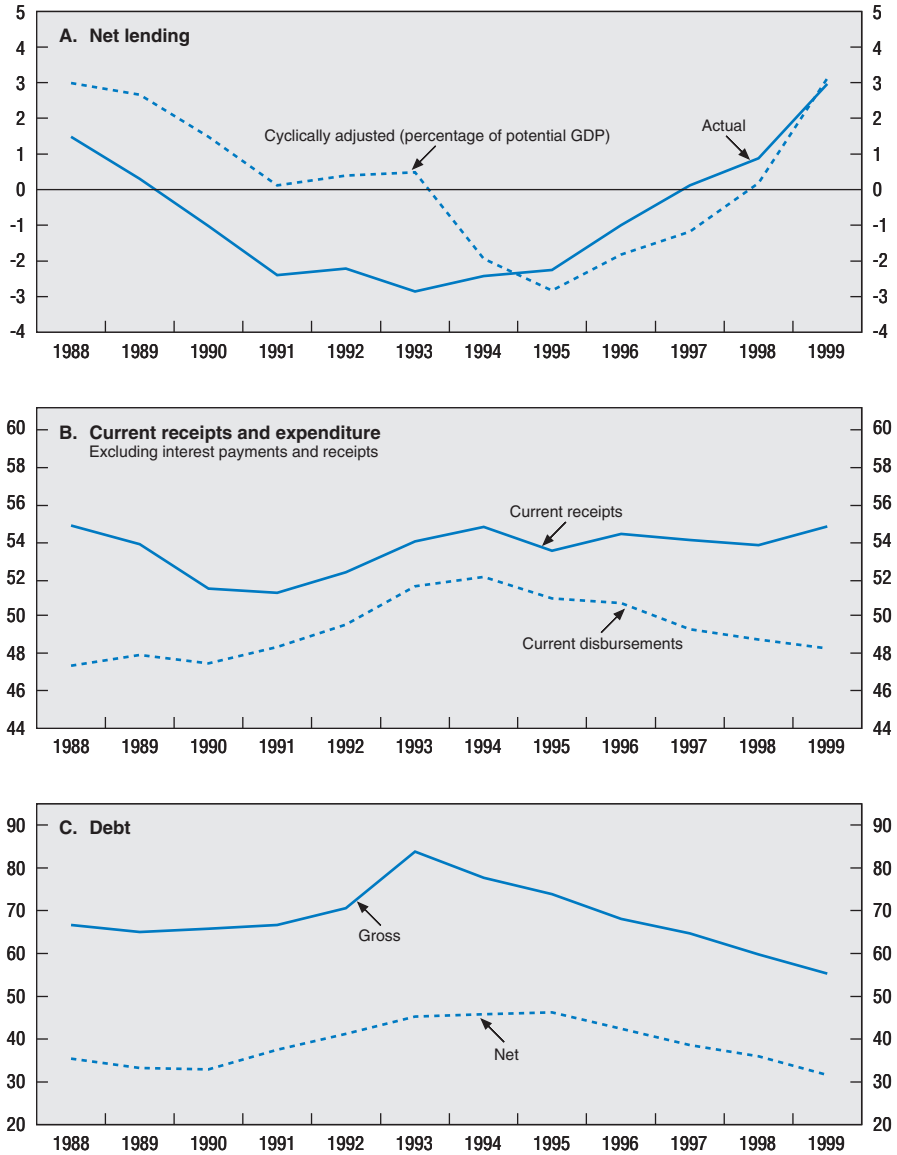
Figure 2. **General government budget balances**
Percentage of GDP



Source: OECD.

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The temporary increase in stamp duty on household borrowing against collateral in owner-occupied housing was continued; this was intended to further restrain demand. Measures were also put in place to limit publicly-controlled residential construction works. Public consumption, however, was projected to grow by 1.4 per cent, as a result of growth in local-government employment and expenditures linked to the third phase of the labour-market reforms.

The out-turn for 1999 in fact produced a fiscal surplus for general government of 3 per cent of GDP, ½ percentage point better than budgeted (Table 1). This was due to stronger revenue across a range of sources, with only VAT receipts relative to GDP lower than expected at the beginning of the year. Corporate taxation revenues were boosted in 1999 in part because of the one-time effect of a shift to provisional tax payments. Public consumption grew by 1.2 per cent, less than budgeted for and a significant slowdown from previous years, but still higher than the government's medium-term target of 1 per cent annual growth (see Chapter II). Although transfer payments were slightly higher than budgeted, they were still almost ½ percentage point of GDP lower than in 1998, reflecting continued

Table 1. **General government budget and out-turn**
As per cent of GDP

	1998 Out-turn	1999		2000 Budget
		Budget	Out-turn	
Consumption	25.7	25.7	25.7	25.7
Transfer payments	18.1	17.6	17.7	17.7
Investments	1.7	1.7	1.7	1.7
Interest expenditures	5.3	4.7	4.6	4.3
Other expenditures	5.3	5.0	5.1	5.2
Total expenditure	56.1	54.7	54.8	54.6
Personal income taxes, etc.[1]	21.8	21.7	21.9	21.4
Social security contributions	4.4	4.5	4.5	4.5
Real interest tax	0.8	0.9	1.0	0.8
Corporate tax	2.8	2.9	3.0	2.5
Value added tax	9.7	9.7	9.6	9.6
Excise duties	8.5	8.3	8.4	8.8
Other taxes	1.5	2.2	2.2	2.4
Total tax	49.6	50.2	50.8	50.0
Interest revenue	3.0	2.4	2.4	2.3
Other non-tax revenue	4.9	4.7	4.7	4.6
Total revenues	57.3	57.2	57.6	56.8

Note: Total revenues, according to the new national account principles, are no longer the sum of the taxes and interest revenue and other non-tax revenue. This is due to a tariff revenue, which according to the new national account principles is categorised as taxes, but not as public revenues, because it is an EU-arrangement.

1. Personal taxes etc. include: withholding taxes, specific taxes from households, inheritance tax, gift tax, mandatory fees and other personal taxes.

Source: Ministry of Finance.

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declines in unemployment and reforms to other transfer schemes (see Chapter III). The combined effects of lower expenditure growth and a higher tax-to-GDP ratio produced a significant improvement in public finances over the year, amounting to 1.8 percentage points of GDP. With the economy performing at close to its potential, the OECD estimates that the cyclically-adjusted budget balance was also around 3 per cent of GDP (which is slightly higher than the calculations of the Danish authorities).

For 2000, the budget provides for the general-government surplus to fall to 2.1 per cent of GDP. This is consistent with the government's stated goal of maintaining a surplus above 2 per cent of GDP out to 2005, so as to create room to meet rising expenditures on the elderly beyond that point (see Chapter II). However, this expected decline is almost entirely due to cyclical developments and based on the government's economic forecasts in December 1999. With no significant spending or tax initiatives incorporated in the budget, the structural surplus is expected to remain steady, providing Denmark with a reasonable fiscal policy cushion in the event of adverse economic shocks. Public consumption is projected to grow by 1¼ per cent – again above the government's medium-term target. Overall, the expenditure-to-GDP ratio is projected to fall slightly, because debt servicing is expected to continue to fall. On the revenue side, the tax burden would ease back ¾ percentage point from 1999's peak, to exactly 50 per cent of GDP, both for cyclical reasons and because of a reduction in the payments of corporate tax and tax on the return on savings in pension funds. The 2000 budget did not contain any major structural reforms (such reforms are usually included in the negotiations between the political parties that culminate in the budget bill). However, the outline of a framework for a new early retirement (disability) pension scheme was agreed, as was a new financing scheme for adult and vocational training (see Chapter III).

Monetary conditions facilitated the slowdown

Monetary policy in Denmark has long been oriented towards the sole objective of maintaining the Danish krone not only within narrow bands in the context of the European Exchange Rate (ERM) mechanism, but targeted on the central parity rate with the Deutschemark. Once parity had been achieved, it was strictly adhered to, with deviations limited to less than ½ per cent. Although Denmark had obtained an "opt out" of the Euro in 1992, it was nevertheless affected by the adoption of the Euro, by virtue of its membership of the ERM. Until the Euro was actually introduced on 1 January 1999, there remained uncertainties about how smooth the transition would be, and the situation was complicated by the financial turbulence associated with the Asian and Russian crises. In September 1998, Denmark reached agreement with the European Central Bank to participate in a new fixed exchange rate mechanism, ERM2. The agreement

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narrowed the bands of allowed exchange rate fluctuations between the Euro and the krone to ±2¼ per cent around the central parity rate. It provides the same support arrangements in the event of exchange rate pressures as were embodied in its predecessor. The introduction of the Euro in fact went smoothly, and short-term interest-rate differentials between Denmark and Germany have narrowed even further, and were 35 basis points at the end of March, indicating that the arrangements are considered fully credible by financial markets.[2]

Given the fixed-exchange-rate policy, short-term interest rates and exchange rates are effectively exogenous to the Danish economy. The Nationalbank has followed the official interest-rate decisions of the Bundesbank and subsequently the European Central Bank (ECB), except when unilateral shifts have been necessary to maintain exchange-rate parity, as happened in 1998. Falling interest rates had provided a good deal of impetus to the recovery, particularly from 1995 onwards. By the beginning of 1998 short-term market interest rates were down to around 4 per cent, although they rose in line with official movements necessitated by the turbulence in world currency markets, following the depreciation of the Russian rouble (Figure 3). By late 1998, however, the period of instability had subsided, and the Nationalbank was able to reduce the discount rate from 4¼ to 4 per cent in early November and to 3½ per cent in early December, after a co-ordinated move among Euro-area central banks to lower rates. Denmark lowered its official interest rates again at the beginning of February 1999, as the krone strengthened against the Euro and foreign-exchange reserves were mounting. In April the ECB eased monetary policy, as concerns in the Euro area about inflation had given way to disquiet about slow growth, and Danish monetary policy remained similarly accommodative until November, when the ECB tightened by reversing the April move. Since then, Denmark also followed further ECB moves towards tightening in February, March and April this year of ¼ percentage point each.

The pattern of long-term interest rates has been quite different. Yields on 10-year government bonds continued their decline throughout 1998 before rising sharply from the second quarter of 1999, probably in part reflecting "year 2000" uncertainties, since they have fallen back around 30 basis points since the beginning of the year. In any case, the movements in Danish yields have closely mirrored international trends over this period. In contrast, yields on 30-year mortgage-credit bonds continued their trend decline from their peak of 10 per cent at the beginning of 1995, to reach 7 per cent by the middle of 1998. Although they bounced up around the time of the global financial stress in October 1998, they then resumed their downward path, until bottoming out in the summer of 1999. During the second half of 1999, they edged up again, although most recent data show a declining pattern again. These long-term mortgage rates are particularly important in the transmission mechanism from the financial market to real activity in Denmark (see Box 1). House values have a strong effect on households' spending decisions, and lower mortgage interest

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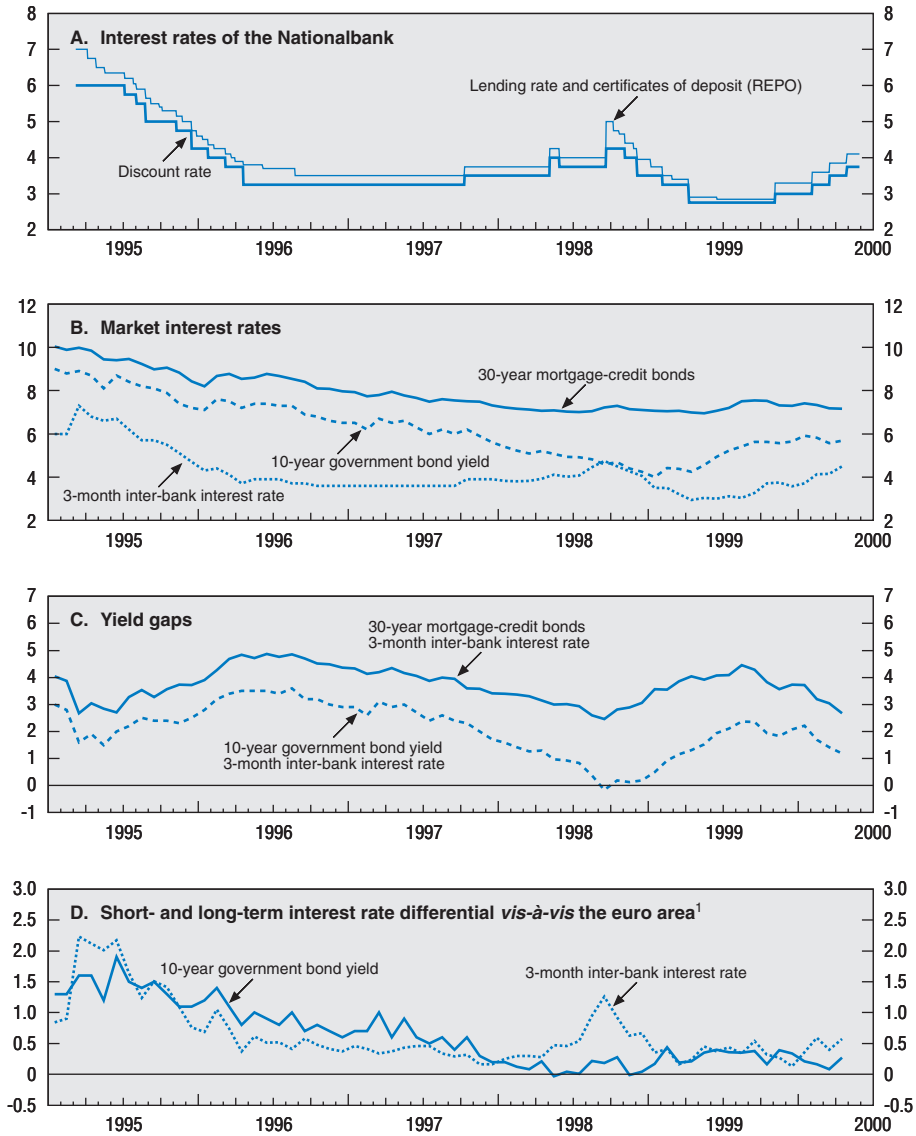
Figure 3. **Interest rates**

Per cent



1. For the euro area, before 1999, German government interest rates are used.

Source: Statistics Denmark; OECD.

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Box 1. Financing owner-occupied housing

Financing of owner-occupied housing has played a major role in macroeconomic developments in Denmark. Owner-occupied housing is generally financed through nominal mortgage-credit bonds with a maturity of 30 years and with maximum loan-to-value ratios of 80 per cent. These mortgages typically involve fixed quarterly payments, where the proportion of principal repayment increases over time. Tax deductibility of interest payments provides an incentive for borrowers to front-load interest payments. These mortgage bonds also typically give the borrower a call option to repurchase the bond at par before maturity. This feature is reflected in issue price of the bond which is set at a discount to generate a "call premium" to lenders. In times of falling interest rates and bond yields, households would generally exercise these call options, redeeming their bonds and refinancing at lower yields.

Given the fall in bond yields since 1993, many households have found themselves able to refinance at lower interest rates. But the redemption rate was also particularly high in late 1993 and 1994 as a result of policy changes. Denmark has a history of influencing economic activity through regulating the mortgage market. In 1987, the issue of 30-year mortgage-credit bonds with fixed total quarterly payments were prohibited; all new mortgages were limited to 20-years' duration and with a requirement that 40 per cent of the loan be amortised in equal increments over the life of the loan. This measure significantly increased the after-tax payments on new loans. In 1993, these regulations were reversed, and the previous financing arrangements were again permitted. In addition, capital losses related to the older stock of cash loans remained tax deductible even after their conversion. For households locked into loans from the early 1980s when the interest rate was 20 per cent, the incentive to refinance was particularly attractive.

A further feature is the financial innovation offered by mortgage institutions in the last two or three years, amounting to floating-rate loans – by providing refinancing through shorter-term (one- to two-year) mortgage bonds within an overall long-term financing arrangement with a mortgage-credit institution. These have rapidly become more popular, and by the third quarter of 1999 around half of all new mortgage loans were of this type. This has made it possible for homeowners to take advantage of the currently large gap between short- and long-term interest rates. With a greater reliance on floating interest rates, the transmission mechanism from monetary policy to real activity is likely to change and probably make short-term rates more important in the future.

rates affect consumption not only through a direct income effect but also by their influence on house prices and therefore wealth. In addition, with the increased popularity of floating-rate mortgages, a considerable additional stimulus to the domestic economy came from the scope to exploit the significant margin between rates on longer- and shorter-term financial instruments.

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Although the Danish krone is fixed to the Euro, the effective exchange rate played a helpful, albeit exogenous, role in stabilising the economy, first by bringing about the necessary slowdown in the economy, and then by encouraging a switch towards tradable-goods sector as the economy cooled, making it easier to avoid a hard landing. With less than half of goods sales abroad and only one third of services exports going to the Euro area, shifts in the exchange rate against currencies other than the Euro can have significant effects (Figure 4). The appreciation of around 8 per cent in the effective exchange rate between mid-1997 and the end of 1998 acted to take some of the steam out of the economy, as well as tempering the effect of import prices on inflation. However, the exchange rate peaked in the last quarter of 1998, and was around 7 per cent lower by the first quarter of 2000, helping to support the economy.

Growth weakened but is picking up again

The Danish economy was at its weakest around a year ago, with no growth in the last quarter of 1998 and a fall in the first quarter of 1999, although this brief dip in activity is obscured in the annual results which show 2.5 per cent growth for 1998 as a whole and 1.6 per cent for 1999 (Figure 5). Furthermore, the effects of the May 1998 conflict, estimated to have reduced GDP in 1998 by ½ per cent, mask the sharpness of the slowdown. Domestic demand growth remained high at more than 4 per cent in 1998 but evaporated in 1999, while net exports of goods and services brought down overall GDP growth by almost 1.8 percentage points in 1998 before producing an even larger positive contribution of 2.2 percentage points in 1999.

Exports and imports

External conditions were not favourable to the expansion of exports in 1998, given the slowdown in global trade and the appreciation of the effective exchange rate. Exports of goods, which make up 80 per cent of total exports, grew by only 2 per cent in 1998, much less than export market growth of 8 per cent, leading to a sharp decline in export performance (Table 2). This was partly due to the labour market conflict in spring 1998. Exports of services showed a smaller loss in performance in 1998, both because they increased by more than goods and because market growth was lower. In contrast, and capitalising on the improvement in competitiveness, exports of both goods and services expanded at a strong and steady pace during 1999, reaching an overall growth rate of 7 per cent, exceeding export market growth and thus regaining some market share which had been lost partly as a result of the disruption to production in the previous spring. Such high growth rates of exports have not been seen since the beginning of the recovery in 1994. Imports of goods and services showed a reverse pattern, as would be expected given the reinforcing effects of both domestic demand developments and the effective exchange rate movements. Imports of services seem to

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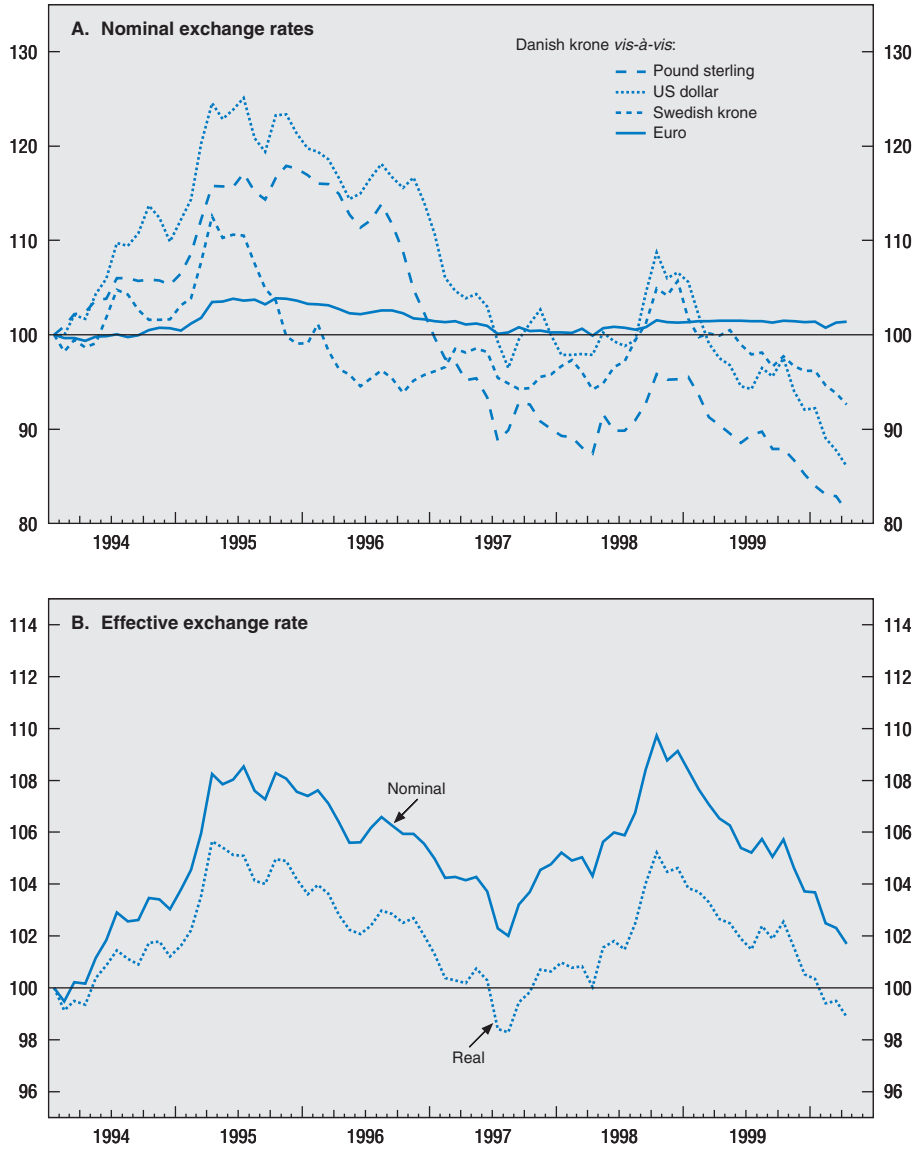
Figure 4. **Exchange rates**
January 1994 = 100



Source: OECD.

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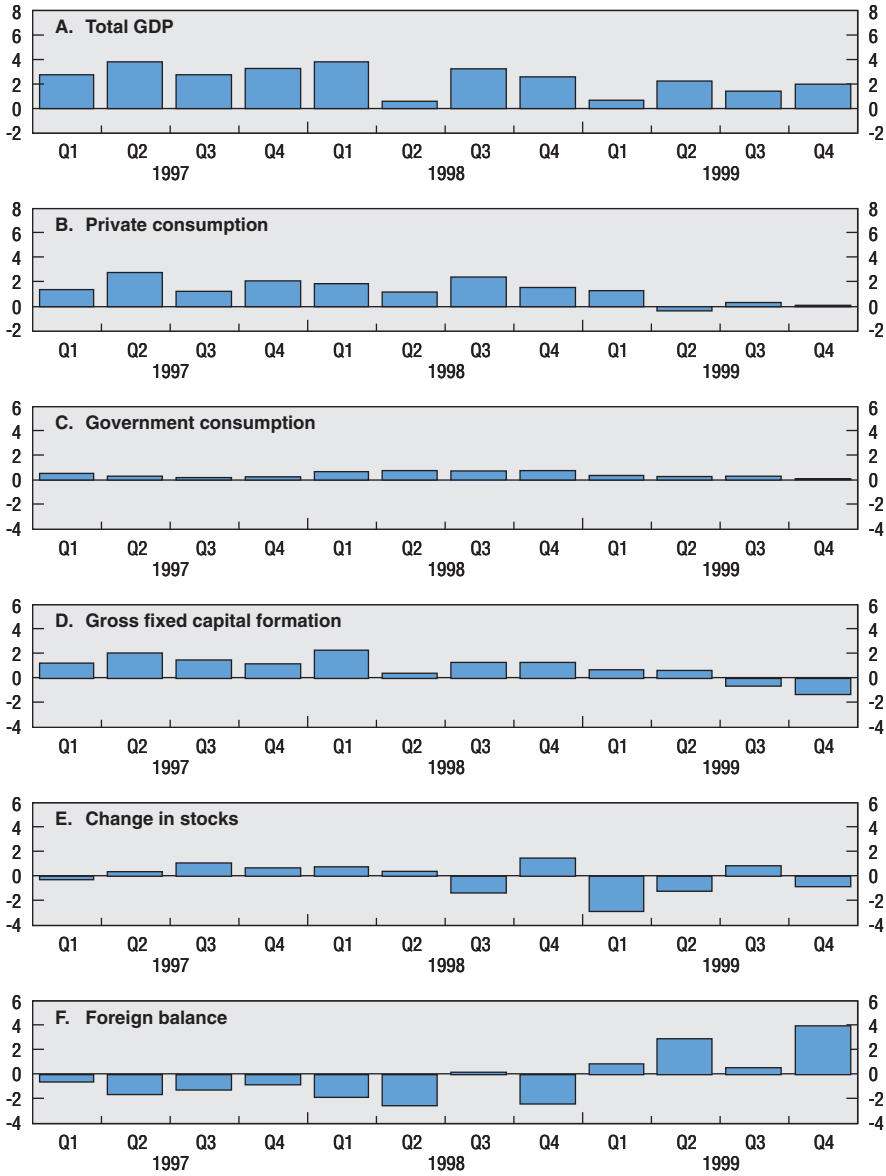
Figure 5. **Contributions to changes in GDP**

Per cent change over 4 quarters



Source: OECD.

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Table 2. **External indicators**
Percentage change over previous year

	1996	1997	1998	1999
Exports				
Goods				
Volume	2.8	5.8	2.0	6.4
Market growth	6.2	7.4	8.1	2.0
Performance	–3.3	–1.6	–6.1	4.4
Price changes	1.5	1.8	–1.2	0.9
Services				
Volume	10.1	–2.4	2.6	9.7
Market growth	5.2	10.2	3.8	3.4
Performance	4.9	–12.6	–1.2	6.3
Price changes	2.9	6.8	0.2	1.5
Goods and services				
Volume growth	4.3	4.1	2.2	7.0
Price changes	1.8	2.9	–0.9	1.1
Imports				
Goods				
Volume growth	3.1	7.4	6.3	2.7
Price changes	–1.0	4.2	–1.2	–1.7
Services				
Volume growth	5.6	10.7	11.4	–4.2
Price changes	4.2	3.6	–2.4	6.1
Goods and services				
Volume growth	3.5	8.0	7.3	1.3
Price changes	–0.1	4.1	–1.4	–0.1
Effective exchange rate	–0.8	–1.9	1.5	–1.0
Relative export prices (manufacturing)	–0.5	–1.8	3.2	–0.3
Relative unit labour costs (manufacturing)	–4.6	1.6	2.6	–0.2

Source: OECD.

have been particularly sensitive to these two factors, growing more 11 per cent in 1998 before falling by 4 per cent in 1999. While services remain less than a quarter of the total, these shifts, together with slower growth in goods imports, resulted in minimal growth in overall imports in 1999.

Private consumption

Consumer confidence fell sharply in the second quarter of 1998, and generally worsened as the year went on, reflecting both the impact of the tax reforms agreed in the early summer, and perhaps some worries about international developments such as the Russian debt crisis and uncertainties about interest-rate prospects. The combined effect of the Whitsun package and slow employment increases, together with the rise in inflation led to a sharp decline in the growth of real household disposable incomes in 1999. While to some extent this was

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buffered by a fall in the household savings rate, the result was nevertheless a fall in private consumption in the first half of 1999 before a weak rebound in the second half, so that for the year as a whole, consumption edged up by 0.7 per cent, well below the growth rates of 3½ per cent in the previous two years. Consumers became gradually more optimistic during most of 1999, but their confidence weakened again towards the end of the year.

Capital formation

After a very strong increase in 1997, the growth in gross fixed capital formation slowed to 6.7 per cent in 1998 and then it actually fell in 1999. Some of this decline can be attributed to "lumpy projects" although with significant expansion in capacity over the course of the recovery, a slowdown was probably inevitable, even if the economy had not weakened as it did. Residential construction increased by 5 per cent in 1998 and remained flat in 1999. Non-residential building and construction grew by almost 4 per cent in 1998 and then fell dramatically, reflecting completion of major construction projects, such as the bridge over the Great Belt, and scaling back of investment in the North Sea oil fields. Machinery and equipment investment growth also slowed after its 1997 peak, but it remained strong in 1998 at 9 per cent and 1999 at 5 per cent. These rates suggest that considerable additional capacity has been added to the economy. Public construction fell sharply in 1998, leading to a slight dip in overall public investment, reflecting a deliberate effort to dampen overheating in the construction sector. Other public investment then fell in 1999, reducing overall public investment further.

Production

Value added by sector has shown a mixed picture during the last two years. Although output in the economy as a whole increased by 2.7 per cent in 1998 and by another 1.2 per cent in 1999, the sector breakdown reveals significant differences (Table 3). The manufacturing sector had a difficult year in 1998, with weak exports and slowing private consumption along with the industrial action in the Spring. Production fell sharply in the fourth quarter of 1998 and then slightly further in the first quarter of 1999, before rebounding again. Demand however was even weaker than production in 1998 – orders had already peaked in January 1998, before declining by 8 per cent by the end of the year (Figure 6). Consequently, stocks built up quite sharply. Business confidence weakened significantly in late-1998 and worsened further in early 1999, and a particularly sharp downward adjustment in inventories was made in the first and second quarters of 1999. Order for exports started to recover at the beginning of 1999, although orders for the domestic market showed a clear improvement only in the second half of 1999. With the earlier inventory build-up largely eliminated, manufacturing sales volumes picked up again quite sharply in the second half of the year, reflecting in

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Table 3. **Value added and employment growth by sector**
Percentage change from previous period

	Share in 1995	1996	1997	1998	1999
		Real value added by sector			
Agriculture and mining	4.6	1.9	4.0	2.8	1.4
Manufacturing	25.9	0.0	3.0	2.6	–1.2
Electricity and water supply	2.0	12.6	–3.6	–5.6	–7.4
Building and construction	6.9	7.1	3.3	2.2	–6.2
Wholesale, retail trade, hotels and restaurants	13.9	4.5	4.4	1.1	4.2
Transport, storage and communication	8.9	6.1	2.1	5.0	3.4
Financial intermediation, housing and business activities	17.7	0.4	3.6	3.7	5.8
Public and personal services	20.1	3.0	2.1	3.0	0.4
Total, all sectors	100.0	2.7	3.0	2.7	1.2
		Employment (National Accounts basis)			
Agriculture and mining	4.6	–4.0	–2.9	–3.0	–1.4
Manufacturing	17.7	0.2	–0.4	2.2	–2.3
Electricity and water supply	0.7	0.0	0.0	–4.4	–6.2
Building and construction	5.8	0.8	2.3	3.9	1.6
Wholesale, retail trade, hotels and restaurants	18.5	4.8	2.0	1.7	1.0
Transport, storage and communication	6.7	0.6	1.5	1.0	0.4
Financial intermediation, housing and business activities	10.8	–0.1	2.9	4.2	4.9
Public and personal services	35.1	1.5	1.5	2.2	1.2
Total, all sectors	100.0	1.3	1.2	2.1	0.7

Source: Statistics Denmark.

particular an improvement in export sales to the rest of Europe. Business confidence has been picking up steadily since the first quarter of 1999, indicating an expectation that better conditions will continue.

Other sectors of the economy have had mixed fortunes. Construction showed a particularly steep decline in output in 1999, reflecting the pattern of gross fixed capital formation. Services, in contrast, generally fared better, led by transport and communications. A large part of this is related to exports of shipping and related services, an area where Danish companies are strong. Financial and business services also enjoyed buoyant growth. Expansion of computer services and the high rate of investment in computing technology would go some way to accounting for this. Trade, hotels and restaurants had lacklustre growth in 1998, but surprisingly, given the private consumption profile, activity in the sector picked up sharply in the first quarter of 1999 and remained robust throughout the year. Foreign visitors, taking advantage of more favourable bilateral exchange rates, could explain this performance.

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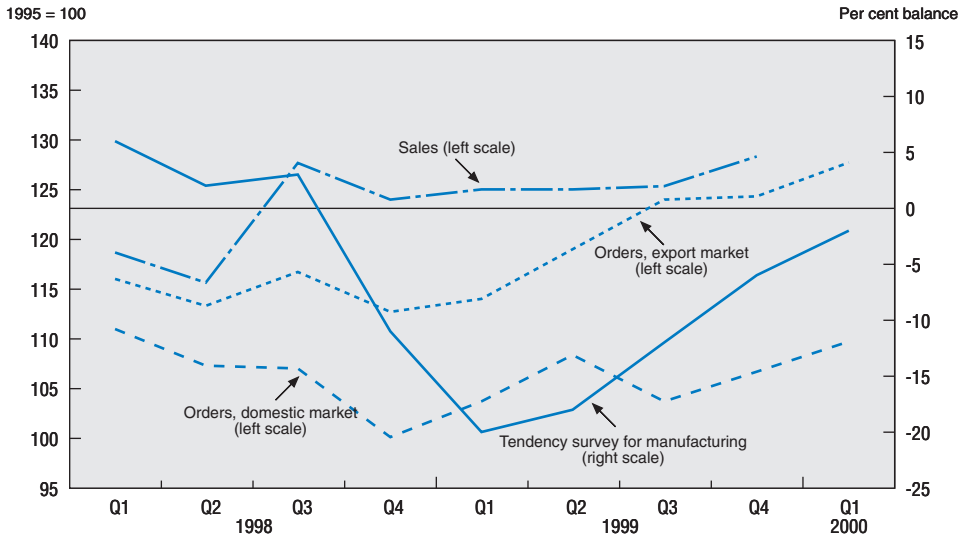
Figure 6. **Sales, orders and business confidence**



Source: Statistics Denmark.

Labour-market developments

Given the developments in the real economy, the employment and unemployment data show some puzzling features. Employment (measured on a national accounts basis) grew by 2.1 per cent in 1998, considerably stronger than the previous two years (Figure 7). Despite slower growth in production in 1999, private-sector employment grew by almost 1 per cent. However, these quarterly national accounts data are based on the *Arbejdsmarkedets Tillaegspension* (ATP) statistics (which are based on payment of statutory pension contributions) and show some surprising movements during 1999. But while the third-quarter ATP figures still showed a year-over-year gain of 1 per cent, by the fourth quarter employment had fallen 0.6 per cent from the previous year. The flip side of this employment pattern however has been a dramatic slowdown in productivity growth (output per person) to only around ½ per cent in each of the last two years.

The evolution of unemployment shows a different pattern depending on which measure is used. Despite the weak output growth, registered unemployment continued to show a steady decline throughout 1999 to reach 150 000 persons (5.4 per cent of the labour force) by December, a fall of around 15 000 (0.6 percentage point) from 12 months earlier, while the most recent data, for

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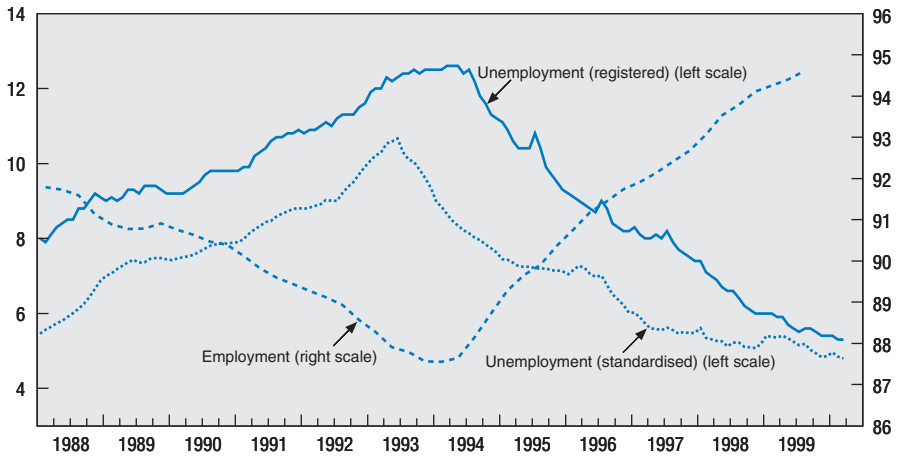
Figure 7. **Employment and unemployment rates**
Percentage of labour force



Source: Statistics Denmark; OECD.

February 2000, show the rate continuing to edge down. In contrast, the survey-based standardised unemployment rates show a more gentle decline of around 0.2 percentage point between the fourth quarter of 1998 and the fourth quarter of 1999. This suggests that some of the fall in registered unemployment may reflect policies put in place to tighten eligibility rather than being a reliable indicator of the degree of labour market slack and therefore of pressure on wages. Indeed, the narrowing of the gap between the two measures maintain a trend that began in the mid-1990s.

Price and wage developments

Nonetheless, wage inflation has been a major concern for policy makers, as an acceleration was observed in hourly earnings, which had drifted up from 3.9 per cent growth in 1997 to 4.3 per cent on average in 1998 and a peak rise of 4.7 per cent in the first quarter of 1999 (Figure 8). Compensation per employee also rose sharply as did unit labour costs, although their growth rates had both peaked by mid-1998. In the event, year-on-year hourly earnings growth moderated quickly, returning to around 4 per cent during the rest of 1999. Given the link between strong wage growth and domestic inflation, accelerating price inflation looked likely throughout 1998 and most of 1999. Headline inflation, as measured

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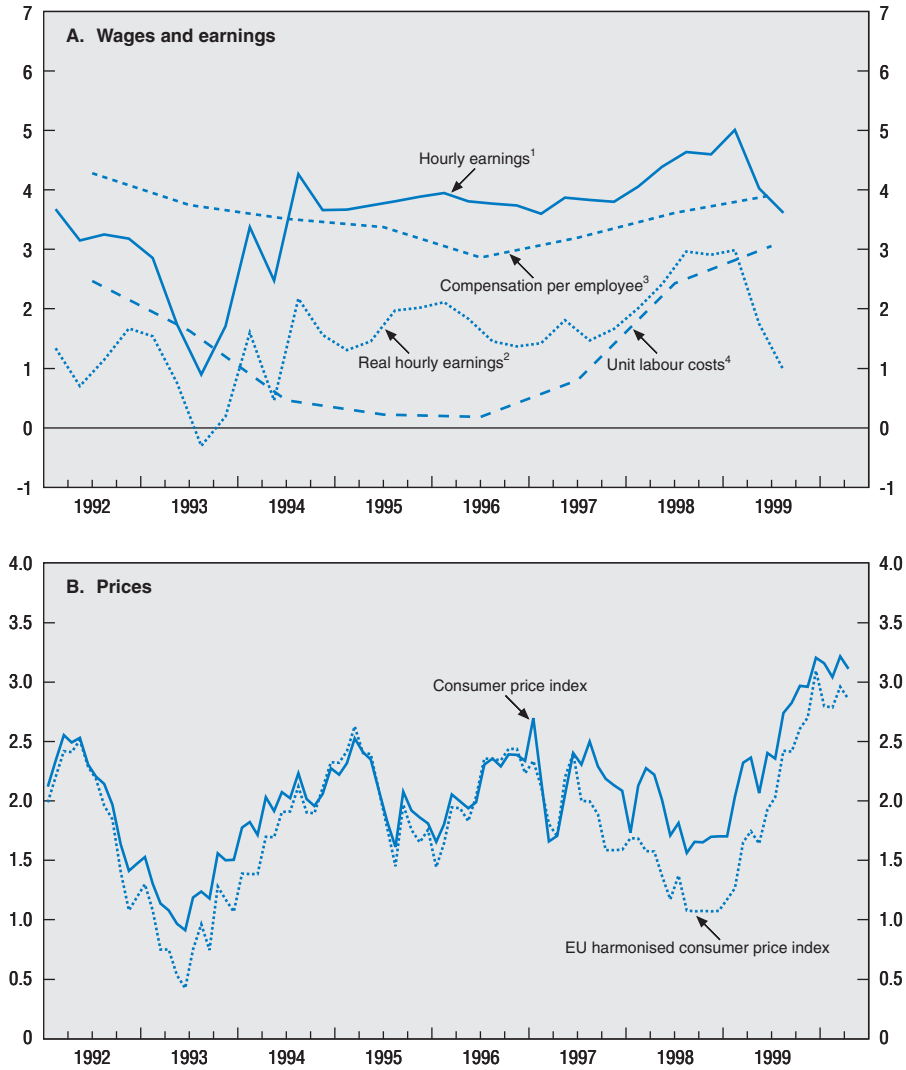
Figure 8. **Wage and price inflation**
Year-on-year percentage change



1. Manufacturing.
2. Deflated by the CPI.
3. Business sector.
4. Total economy.

Source: OECD, *Main Economic Indicators, National Accounts.*

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by the consumer price index, continued to rise throughout 1999, ending the year at 3.2 per cent on a year-on-year basis. Oil prices and tax increases played a significant part here, as did rising food prices, especially for pork products. On the other hand, falling prices for telecommunications have helped to keep the CPI increases a little lower than they would otherwise have been. Import prices fell in 1999, albeit less than in 1998, and this also helped to temper the overall result. Domestic, market-determined inflation, which to an extent reflects the flow-on effects of wages into prices, had peaked and started to slow by mid-1999. Harmonised consumer prices breached the Euro convergence criterion in January 2000 but are expected to again fall to below the ceiling rate later this year. These developments were foreseen and should not imply any impediment to possible Euro membership.

House prices

Real house prices continued to rise in 1999 (Figure 9). In the third quarter of the year (the latest figures available), they still increased by 1.5 per cent and were more than 6 per cent above year-earlier levels. With very weak private consumption from the beginning of the year, the Whitsun package, and long-term mortgage-credit bond yields edging up around mid-year, house prices were widely expected to level out or even fall. The main explanation seems to be that households have been able to take advantage of the gap that reappeared between short- and long-term interest rates in their mortgage-financing decisions. The Nationalbank observed that towards the end of 1999 this made it possible for a household to reduce its first-year mortgage payments on a purchase by up to a quarter compared with financing with a traditional long-term fixed-rate loan. Recent rises in short-term interest rates should mean that this opportunity has now diminished, although confirmation in the form of stabilising house prices is not yet discernible.

The balance of payments

For the first time since the 1980s, a current-account deficit developed in 1998, despite the steady improvement in the government's budget position. It reflects the net effect of private-sector investment and savings decisions, which in turn reflect to some extent, normal cyclical developments (Figure 10). The private savings rate fell by 1.5 percentage points in 1998 and by a similar amount in 1999, while the investment rate rose by 0.8 point in 1998 and fell by 0.6 point in 1999. Altogether, these shifts left a gap between national savings and investment of ½ per cent of GDP in 1998, but lower investment reversed the gap in 1999. Several factors have facilitated a particularly rapid return to a surplus of more than 1 per cent of GDP in 1999. Oil exports rose in volume terms, as North Sea production increased by around 30 per cent, and these also benefited from the substantial

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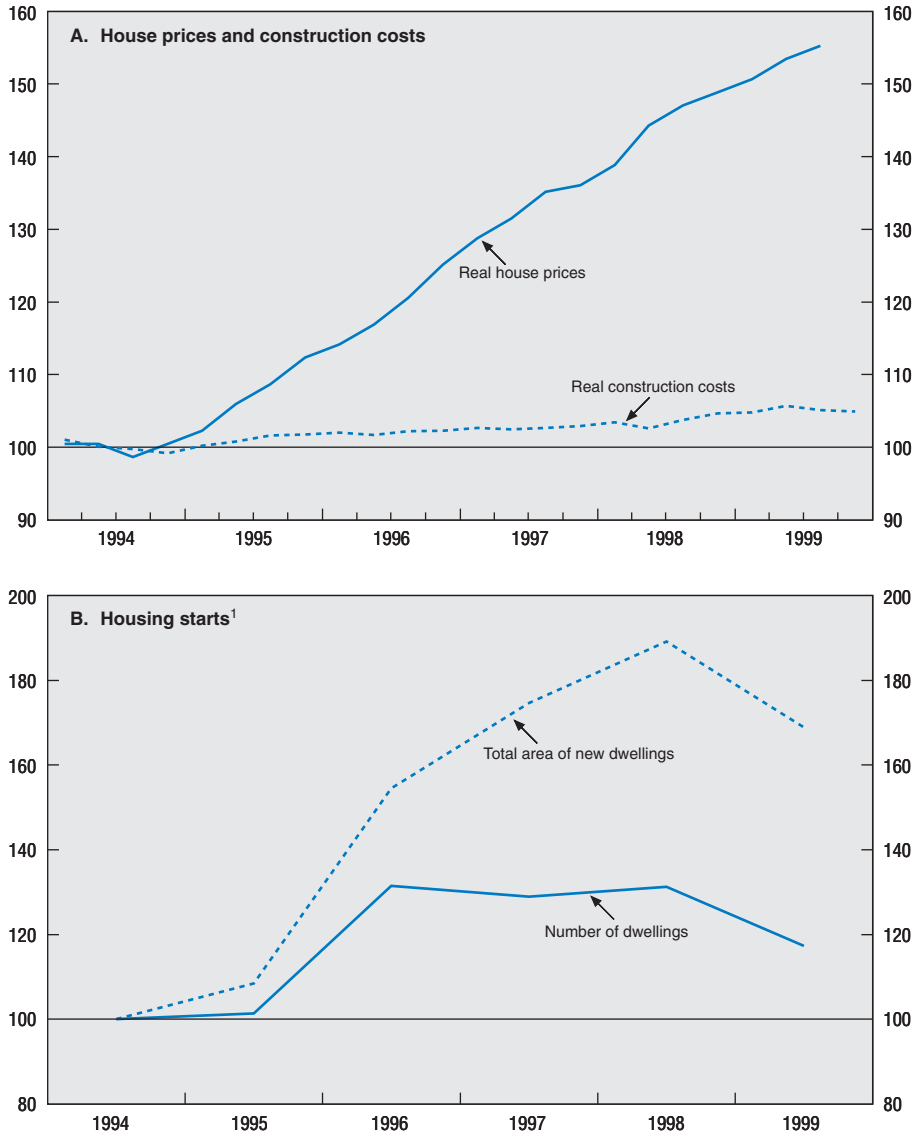
Figure 9. **Housing indicators**
1994 = 100



1. Estimates for 1999.
Source: Statistics Denmark.

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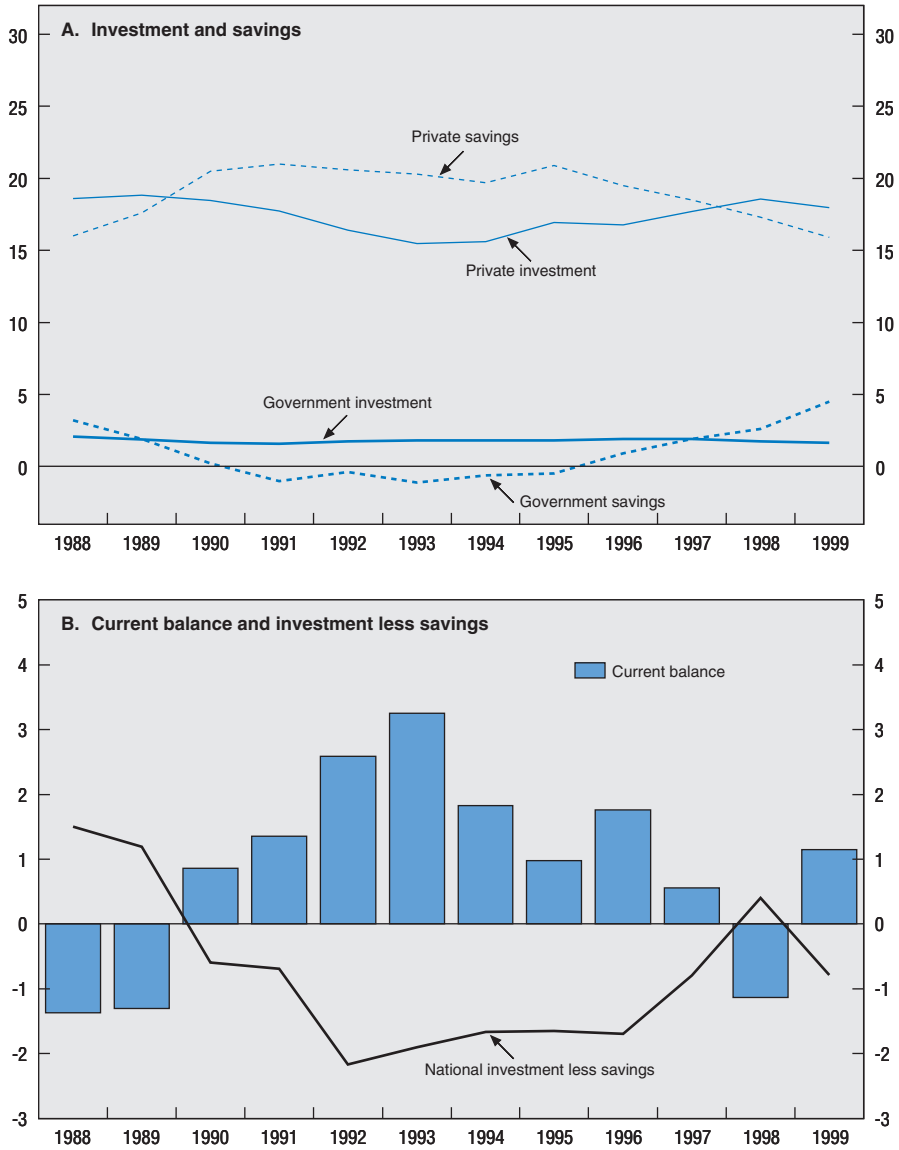
Figure 10. **Current account balance, investment and savings rates**
Percentage of GDP



Source: Statistics Denmark; OECD.

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rises in oil prices, especially in the last quarter of the year. In contrast, Denmark imports refined products, and refining costs have not changed significantly, which helped to keep import prices lower, providing a significant terms-of-trade improvement for oil products as a whole. A further boost to export values came from a turnaround in pork prices. Manufactured goods prices fell by 0.8 per cent, while the price decline was slightly larger for manufactured imports. Services showed a particularly marked terms-of-trade improvement, as prices for exports firmed by 1.7 per cent and fell for imports by the same amount.

The short-term outlook

Conditions are in place for gradual increase in growth

External help

With a soft landing achieved, there is a high degree of consensus in Denmark that the conditions are in place for a gradual and modest pick-up in activity (Table 4), with the OECD projections showing GDP growth of 2¼ per cent in 2000 and 2½ per cent in 2001. The economy will clearly benefit from the stronger growth emerging in Europe and particularly from the current strength of the German and Swedish economies, Denmark's two largest trading partners. This effect can already be seen in the export data for the last months of 1999, and in the return of the current account to a surplus position. Recent declines in the effective exchange rate will also give a significant boost to exports – even on the assumption of a constant effective exchange rate for the rest of the year. Overall, exports are projected to grow by more than 6 per cent in both 2000 and 2001. Imports are also projected to rebound from their exceptionally low growth rate in 1999, to rise by 4½ per cent in 2000 and 2001. Nonetheless, with the net foreign balance contributing around 1 per cent of GDP growth, the current account surplus is likely to increase to almost 2¾ per cent of GDP in 2001.

Domestic demand, however, is likely to remain somewhat subdued, although it is projected to gradually pick up. Slower real wage growth and modest growth in employment should temper household income growth, while the Whitsun package reduces further the tax deductibility of mortgage interest payments. In addition, house prices are expected to fall from their presumed peak in late 1999, which would act to constrain private consumption growth, since they have a significant wealth effect, making households more hesitant to spend. Higher interest rates (driven by interest-rate developments in the Euro area) will also play a role and perhaps have a stronger negative impact on activity than in earlier years because of the increased proportion of floating-rate mortgages. Nevertheless the strong second half of 1999 provides some momentum to private consumption in 2000, when it is projected to rise by almost 1½ per cent, with a further

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Table 4. **Outlook for 2000 and 2001**
Percentage change from previous year (unless otherwise noted)

	1998	1999	2000	2001
Private consumption	3.5	0.7	1.4	1.8
Government consumption	2.9	1.1	1.1	1.0
Gross fixed investment	6.7	–0.7	1.2	1.6
Housing	4.9	0.0	0.0	–2.7
Business	8.3	–0.5	0.4	2.6
Government	–2.7	–4.0	10.9	4.0
Final domestic demand	4.1	0.5	1.3	1.6
Change in stockbuilding[1]	0.4	–1.0	0.0	0.0
Total domestic demand	4.5	–0.5	1.3	1.6
Export of goods and services	2.2	7.0	6.5	6.3
Imports of goods and services	7.3	1.3	4.4	4.6
Net foreign balance[1]	–1.8	2.1	1.0	0.9
Gross domestic product (constant, market prices)	2.5	1.6	2.2	2.4
Memorandum items				
Private consumption deflator	1.6	2.5	2.8	2.5
GDP price deflator	2.0	2.6	2.5	2.5
Current account, per cent of GDP	–1.1	1.1	2.1	2.7
Total employment	2.1	0.8	0.8	0.8
Labour force	0.7	–0.2	0.7	0.7
Unemployment rate, per cent of labour force	6.4	5.5	5.4	5.4
Effective exchange rate	1.5	–1.0	–3.3	–0.2
Interest rates :				
Short, per cent	4.1	3.4	4.7	5.5
Long, per cent	4.9	5.0	6.2	6.5
Oil prices, US$ per barrel[2]	12.58	17.3	25.2	22.7
Government balance, per cent of GDP	0.9	3.0	2.8	2.8

1. As a per cent of GDP in the previous year.
2. Crude oil prices are OECD average import cif unit prices, including cost, insurance and freight, but excluding import duties.

Source: OECD.

slight pickup in 2001. At the same time, residential investment is expected to be flat this year and contribute negatively to growth next year, as housing starts continue to fall and existing house prices soften. Business investment is projected to pick up only modestly. In any case, capacity has been significantly expanded in recent years, and there would currently seem to be scope for expansion without capacity constraints on the production side becoming a problem. Public consumption is projected to rise by slightly more than 1 per cent, reflecting public employment growth. In 2001, it is assumed that the government will take the steps necessary to meet its target of limiting consumption growth to 1 per cent of GDP.

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In line with the outlook for activity, employment is expected to continue expanding at around ¾ per cent in 2000 and 2001, with private-sector increases slightly slower and public-sector growth slightly faster, reversing the 1999 outcomes. With weak productivity gains in 1998 and 1999, it seems likely that labour is to some extent currently under-employed, allowing for some expansion to take place without a sharp pick-up in employment. Thus, output increases should be met by steady improvements in labour productivity growth. Unemployment is expected to remain close to current levels with the average rate over the projection period just under 5½ per cent, slightly lower than the average rate for 1999, since unemployment fell so sharply as that year progressed.

Wage agreements

The centralised wage agreements reached in January 2000 bode well for more modest wage outcomes in 2000 and improving international competitiveness, compared with Denmark's trading partners. A new element in the centralised bargaining process was the initial "climate talks". Representatives of the unions, employers and the government reviewed likely wage outcomes in other countries and provided an assessment of what a reasonable outcome of these wage negotiations would be. This process seems to have played a constructive role. It helped to moderate expectations of wage settlements and to restore a more co-operative approach after the bruising experience of the 1998 negotiations. The main elements of the agreements reached were that they would cover a four-year period and provide for a full sixth week of holidays. The extra days already introduced in the 1998 settlement would be netted out, with no changes in the first year, most employees receiving four extra days in the second year and a fifth day added in 2003. The net effect is to reduce annual working time on average by around three days when fully implemented. In addition, supplementary pension contributions were raised, to 9 per cent for most employees by 2003. It was also agreed to raise the minimum wage by around 2½ per cent in each of the first three years of the agreement, and it will be renegotiated for the fourth year without the right to strike. Against that backdrop, localised wage bargaining is expected to produce reasonably modest wage settlements and lead to moderate increases in private sector compensation per employee of 4 per cent in 2000 and slower growth in 2001 of around 3½ per cent reflecting the lower working hours. Public sector wages are expected to increase, at a rate closer to 3½ per cent in 2000 and 4 per cent in 2001. But there are uncertainties associated with the new framework for linking public sector wage growth more closely to performance, which may make it harder to contain aggregate wage formation and in 2002, an expected corresponding reduction in annual working hours to match the private-sector settlements.

Although the harmonised price index[3] breached the EU convergence criterion for inflation in January, it is expected to decelerate in the first half of 2000.

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(In any case, the convergence criteria will become easier to meet, as inflation tends to pick up slightly elsewhere as Europe's prospects brighten.) Inflation, as measured by the CPI, is likely to slow as the effects of higher oil prices and indirect taxes pass, and the most recent data is encouraging. More moderate wage growth will help to slow the inflation rate over the short term. In any case, the economy is projected to continue performing at close to its potential output level, and with no obvious source of overheating, price pressures should ease and inflation slowly return to lower rates.

Risks and policy challenges

Overall, the Danish economy seems to be in good shape and to have good prospects over the coming two years for moderate, balanced growth and movement back towards price stability. The main macroeconomic policy challenge will therefore be to keep the economy operating on a reasonably even keel, which current policy settings should permit. However, there are a number of risks and policy challenges that may affect the outlook. Future outcomes depend heavily on external developments and particularly the sustained recovery in exports. The projections have been built on the assumption of no changes in the effective exchange rate. If however, the effective exchange rate strengthens, slower growth in exports, mainly in 2001, would be likely. An adjustment to production or a build-up of inventories would ensue, since slow-growing domestic demand would not facilitate significant switching to the local market. International developments could also affect interest rates, since if the European Central Bank were to raise interest rates by more than already assumed in the projections, the Danish Nationalbank would almost certainly feel obliged to follow suit. With a higher proportion of floating-rate mortgages, higher short-term rates would be more likely to feed negatively into households' spending decisions.

The two main domestically generated risks would seem to be a sharp correction in house prices and higher-than-expected wage settlements. The last two recessions in Denmark each saw real house prices fall by around 30 per cent, although the overall economic circumstances were very different. In addition, the measures taken in the Whitsun package should have reduced the risk of a sharp correction, and only a very small decline is assumed in the projections. However, a more significant fall could have depressing effects on consumption, particularly if it put some households into a negative-equity situation. Were house prices to fall very sharply, it could possibly be enough to trigger a full recession. The risk of excessive wage settlements has receded with the modest agreements struck in January at the national level, and labour demand remains relatively weak. However, the more positive economic data that have recently become available and buoyant export markets may lead to additional pressure in local bargaining and an upward drift in average wages, weakening competitiveness

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and profitability, thereby feeding through into slower employment growth and thus higher unemployment.

The referendum on joining the euro, scheduled for 28 September, presents a degree of risk. Current poll indications suggest that public opinion is finely balanced, and the outcome is unpredictable. An affirmative outcome of the referendum would immediately eliminate the risks of speculative attacks on the krone and consequently all currency risk premia against the euro in interest rates and prices of financial assets. But if adoption of the euro were rejected, this could raise questions about the extent to which the electorate supported continued participation in ERM2. This would likely be tested by financial markets, and a period of financial-market turbulence could ensue. In such a situation the Nationalbank might need to raise interest rates in order to demonstrate Denmark's continued commitment to its fixed exchange-rate regime.

Despite these risks, Denmark is unusually well placed at the present moment, because it has a rare opportunity to get into "phase" with developments in the Euro area, after years of being at a different stage of the cycle, which has at times made macroeconomic policy settings more difficult to manage. Keeping the economy on a steady path will not only produce greater convergence with Europe, but also avoid painful economic adjustments.

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II. Maintaining sound public finances

Denmark has achieved considerable fiscal consolidation in recent years; the general government financial balance has improved from a deficit of close to 3 per cent of GDP in 1993 to a surplus of 3 per cent in 1999. With output being close to potential, a surplus near this size should be sustainable over the course of the economic cycle. It should also provide a reasonable cushion under the arrangements of the Stability and Growth Pact, even if the automatic stabilisers are estimated to be relatively strong in Denmark (Dalsgaard and De Serres, 2000). The improvement in public finances is in large part due to lower structural unemployment and lower debt-servicing costs. Expenditure decisions, on the other hand, have been expansionary during the latest upswing but, according to official estimates, have been offset by tax increases. Thus, while discretionary fiscal policy has been broadly neutral over the period 1993 to 1999 as a whole, it has exerted upward pressure on the expenditure- and the tax-to-GDP ratios.

Fiscal policy now faces the challenge of maintaining the current structural surplus, which may prove equally difficult as the past process of reducing the structural deficit. With positive contributions from falling structural unemployment and interest rates coming to an end, this requires a more rigorous fiscal policy stance in the years to come than during the past upswing when spending concessions were routinely granted. As tax rates and the overall tax burden are probably already at levels restricting the supply potential of the economy, efforts to ensure that fiscal policy is sustainable in the long term should concentrate on the expenditure side of the budget.

Notwithstanding the current favourable budget position, a number of risks should be mentioned. First, the current budget surplus is crucially dependent on keeping structural unemployment low. Second, the still rather high gross public debt ratio and high marginal tax rates imply that the starting point may be less favourable than appears at first glance. Third, as in most other OECD countries, changes in the age composition of the population will gradually increase dependency ratios. Finally, the past record of inadequate expenditure control and the continued pressure to extend coverage and standards for a broad range of services put into question the will and ability of policy makers to undertake the appropriate adjustments in expenditure policy.

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The focus of this chapter is primarily the potential problems over the medium and longer terms of maintaining full public financing of generous transfer schemes and universal welfare services such as childcare, elderly care, education and health care. These services are predominantly financed through taxation and provided free, or at low user charges, to the public. A number of forces may work to increase expenditures or reduce tax bases in the future, thereby putting further upward pressure on already high marginal tax rates. First, with limited user charges for most services, upward expenditure pressures are likely to persist, leading to a need for non-price rationing. Second, this could be reinforced if below-average productivity gains in the production of public services cause relative costs to continue trending upwards. Third, the decentralised governance structure of the Danish public sector and the institutional settings regulating economic relationships between central and local governments may, judged by past experience, impede expenditure control. Fourth, the ageing of the population will gradually put pressure on both revenues and expenditures. And finally, a continued decline in average working time could further squeeze the tax base. The chapter addresses these problems and points to policy adjustments that could enhance expenditure control as well as efficiency in the provision of services.

Trends in public finances in recent decades

Since 1960, Denmark has been in a period of favourable demographic developments. The increase in the working-age population and the simultaneous change in its age composition have boosted labour supply by an average of ½ percentage point per year (Figure 11). Since tax revenue is closely linked to labour-supply growth, this has the same effect as a broadening of the tax base. However, the increase in numbers of workers has been more than offset by a reduction in average working hours. Total hours worked across the economy as a whole have fallen by around 8 per cent over the past 40 years. Although the reductions in annual working hours resulted from bargained agreements, they have still worked to narrow the tax base, with resulting negative consequences for public finances (see Box 2). Even if fully matched by productivity gains in the private sector, such gains would have virtually no beneficial impact on the general-government financial balance (because public-sector wages and transfers are effectively indexed to private-sector wages, offsetting the favourable tax revenue effect) and could not, therefore, compensate for the loss of tax revenue induced by shorter working hours. Under these conditions the long-term evolution of the tax burden closely mirrors what happens to the relationship between the total of public employment and transfer recipients on the one hand, and total employment on the other.

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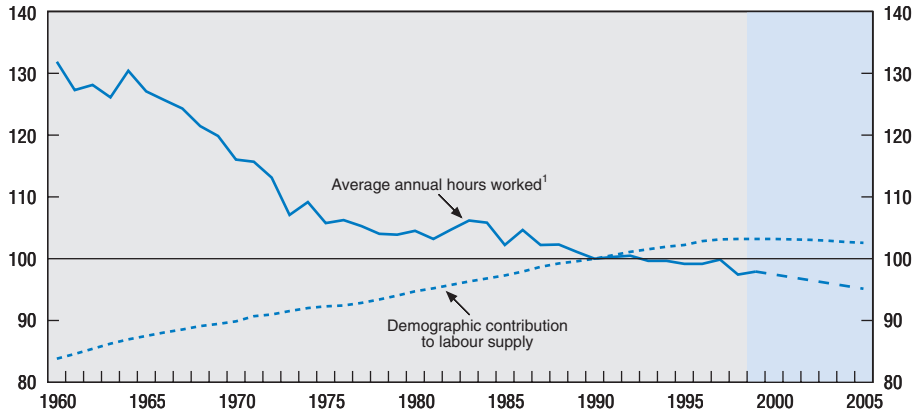
Figure 11. **Hours worked and demographic contributions to labour supply**
Indices, 1990 = 100



1. The dashed line in the period 1999 to 2005 indicates the effect of the extra week of holiday introduced in the recent wage settlements, assuming full carry-over to average hours worked.

Source: Ministry of Finance.

Despite the favourable demographic conditions, the tax burden has increased sharply in recent decades and, in addition, a relatively large gross public debt has accumulated. The tax burden has doubled since 1960, from 25 per cent of GDP to approximately 50 per cent. Although gross public debt has been reduced somewhat since 1994, it still amounts to approximately 50 per cent of GDP. These developments reflect the expansion of the public sector and the introduction of more generous transfer schemes, both in terms of broader accessibility and higher replacement rates. During the last four decades Denmark has experienced a strong and steady increase in the share of the working-age population receiving tax-financed public transfers as their main source of income; from 6 per cent in 1960 to 24 per cent in 1999 (Figure 12). In the same period the employment rate has remained remarkably stable around 72-73 per cent of the working-age population, abstracting from cyclical fluctuations. Thus, there has been a significant reversal within the group of non-employed: in 1960 the vast majority of the non-employed were self-supported (*i.e.* they lived in one-earner families), while today they receive transfer income of some kind. There has also been a change within the group of employed, with a smaller share working in the private sector and an increasing proportion working in the public sector.

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Box 2. Working time and public finances

As in several other OECD countries, most notably France, Italy and Sweden, demand for more leisure time has recently reappeared as a policy issue in Denmark. Reduced working time has adverse long-term impacts on public finances quite similar to those of increasing dependency ratios. The main difference is that, while shorter working hours impinge only on tax bases, higher dependency ratios would normally be accompanied by increased expenditure pressures as well.

The 2 per cent reduction of average annual working hours following from the extra week of holiday negotiated earlier this year (see Chapter I) will reduce production and income correspondingly in the long term. The short-term effects may be different. To the extent that companies and public-sector workplaces respond by recruiting additional staff, production may not fall commensurately in the short term and unemployment may be reduced. However, unless additional capacity is encouraged, this is likely to increase wage pressures, eventually bringing employment and unemployment back to their starting points, so that the fall in production will fully reflect the working-time reduction in percentage terms.

Therefore, total consumption, that is the sum of private and public consumption, will also have to be reduced by approximately 2 per cent. The reduction of private consumption will occur endogenously as households' disposable income decreases. By contrast, it may be a difficult political process to reduce public consumption or raise taxes, especially if the link between working time and public finances is not widely understood. In general, therefore, the policy implications of reduced working hours are particularly pronounced in countries with a large public sector, especially where budgetary flexibility is limited.

Future challenges facing public finances

In contrast to the favourable developments hitherto, the working-age population has now stopped growing and will eventually start to shrink, and dependency ratios will gradually start to increase, particularly after 2005. Thus, the positive demographic impact on tax bases in the past has now come to an end, and further pressures on public expenditures are certain to build up. Consumption of public services is relatively high until people reach their forties, reflecting the child-rearing years, and then tapers off. It picks up again as people reach their late sixties and rises again to around the levels of the child-rearing years (Figure 13). The age profile of the population in recent decades has been favourable to public finances, as the fraction of prime-aged adults in the total population has increased while that of people aged 60 or more has declined.

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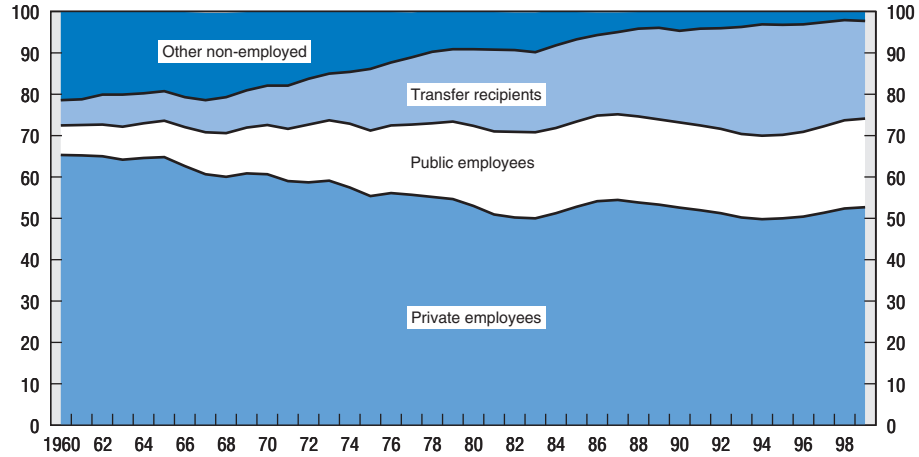
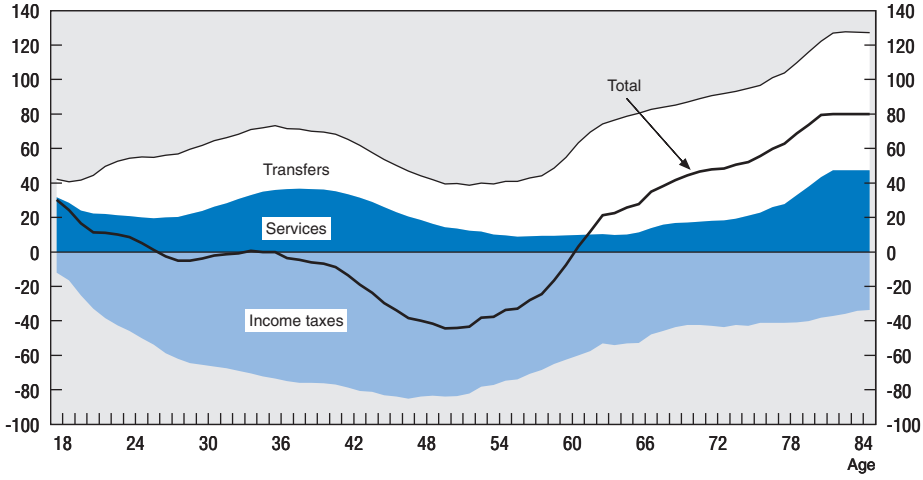
Figure 12. **Working-age population by labour market status**
Percentage of working-age population



Source: Ministry of Finance.

Figure 13. **Age profile of income taxes, transfers and services**
Thousand DKr



Source: Ministry of Finance.

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The long-term impact on public finances from ageing is not likely to be as severe in Denmark as in many other OECD countries, in part because of deferred taxes related to private pension schemes. A recent assessment by the Danish authorities estimated the induced increase in the tax burden necessary to offset the effects of ageing at 4 percentage points of GDP by 2035 (Regeringen, 2000).[4] Such estimates are sensitive to the underlying assumptions, especially about the time profile for the public-sector financial balance and the starting point for structural employment and unemployment rates. While the impact up until 2005 is likely to be relatively modest when assessed against a demographically "neutral" baseline, compared with the positive contributions from demographics in the past, the shift is more significant. In addition, further pressures could build up if the trend decline in the average age of retirement continues and/or average annual working hours stay on a descending curve.

The introduction of the sixth week of holiday makes it more difficult to achieve the government's medium-term objectives for labour supply (see Chapter III). Moreover, it hastens the need to curb the growth in public consumption and achieve better control over consumption expenditures than in the past. Thus the remainder of this chapter deals mainly with policy issues relating to the scope, provision and financing of public services, in particular individual basic public services like education, health care, childcare and elderly care.

High expenditures and weak control

Public wage consumption expenditures are above average

From 1995 to 1999 real general government consumption increased by 0.9 per cent per year, while in the Euro area it grew by only 0.5 per cent per year (Table 5). Even though GDP growth was strong, the ratio of public consumption to GDP has remained unchanged in Denmark. This contrasts with outcomes in other countries with high growth during the 1990s, for example, the United Kingdom, Ireland, Netherlands and the United States, where the ratio fell. This relatively strong public consumption growth follows the normal pattern of previous decades; since 1970 public consumption as a share of GDP has increased by 5.4 percentage points in Denmark, nearly double the increase in the Euro countries.

Next to Sweden, Denmark's public-consumption-to-GDP ratio (25.8 per cent in 1999) is the highest in any OECD country, almost 6 percentage points higher than the Euro area average. The differential fully reflects higher public-sector wage expenditures. The large public-sector labour requirement is even clearer in employment terms. Public-sector employment was 30 per cent of total employment in 1999, almost double the Euro-area average of 16.5 per cent. By implication, the proportion of non-wage to wage expenditures in total

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Table 5. **Trends in public consumption**

	Public consumption, 1999			Public employment	Change in public consumption to GDP ratio		Public consumption annual real growth
	Total	Wages	Other	1999	1970-1999	1995-1999	1995-1999
	Per cent of GDP			Per cent of total employment	Percentage points		Per cent
USA	14.4	9.2	5.2	15.1	–4.1	–0.9	0.6
Japan	10.1	7.3	2.8	8.3	2.7	0.3	0.6
Germany	19.0	8.4	10.6	12.6	3.5	–0.8	0.2
France	23.6	13.7	9.9	24.8	5.5	–0.3	0.6
Italy	18.4	11.4	7.0	17.5	3.1	0.5	0.4
United Kingdom	18.2	7.4	10.7	13.8	0.4	–1.5	0.5
Canada	19.3	7.7	11.7	20.7	0.0	0.0	0.4
Austria	18.7	12.0	6.8	16.4	3.9	–1.4	–0.1
Belgium	21.1	18.6	2.5	18.2	2.9	–0.3	0.7
Denmark	**25.8**	**17.6**	**8.2**	**30.0**	**5.4**	**0.0**	**0.9**
Finland	21.1	14.0	7.2	24.9	6.3	–1.7	0.7
Ireland	13.1	8.0	5.1	11.2	–1.2	–1.9	1.7
Netherlands	23.2	10.2	13.0	10.7	–1.7	–0.8	0.9
Norway	21.2	14.1	7.1	30.7	4.8	0.2	1.1
Portugal	18.9	14.9	4.1	18.4	6.3	1.3	1.1
Spain	17.1	10.5	6.6	14.8	6.8	–1.0	0.8
Sweden	26.4	16.5	9.9	31.5	3.8	0.0	0.2
Switzerland	14.3	10.4	3.9	13.9	3.1	–0.8	0.3
EU11	**20.0**	**11.1**	**8.9**	**16.5**	**2.8**	**–0.5**	**0.5**

Source: OECD.

public consumption is below average in Denmark, which is consistent with a less pronounced use of outsourcing in Denmark than in many other countries (see below).

Standards and coverage rates have increased

The enlargement of the public sector during the recent upswing has been concentrated on social services (including childcare and elderly care) and education. The increase in public employment since 1992 can be fully ascribed to the expansion in these two areas, with less significant changes in other sectors cancelling out. The increase in public employment was particularly strong from 1996 to 1998, coinciding with high growth in private-sector demand and increasing labour-shortage problems. Thus, the growth in public employment has not been well timed from a cyclical point of view, adding to a level of spending already above potential output.

The growth in public consumption and employment in recent years reflects only very modest shifts in age composition. Far more significant has been

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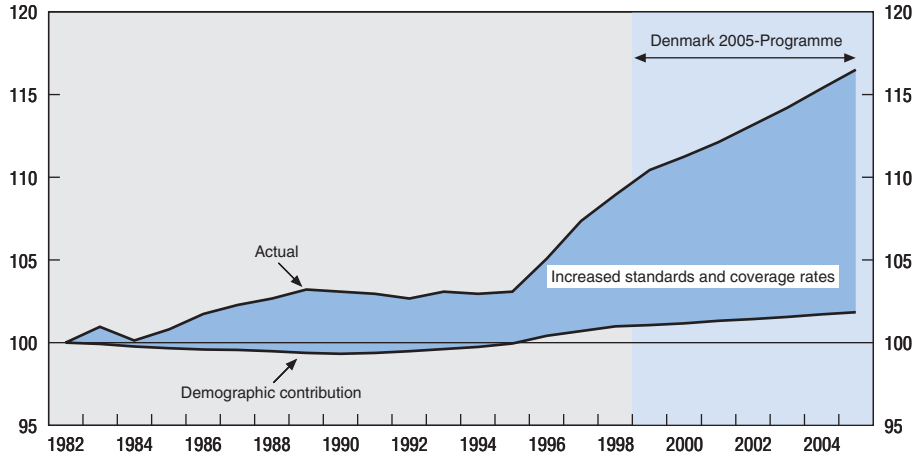
Figure 14. **Public employment, 1982-2005**
Indices, 1982 = 100



Source: Ministry of Finance.

the increase in standards and coverage rates for social services (Figure 14). This is in sharp contrast to the 1980s and the first half of the 1990s. Moreover, the medium-term programme *Denmark* 2005[5] assumes an annual growth rate in real public consumption of 1 per cent, and a corresponding increase in public employment. This compares with an average yearly growth rate of public employment of 0.8 per cent from 1993 to 1999.[6] The authorities argue that even limiting average annual growth to 1 per cent in real terms will be difficult, given public expectations, even though this would be enough to continue the increase in standards and coverage. In any case it would pre-empt most of the expected increase in labour supply (see Chapter III).

The increase in standards and coverage rates in part reflects shortcomings in expenditure control, rather than *ex ante* policy objectives. Municipalities and counties are responsible for the delivery of almost all social services in Denmark.[7] They have repeatedly exceeded expenditure levels agreed with central government within the framework established in the 1980s (see Box 3), although the adopted municipal budgets for 2000 are in line with the previous year's agreement. From 1995 to 1999, the average annual increase in real expenditures aimed for in the agreements was 1.6 per cent, while the average real increase subsequently budgeted by counties and municipalities was 2 per cent (Table 6). Thus, the excess spending by counties and municipalities added a ½ percentage point

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Box 3. The relationships between central and local governments

Following the Municipal Reform in 1970, the next 20 years was a period of significant decentralisation within the public sector, with still wider responsibilities being transferred from the central government to the counties and municipalities. Today some 80 per cent of public consumption is provided by counties and municipalities, and, in addition, the latter administer all social transfer schemes (pensions, sickness and maternity benefit, cash benefit, etc.). The economic relationships between central and local governments consist of three main elements, the annual agreement framework, the so-called DUT-provisions and the budget guarantee. The overall intention of the system is to eliminate cyclical dependencies of the local governments' budgets and to ensure compensation for impositions from the central government.

The *agreement framework* was established in 1980. Annual agreements between the central government and the counties and municipalities associations are made in late spring for the following calendar year. The key element of the agreement is the following year's block grant. This is equivalent to around 50 per cent of the local government's total revenues and is determined in the light of tentatively agreed expenditure levels (operating costs), assessed tax bases and changes in the municipal tax rates. The projected expenditures and tax rates are not known until budgets are adopted in October, so there is an inherent uncertainty at the time of the agreements. In recent years expenditures and tax rates have repeatedly exceeded the agreed levels, questioning the suitability of the system (see the main text).

The DUT-*provisions* (Det Udvidede Totalrammeprincip) were laid down in the mid-1980s. They extend the principle of closed envelopes to cover operating costs, as applied at the central level, to local governments' operating costs, where they are influenced by central-government decisions. Thus they aim to ensure neutrality for counties and municipalities with respect to changes in their responsibilities imposed by the central government. For example, the imposition of the brothers-and-sisters discount within publicly provided childcare automatically led to additional transfers from central government to compensate for lost revenues.

T*he budget guarantee* covers most transfer schemes administered by the municipalities. It compensates the counties and municipalities in aggregate, for fluctuations in the number of transfer recipients. No distinction is made whether the change in numbers is the result of cyclical factors (for example, sickness and cash benefit), demographic changes (such as pensions and maternity benefit) or other influences, including changes in local government administrative practices. The reimbursement is divided in two: a direct partial reimbursement of each municipality's expenditures and a residual reimbursement to the municipalities as a whole through the budget guarantee. For example, cash benefit expenditures are reimbursed at a 50 per cent rate to each municipality, implying that the municipality pays half of the marginal change in expenditures. Afterwards, national changes in the number of cash benefit recipients are reimbursed at a 50 per cent rate and divided up amongst the municipalities in proportion to their block grants. Consequently, the system combines full central-government financing of

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Box 3. **The relationships between central and local governments** (*cont.*)

transfers with incentives for each municipality to keep expenditures low. Experience shows that changing the direct reimbursement percentage significantly affects municipalities' administrative practices. A recent example is the reduction of the reimbursement percentage in the early retirement pension scheme from 50 to 35 per cent as from 1999, which reduced the inflow of pensioners markedly.

Finally, the revenues and expenditures of each individual municipality are affected by the *inter-municipal equalisation scheme*. Using objective criteria, the equalisation scheme transfers money from municipalities (primarily in the greater Copenhagen area) with large tax bases and/or low expenditures (for example, due to a relatively small proportion of pensioners) to municipalities in the opposite situation. While the equalisation scheme is a logical and necessary consequence of the large share of tax revenues in municipalities' total revenues, it also affects each one's incentives to broaden tax bases or reduce expenditures.

to the real annual growth in public consumption over this period.[8] Further expenditure growth resulted from other factors, including imposed expenditures resulting from the central government budget by virtue of, for example, decisions taken in connection with the agreement on the fiscal bill. On several occasions the central government has also backtracked from agreements with the local government associations – the decision not to implement the agreed user charges on elderly care was a recent example. Thus, on top of municipal non-compliance with the agreements, final municipal accounts have also repeatedly shown higher spending than was envisaged in the budgets. From 1995 to 1999 such overruns have, on average, been around 0.5 per cent and in the most recent years somewhat higher.

The growth in real public consumption has followed a pro-cyclical pattern since the late 1980s (Figure 15). This is in stark contrast with the traditional desire of the Danish government to aim for an overall counter-cyclical fiscal policy stance and is, therefore, probably unintended. It may reflect deficiencies within the institutional framework itself, or it may illustrate the political difficulty of holding back expenditures under favourable economic conditions. Overall, the record of pro-cyclical developments, supplementary expenditures by the central government, and systematic breaching of agreements by the municipalities points to deficiencies in the present arrangements.

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Table 6. **Planned fiscal policy and actual outcome**

	1995	1996	1997	1998	1999	2000	Average change 1995-1999
	Per cent						
Real net municipal expenditure growth[1]							
Agreed[2]	1.4	1.1	1.8	2.2	1.3	1.1	1.6
Budgeted[2]	2.0	1.6	2.2	2.8	1.8	1.1	2.1
	Overruns, per cent						
Net municipal spending relative to budgets							
Adjusted[3]	–0.7	1.1	0.9	0.9	1.0	–	0.6
Unadjusted[3]	–0.9	0.7	0.7	0.9	1.1	–	0.5
	Per cent of GDP						
Fiscal effect:[4]							
Estimated[5]	–0.3	–0.5	–0.5	–0.2	0.1	–	–0.3
Actual[6]	0.1	0.6	–0.5	0.0	0.2	–	0.0
Difference[7]	0.4	1.1	0.0	0.2	–0.3	–	0.3

1. Excluding investment expenditures, transfers and interest payments.
2. *Agreed* increases are those aimed for in the spring agreements between central and local governments. *Budgeted* increases are those contained in counties' and municipalities' budgets, which are adopted in October.
3. Overruns are presented with ("adjusted") and without ("unadjusted") correction for changes in municipal responsibilities between budgets and accounts.
4. The fiscal effect measure is a broad indicator for the overall stance of fiscal policy used by the Danish authorities. It includes, in principle, the first-year GDP effects of discretionary decisions at all levels of government. Thus the activity effects from increases in real municipal expenditure growth (the upper section of the table) are included in the fiscal effect
5. Estimations presented by the government in August of the preceding year, that is together with the fiscal bill and just after finalising the negotiations with the counties and municipalities.
6. Differs from estimated because of spending overruns at both municipal and national levels as well as, less commonly, unforeseen discretionary tax changes. Data for 1998-1999 are still preliminary.
7. The recorded differences are slightly biased downwards. Actual fiscal effects have been calculated *ex post*, applying definitions and multipliers that result, on average, in a tighter fiscal policy stance, than those used at the time the *ex ante* estimations were made.

Source: Ministry of Finance, Ministry of the Interior and OECD.

Reconciling efficiency and expenditure control with decentralisation

For a small and relatively densely populated country like Denmark, an administrative division into fourteen counties and 275 municipalities implies a very decentralised structure of the public sector. Around 40 per cent of all municipalities have less than 10 000 inhabitants. A significant difference between the Danish public sector and that of most other countries, however, is the extensive decentralisation of the decision-making powers and responsibility for the conduct of policies in high-priority areas. On the revenue side, counties and municipalities have the right to set their own income taxes rates without a ceiling. The average municipal rate was 20.5 per cent in 1999 (on top of an average county tax of

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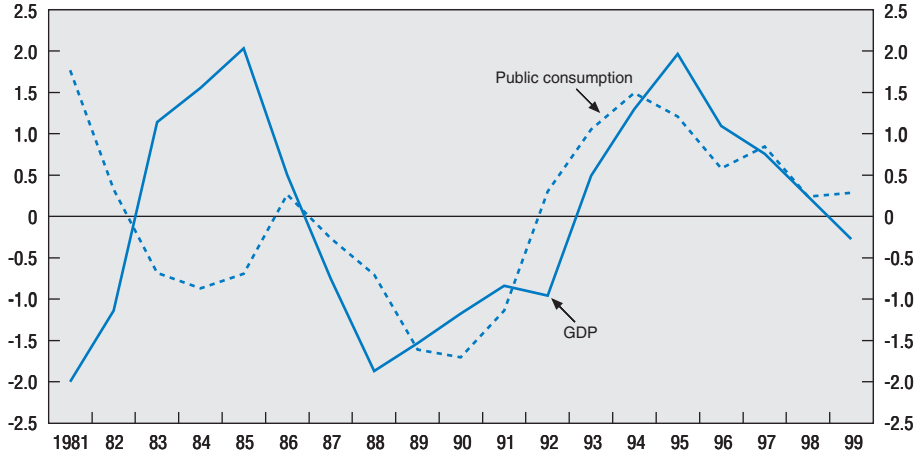
Figure 15. **Pro-cyclical development in public consumption**
Deviation from trend (3 years centered moving average), per cent



Source: OECD.

11.5 per cent), and the spread between the highest and lowest rates was approximately 5½ percentage points, despite the reduced need for less well endowed municipalities to have higher rates because of the inter-municipal equalisation scheme (described in Box 3). On top of the income taxes is a range of other municipal taxes, in particular property taxes, which can be decided within some limits by the individual municipalities. Finally, the municipalities also manage the tax auditing of individuals, individually-owned firms and – until recently – corporate entities.[9]

Counties and municipalities are also responsible for determining the standards and coverage rates of many services. These would include, for example, opening hours in childcare institutions, the number of teaching hours in primary schools (within limits) and entitlement criteria for practical support (cleaning, shopping etc.) for the elderly. Moreover, while provisions regarding the financing of services are typically determined by law or executive order, counties and municipalities can most often decide whether to provide the services themselves or make use of outsourcing and tendering. The municipalities also play a central role in the conduct of active labour market policies, as the largest suppliers of job-training offers. They also have a significant influence on policy outcomes with respect to the activation of cash-benefit recipients. Finally,[10] the local authorities have a central role in the administration of transfer schemes where the entitlement

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is dependent on discretionary assessment, *e.g.* early retirement (disability) pension, rehabilitation and "Flexjobs".[11]

The advantages of decentralisation most often emphasised are the better ability to adapt policies to local conditions and the strengthening of local democracy. However, in Denmark's case, the underlying variations in local conditions are not that large, and, in any case, the advantages of a highly disaggregated approach should be assessed against the costs. Benchmark studies comparing unit costs across municipalities have been conducted for a broad range of services in recent years. Even taking due account of the uncertainties involved in such analysis, the unambiguous result is that there are large differences in efficiency (Table 7). Another interesting finding is that the general satisfaction with the services provided is not correlated with the amount of allocated resources; for primary schools, several studies have even found an inverse relationship. The implication of these analyses is that there seems to be substantial potential for improving efficiency without reducing the general level of satisfaction, simply by applying best practices across more municipalities. However, whether or not the existing inefficiencies can be traced back to the municipalities' small average size, the highly decentralised governance structure makes realising potential efficiency gains a rather slow and painstaking process.

The principle of municipal self-governance reflected in the decentralised structure is deeply rooted and has a long history. The tendency in recent years for the central government to interfere more directly in the formulation of objectives and/or allocation of resources to areas for which the municipalities are responsible

Table 7. **Variations in cost efficiency within the public sector**

	Production unit	DEA – analysis,[1] Proportion of units with unit costs 20 per cent or more above the most efficient units	Regression analysis Proportion of units with excess costs of 10 per cent or more compared with the "expected"[2] level
Primary schools	Municipalities	80 per cent or more	7 per cent
Hospitals	68 individual hospitals	40 per cent or more	12 per cent
Elderly care (home help)	Municipalities	50 per cent	–
		Spread in unit cost	
Public libraries	Municipalities	Costs per loan: Bottom quintile = 16 DKr, average = 24 DKr, Top quintile = 34 DKr	

1. Data Envelopment Analysis.
2. The expected level is that resulting from the regression analysis.
Source: Ministry of Finance. Budgetredegørelse, various volumes.

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may blur the picture somewhat. It has certainly impeded expenditure control. Central government interference typically takes the form of declarations of intent, often accompanied by additional resources. However, the allocation of additional resources to individual municipalities usually reflects the distribution of the general block grants, and they are not required to spend the resources on the areas that the central government wanted to promote. Even in these circumstances, the municipalities remain entirely independent. Moreover, it has proved very difficult politically to change the existing administrative structures. This was manifestly demonstrated by the broad rejection of a government proposal in 1998 to put a 25 per cent tax on municipal tax increases. The recent failed effort to establish a more efficient structure for hospital services in the greater Copenhagen area is another example. Changes in the administrative structures seem to rely solely on voluntary mergers of municipalities, and these are very rare.

Thus, the present institutional settings seem to have some inherent deficiencies in terms of both expenditure control and average efficiency. With the very comprehensive set of local government responsibilities it is questionable whether these are best provided within the existing structure, dominated by many small units. Given the weak incentives for individual municipalities to seek better institutional arrangements or to amalgamate, the scope for improved performance from institutional reform must be assessed by policy makers at the central level. However, the structure is not likely to change much, at least for the short term, and policies should therefore focus on developing performance-enhancing measures within the current framework. Improving efficiency through strengthened managerial incentives and outsourcing and achieving better signalling of demand through user charges are two instruments examined more closely below.

Enhancing efficiency and expenditure control in the future

New public management tools

Improving public management as a means of reaping potential savings and/or increasing the quality and coverage of services within current expenditure envelopes has received greater attention in recent years. “New” management tools such as company-style financial accounts for public agencies, contracts for directors and (cautiously) linking public agencies’ appropriations and wages to performance have been introduced on a wider scale. Most of these instruments have been implemented too recently to allow for a through evaluation yet, although early indications point to good results. However, the successful use of such instruments depends on what is done with the realised savings. There are trade-offs involved. From a global perspective, all the savings should be pooled and the freed resources reallocated to the highest-priority areas. But at the micro level, allowing none of the savings to be kept by the agency would undermine the

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incentives for better performance and reduce their overall level. In recent years demands for productivity gains (imposed by means of automatic annual reductions in appropriations) have been applied less vigorously than during the 1980s and early 1990s. Therefore, while the more widespread use of new management tools has contributed to improved efficiency at the micro level, it has had less impact in terms of curbing overall expenditures.

A new framework for public-sector wage-setting was introduced in 1997 and extended further in 1999, so that it now covers virtually all employees of county and municipal governments. The new scheme implies a considerable decentralisation of public-sector wage formation and aims to link individual employees' pay more closely to performance, while scaling back automatic seniority-related wage increases. The potential benefits from the new wage system in terms of improved efficiency cannot yet be assessed. On the other hand, control over aggregate wage formation in the public sector may weaken. This risk relates to the fact that individual workplaces in the public sector are now allowed to transform productivity gains into higher wages. This may help to alleviate labour-shortage problems, but if higher wage increases spread to the private sector, the disadvantages may outweigh the benefits. Recent performance has been mixed. Public-sector wage increases accelerated from 2½-3 per cent in 1996/97 to 4½-5 per cent in 1998 and the first half of 1999, but they came down considerably in the second half of last year (Figure 16). With the new four-year wage agreements in the private sector, the risk of contagion effects have diminished for the moment, but developments still need to be monitored closely.

Wider use of outsourcing would improve efficiency

Within the existing administrative structure, economies of scale in the provision of services could be reaped through more widespread inter-municipal co-operation, voluntary municipal mergers, or extended use of outsourcing. The first two do not occur in practice, and contracting out has been used only hesitantly. For example, central government institutions do not comply even with the existing rules. An executive order from the Ministry of Finance obliges all central government institutions to make use of competitive tendering if it is likely to reduce costs. In practice, however, very few do so (Udliciteringsraadet, 2000). Even where economies of scale are not present, subjecting services to a competitive process can help to reduce costs, through choosing the most efficient supplier. In one concrete example, building cleaning services at the National Hospital were contracted out in 1992, and the savings realised amounted to more than DKr 100 million over four years, with an improvement in quality standards (OECD, 1997). Outsourcing has become common in a number of OECD countries, and the experience has been highly positive. Accordingly, OECD best practice guidelines have been developed.

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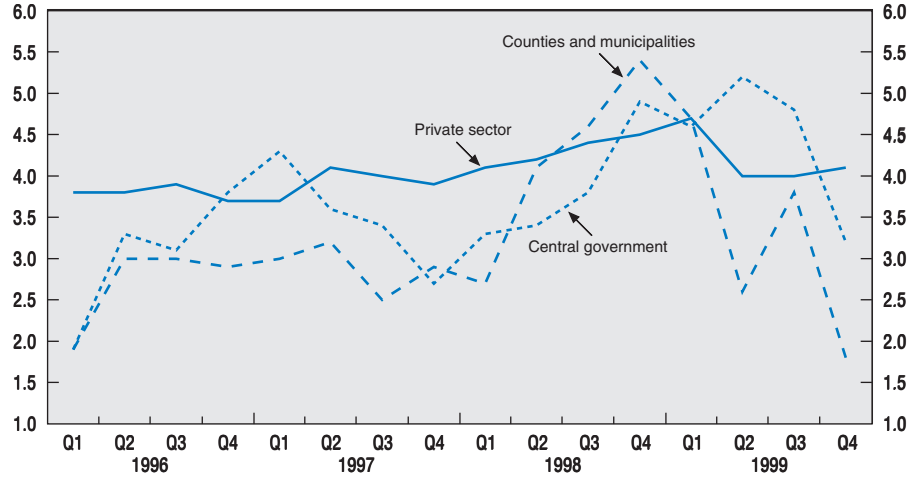
Figure 16. **Private and public sector wage increases**
Year-on-year percentage change



Source: Statistics Denmark.

Regulatory provisions prohibit municipalities from outsourcing primary schooling and home nursing, but otherwise limitations are few (Table 8).[12] The use of outsourcing and competitive tendering in counties and municipalities has been stagnant at a low level throughout the 1990s. Around 10 per cent of total costs in the municipalities and around 7 per cent in the counties are covered by such procedures (Table 9). Except for the technical and environmental expenditures among the municipalities and country spending on roads, all other areas have

Table 8. **Legislation regulating municipal outsourcing**

Regulation	
Areas where outsourcing/calls for tender cannot be employed	Authoritative responsibilities Home nursing schemes and visiting nurses Primary education
Areas in which outsourcing/calls for tender are compulsory	City bus services
Areas in which outsourcing/calls for tender are optional for each municipality	All areas not mentioned above

Source: Ministry of Economic Affairs (2000).

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Table 9. **Outsourcing in counties and municipalities, 1989-1998**

Type of expenditure	Municipalities		Counties	
	1989	1998	1989	1998
	Percentage of total costs			
Technical and environment	39.5	47.6	59.3	60.6
Roads	30.4	32.5	50.6	60.6
Public utilities	14.3	14.7	–	–
Education and culture	5.7	6.0	3.7	5.4
Hospitals and national health insurance	–	–	3.9	5.0
Social and health	3.1	4.0	3.4	3.6
Administration	15.3	17.7	14.9	17.6
Total	9.5	10.9	6.3	7.0

Source: Ministry of Economic Affairs (2000).

shown no or only insignificant change in the proportion of expenditures exposed to competition. Experience from areas that have been contracted out in recent years suggests a 15 per cent average cost reduction (Konkurrencestyrelsen, 1998). Even allowing for declining marginal effects of further outsourcing, the potential cost savings could be substantial.

User charges would strengthen price signals

User charges for publicly provided services perform two main functions: to link consumption more closely to cost and to signal information about demand. Services provided free or at very low cost by government embody a high degree of income redistribution. While this is often justified on equity grounds, in fact in a situation like Denmark's, with a relatively compressed income distribution and a high degree of public consumption, much of it is likely to be simply "churning". This occurs where people consume public services equivalent to the taxes they pay, but with the revenue being channelled through the public accounts, incurring processing and dead-weight costs and a loss of transparency.

If the use of outsourcing and public tendering has been hesitant, this is even truer for user charges. For core social services, such charges are generally lower and applied to fewer areas today than 10 to 15 years ago. In particular, user charges have been scaled back for childcare (see Box 4) and abolished for elderly care (practical help). Where user charges have been introduced in new areas, implementation has been hesitant and at insignificant levels. And often user charges have eventually been abolished again. Some of the most important decisions concerning user charges in the last 10 to 20 years are described in Annex 1.

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Box 4. Boom in childcare services

Childcare has been one of the fastest growing areas of expenditure in the past 10 years. The coverage rate, i.e. the proportion of children attending public daycare institutions, has risen sharply for all age groups. For the age group six months to nine years, the coverage rate has increased from 49 per cent in 1987 to around 80 per cent in 1999. In expenditure terms this corresponds to an increase of more than ½ a percentage point of GDP. Moreover, part-time places have been closed down over the last decade, implying that almost all places today are full time. In the age group 3-5 years (kindergarten) the coverage rate is above 90 per cent, while it is around 70 per cent for children aged 1-2 years and 80 per cent for 6-9 year olds (after-school facilities).

From an economic point of view, this development is of dubious benefit, as labour force participation rates for both men and women in the age group 25-39 have declined over the period concerned, and quite markedly so for women. In addition, part-time work is widespread in this age group. Thus, the sharp increase in coverage rates does not reflect labour-market-related needs. And for the large number of children in daycare who are also at school, there is no educational argument that would justify funding either.

More likely, the developments in childcare services reflect a significant reduction in user charges, in particular with the brothers-and-sisters discount introduced in 1991. This obliged municipalities to give a 50 per cent discount on the normal user charges for the second and subsequent children. In general, the pricing structure does not give parents any incentives to adjust their demand for daycare to their actual needs.

The central government has pressed municipalities to guarantee daycare places to all parents, resulting in a ratcheting up between supply and demand. Waiting lists, on the other hand, have been largely unchanged over the period. This example clearly demonstrates the importance of price signals and the long-term unsustainability of rationing the accessibility to services.

The examples in Annex I clearly demonstrate the importance for public expenditures and total activity of the price signals that user charges provide. They also illustrate how user charges are a sensitive political issue. They are widely perceived to "increase inequality" and conflict with the principle of equal access, regardless of income, to services that are seen as crucial in a modern society. The full implications of rejecting user charges – increased taxes and rationing, with uncertain distributional effects – tend to be neglected in the public debate. In any case, the current situation does not reflect a clear strategy; which services carry user charges is often random. Possibly the political resistance stems not so much from the idea of user charges as such, but rather from the visibility of the winners and losers from any changes made at the margin.

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Conclusions

Denmark's tax-to-GDP ratio is among the highest in the OECD, and high marginal tax rates are discouraging labour supply, particularly in terms of rather short average working hours. The high proportion of resources being allocated in the absence of market signals tends to reduce overall efficiency in the provision of services and make expenditure control more difficult. The continued pressure to extend coverage and standards over a broad range of services challenges the will and ability of policy makers to undertake the appropriate and necessary adjustments in expenditure policies in order to meet the challenges of the future. The following main conclusions are drawn from the chapter:

- Expenditure control has been insufficient during the upswing, even if improvements are visible in the most recent years. The fiscal stance has quite systematically been more expansionary than intended, primarily because real growth in county and municipal consumption has been higher than envisaged. This is partly related to excess of local government budgets relative to agreements and partly to expenditures added by the central government. In both cases, greater discipline is needed.
- The institutional set-up of relations between central and local governments seems to have inherent weaknesses, which tend to impede average efficiency as well as expenditure control at the margin. Control and enforcement mechanisms are generally weak, and the division of responsibilities and competencies across the various levels of government is somewhat unclear. These relationships need to be re-examined to find arrangements that generate better outcomes both with respect to delivery of services and overall financial management.
- The future conditions for extending, or even maintaining, public expenditure are likely to be less favourable than in the past. Demographic changes have worked to improve the fiscal balance of the public sector in recent decades, but they will provide a slightly negative contribution henceforth. Moreover, the positive budget implications following from lower structural unemployment and interest rates will not be repeated in the future. This means that, if the fiscal authorities continue to yield to spending demands, the present budgetary surplus will come under increasing pressure. All efforts should be made to "lock in" the budgetary improvements that have been made over the past few years.
- Potential productivity gains in labour-intensive social services are lower than for the economy as a whole. This will tend to produce trend increases in relative costs of these services – irrespective of whether they are produced in the public or private sector. However, with such services (child and elderly care, education, etc.) being provided "free" to consumers and financed through taxes, relative cost developments

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will not restrain demand. The risk is instead that increases in relative costs will drive up expenditure levels and taxes even further. Outsourcing of services and using competition to ensure as much productivity improvement as possible would be beneficial.

– User charges should be extended significantly. Without the price mechanism to regulate demand for services, queuing problems are bound to persist, causing upward pressure on expenditure and tax levels. Using price signals to more clearly indicate demand would provide a useful discipline on the growth in services. It would also ensure that those who consume the service are bearing a direct share of the cost. Given future demographic developments and already high tax burdens, it seems inevitable that the Danish public will need to accept paying a higher proportion of the costs when they use public services. Developing a coherent strategy for introducing a rational and appropriate framework of user charges should be a priority.

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III. Structural surveillance

Sound structural policies play an important role in underpinning long-term economic performance by encouraging flexibility and responsiveness to market signals. While Denmark has managed to maintain high living standards, structural weaknesses such as low national savings and high structural unemployment emerged in particular during the 1970s. Economic policy struggled to overcome these problems during most of the 1980s and the 1990s. Looking at Denmark's performance over recent years, it has clearly been reaping the rewards of sounder structural policies. Economic outcomes have improved significantly in the 1990s. The level of private savings has risen considerably, and, following almost three decades of continuous deficits, a small surplus on the current account has been achieved over the business cycle. Labour-market reforms have resulted in lower structural unemployment in recent years and have underpinned significant growth in private-sector employment. Labour force participation, on the other hand, has edged down, albeit from a high level.

Notwithstanding progress in many structural policy areas, several structural policy challenges remain outstanding. These include:

- Improving the functioning of the labour market further and strengthening the capability of labour-market policies to operate effectively also under less favourable economic conditions than in recent years;
- Enhancing effective labour supply, by reducing high effective marginal tax rates that discourage labour market participation (or additional hours) and addressing disincentives built into transfer schemes;
- Boosting the efficiency and quality of the education system. Although some progress has recently been made, two problems emphasised in the 1997 *Survey*, excessively high public spending and excessive time spent in and between periods of formal education, remain unsolved;
- Improving the operation of the housing market, where little progress has been made since the last *Survey*; and
- Increasing competition in product markets, as prices in Denmark still appear to be higher than elsewhere, in part at least attributable to insufficient competition.

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As several of these issues have already been addressed in previous *Surveys*, this chapter provides a follow-up on previous recommendations. It also provides a broader assessment of structural progress so far and reviews the scope for further action in each of the above areas.

Policies to reduce structural unemployment

Danish labour market performance in the 1990s has been encouraging. The reforms implemented since 1994 focused on making active labour market policies more efficient, imposing stricter provisions on work availability and enforcing them more rigorously, and tightening the eligibility criteria for receiving unemployment benefit (Table 10). The reforms have addressed the incentives of the unemployed to seek work rather than trying to strengthen their employability (by reducing employers' wage costs). This strategy is consistent with the underlying characteristics of the Danish labour market: wage setting is largely decentralised to the enterprise level, and employment protection legislation is light-handed.

Structural unemployment has been reduced significantly since its peak in 1993. The OECD estimates a fall of between 2.1 and 3.3 percentage points, up to the end of 1998, depending on the estimation technique used (OECD, 2000*a*). Registered unemployment continued to fall in 1999, while real wage growth slowed, suggesting that structural unemployment has fallen further. The correlation between follow-through in implementing labour market reforms and changes in the OECD's NAIRU estimates indicates that Denmark is among the leading reformers and has reaped the benefits (Figure 17).

Nevertheless, as the 1999 *Survey* noted, the reforms should be extended, and a broader range of measures is warranted to address the incentive problems. In light of this, recent policy moves and issues are addressed below. An important question is whether the policy settings now in place are capable of dealing effectively with a situation where unemployment, in stark contrast to the decline over the past six years, may start to edge up.

Tightening unemployment benefits

Denmark has an unusual combination of limited employment protection legislation (EPL) and a generous unemployment benefit system, compared with most EU countries.[13] This combination of policies reflects a tradition of setting working conditions via agreements between the social partners rather than by legislation. This implicit contract means that employers are able to adjust quickly to changes in demand conditions, while employees can count on generous benefits that are, at the margin, fully financed through general taxes. Consequently, the labour market shows a high degree of job turnover, with high job creation and destruction rates and very high inflow and outflow from unemployment.

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Table 10. **Labour market policy reforms, 1993-2000**

	1993	2000
Replacement rates	Maximum 90 per cent subject to a ceiling of approximately two-thirds of the average production worker's wage.	In general unchanged, although reduced for unskilled youth below 25 years during activation.
Benefit period	Formally 2½ years, but in reality infinite as job training qualified for a new benefit period.	4 years by the end of 2001. Activation does not qualify for a new benefit period. Extended benefit rights for 55-59 year olds (infinite).
Activation	Standardised activation for 7 months after 2½ years with the primary focus on restoring benefit entitlement.	Compulsory full-time[1] activation after 12 months. Unskilled youth are subject to educational activation after 6 months at a 50 per cent reduced benefit level.
Availability requirement	Weak compared to other countries. Availability assessments by union-run unemployment insurance funds.	Tightened and comparable to international average. No change in the responsibility of availability assessment.
Eligibility criteria	26 weeks within 3 years. Job training qualified for benefits, although not at first entry.	52 weeks unsubsidised work within 3 years.
De(centralisation)	PES had regional structure, but in reality little scope for local prioritisation due to standardised activation offers.	Increased responsibility for local PES to prioritise the use of activation instruments. Early targeted activation introduced. A consistent incentive structure has been introduced only from 2000, requiring the local PES to bear the full costs of all activation instruments.
Educational rights for the unemployed	Wide possibilities for voluntary education, for example in terms of 52 weeks of educational leave.[2]	The duration of voluntary education for the unemployed has been reduced to 6 weeks.

1. The operational definition applied is that activation should make up at least 75 per cent of the benefit period after 12 months of unemployment.
2. 1994 rules.

Source: OECD.

The unemployment benefit system is more generous than that of most other EU countries and, *a fortiori*, that of the United States (OECD, 1999*a*). For those previously earning up to DKr 165 000 (equivalent to two-thirds of the average production worker's wage) the replacement rate is 90 per cent. As in other countries, benefits for housing and daycare need to be taken into account when calculating the effective net replacement rate. High effective marginal tax rates discourage the unemployed from earning more income from work. Nearly one-quarter of the work force has very weak short-term economic incentives to work rather than

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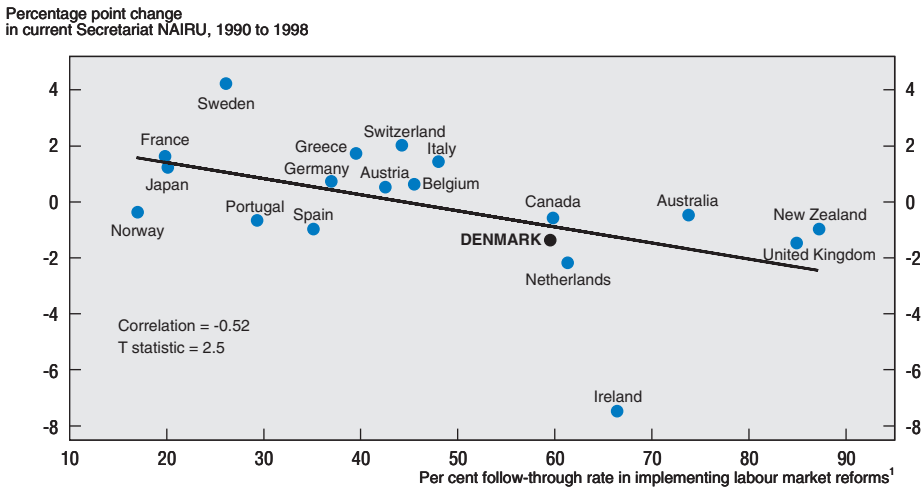
Figure 17. **Implementing labour market reforms and changes in NAIRUs**



1. In assessing progress in implementing the "jobs strategy", a "follow-through rate" has been calculated for each country by assigning different scores to recommendations according to whether they were followed entirely, partially or not pursued at all. See (OECD, 1999*c*) for further details.

Source: OECD.

receive unemployment benefits (Smith and Pedersen, 1998).[14] The replacement rates should not, however, be seen in isolation from the availability rules and mandatory activation. Even though variations in expected lifetime incomes across different occupational groups are to a large extent due to differences in the number and length of unemployment spells experienced, the generosity of the unemployment benefit system remains a significant distortion.

A change in the income basis on which individual unemployment benefits are calculated has been recently considered. At present, an individual's benefit level is calculated using documented wage income in the twelve-week period before becoming unemployed. This requires resources to verify wages paid and makes the system vulnerable to fraud. It also means that seasonal workers have an incentive to shift in and out of unemployment after a period of high earnings rather than accept lower paid work during the off-season. More important, within the present system the benefit level for the long-term unemployed is calculated on wage income earned long ago, which need not reflect current earnings potential – thus, the effective replacement rate may be higher than 90 per cent for some unemployed. In 1998, the government proposed instead calculating the unemployment benefit on annual taxable income primarily with a view to administrative simplification. However, applying the maximum replacement rate of 90 per cent to the previous years' taxable income (rather than the wage income in the 12 weeks

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up prior to the unemployment period) would also gradually scale down benefits for the long-term unemployed previously earning low wages. A committee with participation from the social partners analysed the technical and distributional aspects in further detail under a mandate that the new basis should not lead to a lowering of the general level of benefits. The committee reported, without a unanimous recommendation, in June 1999 (Arbejdsministeriet, 1999*a*). Although the proposed changes would improve work incentives compared with the present system, a decision is not expected in the near future. The committee will resume work when technical problems about electronic registration of wages have been solved.

Making ALMPs *work better*

With the rather weak economic incentives described above, it is up to other policy instruments such as active labour market policies (ALMPs) to ensure thorough job-search behaviour among the unemployed. In particular, enhancing the efficiency of ALMPs has played an important role in the reform process initiated in 1994. Recent reforms have included:

- Making regional labour-market authorities responsible for optimising the effect of ALMP measures within a centrally decided expenditure envelope;
- Shortening the period in unemployment before compulsory "activation" (*i.e.* requiring people to participate in ALMPs in return for receiving unemployment benefit).

Regional labour-market authorities were made responsible for optimising the effect of ALMP measures within a centrally decided expenditure envelope in 1994. However, in practice they have been able to use some instruments, in particular education, without bearing their cost. Consequently, they have tended to use "free" instruments rather than those that might be most effective. As from this year the full expenditures, operating costs as well as benefits, for all activation instruments are covered within the regions' expenditure ceilings. This change should ensure a more consistent set of incentives to encourage the regional public employment services to prioritise between different ALMP instruments, assessing the balance between price and quality. This should enhance the overall efficiency of the system.

Compulsory activation – requiring people to participate in ALMPs in return for receiving unemployment benefit – has been a feature of Danish labour market policy since 1996. One of the aims of this "right and duty" policy is to stimulate adequate job-search behaviour, thereby avoiding the need for compulsory activation. Indeed, a recent study by the Ministry of Labour showed that the probability of finding a job goes up as the point for compulsory activation approaches (Arbejdsministeriet, 2000). The length of unemployment before compulsory activation takes place has gradually been shortened and by 2001 will be after one year for those aged 25 years and older – for those under that age compulsory activation already applies

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after six months. In contrast, Danish studies indicate that the effects of earlier activation (within the first year of unemployment) are negligible or even negative (Larsen and Langager, 1998; Arbejdsministeriet, 2000). This may illustrate that, in practice, it is difficult to identify those unemployed who are at risk of becoming long-term unemployed and that early interventions face the risk of disturbing the normal search processes. Employment effects from the activation of long-term unemployed look more promising, although a firm assessment is difficult to make due to frequent policy changes over recent years as well as problems of distinguishing the effects of policy from those of favourable business cycle developments. The most positive effects are found for private-sector job training, which may, however, be subject to selection bias and only constitutes a small proportion of total activity. For educational and public-sector job training the largest employment effects are measured for members of unemployment insurance funds with a clear dominance of public-sector employees (Arbejdsministeriet, 2000). Thus, while those schemes targeted at subsequent employment in the public sector show rather good employment effects, other programmes have so far given less significant results.

ALMPs are expensive and will become even more so with bringing forward compulsory activation. Despite the sharp decline in unemployment, the numbers participating in ALMP schemes organised by the public employment services rose by almost 15 per cent between 1996 and 1999 (Table 11). Real expenditure on ALMPs increased by almost 25 per cent over the same period. Most of the increase was in education, where unit costs are higher, while job-training activity has been largely unchanged. As a consequence of the policy changes, the activation intensity – the ratio of ALMP participants to the sum of participants and unemployed – almost doubled between 1996 and 1999.

The expansion of ALMPs has put the system under some strain, and concerns have been raised as to the quality of ALMP places. A more active approach has also been adopted towards cash benefit recipients and other social security clients, which has further boosted the demand for ALMPs.[15] Thus, the legitimacy of, and political support for, the compulsory activation approach may be eroded if a certain standard of activation offers cannot be maintained. This year's introduction of the Service Jobs scheme (see below) was very much a response to such criticism. This debate points to a likely weakness of the Danish approach. While ALMPs in general, and compulsory activation in particular, may work well with low and falling unemployment, a crucial question is whether or not the approach can be sustained during periods of more adverse economic development when the necessary expansion of the system could be considerable.

Notwithstanding the overall reduction of structural unemployment, another problem is that the results vary strongly across age groups, being very significant for youth but less so for people aged 50 or more.[16] This is not surprising inasmuch as job-search incentives are weaker – and have been strengthened

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Table 11. **Expenditures and participants in ALMPs, 1996-1999**

	1996	1997	1998	1999	Change
	Million Krone (2000 price level)				
Expenditures					
Public job training	2 276	2 844	2 744	2 616	340
Private job training	316	267	279	272	–44
Startup scheme[1]	525	605	445	198	–327
Education	5 097	5 225	5 836	7 152	2 055
Total	**8 214**	**8 941**	**9 304**	**10 237**	**2 023**
	Thousand persons, full year equivalent				
Participants					
Public job training	14.2	17.7	17.1	16.3	2.1
Private job training	2.7	2.3	2.4	2.4	–0.4
Start-up scheme[1]	7.9	9.1	6.7	3.0	–4.9
Education	25.4	26.1	29.1	35.8	10.3
Total	50.3	55.1	55.4	57.4	7.1
Per cent of labour force	1.8	1.9	1.9	2.0	0.2
Activation intensity[2]	17.0	20.0	23.2	26.6	9.4

Note: The table covers only PES-organised activation of unemployment benefit recipients – not municipal activation of cash-benefit recipients.
1. The start-up scheme (to self-employment) was abolished in 1997.
2. The ratio of activation participants to the sum of participants and unemployed.
Source: Ministry of Labour.

less – for this group. Those aged 55-59 enjoy extended benefit rights,[17] and, in practice, the activation intensity is lower for those unemployed above 50 years of age than for other age groups. The reluctance of policy makers to impose the same demands on the older unemployed as on other age groups may result from the fact that loss of benefit entitlement also implies loss of entitlement to early retirement benefit at age 60: one of the criteria for early retirement benefit is that applicants must be entitled to unemployment benefit at the time of application. An exception applies for persons aged 50 to 54 years, allowing them to receive early retirement benefit even if their unemployment benefit period has run out. As this link impedes the efficiency of ALMPs it should be reconsidered more generally.

The Service Jobs Scheme introduced this year applies to people who are more than 48 years old and have been unemployed for at least 18 months. Municipalities that hire a person from this target group are paid an indefinite wage subsidy of DKr 100 000 per year. The rationale underlying the scheme is the same that led to the 1994 expansion of the early transitional retirement benefit (ETRB) scheme – that the long-term unemployed above a certain age have very little chance to return to employment. The ETRB scheme gave all those above 50 years of age with a preceding unemployment period of at least twelve months the right to a reduced benefit at 82 per cent of the unemployment benefit level until the

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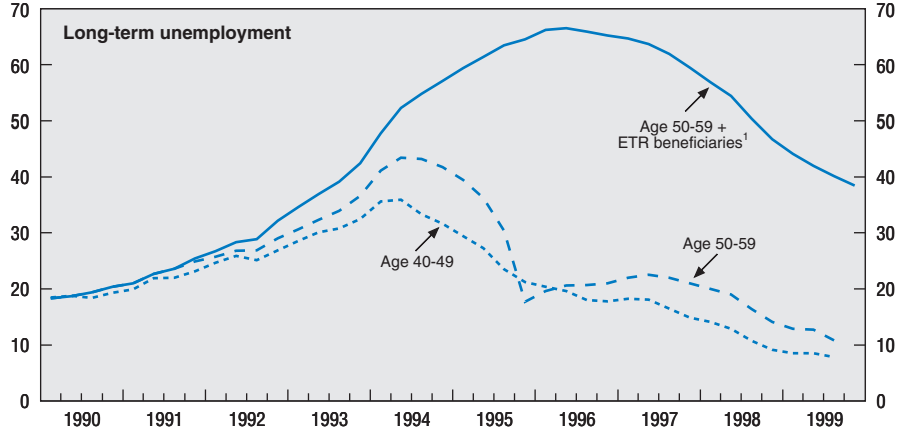
age of 60. The attraction to the unemployed was the accompanying exemption from the normal availability criteria. This was mistakenly not considered a big risk in 1994, as the general perception was that the long-term unemployed in this age group held only a small chance of finding a job in any case. The result, however, was that the normal outflow of the older jobless back to employment was to a significant extent replaced by inflow into the ETRB programme (Figure 18), while long-term unemployment of older workers has not come down any faster. Partly for that reason, the ETRB scheme was abolished in 1996. The Service Jobs Scheme is a wage subsidy to public-sector employers, while the ETRB scheme gave a subsidy to early withdrawal from the labour market. Therefore, it is unlikely that the Service Jobs Scheme will explode the way the ETRB scheme did. Nonetheless, experience suggests there is a risk of adding to already high public employment, without achieving the durable fall in unemployment hoped for. The Service Jobs Scheme is also problematic from an ALMP perspective, because of the indefinite character of the jobs and because they are remunerated at the going public wage rate rather than at the unemployment benefit level, as is the case with ordinary job training. These features significantly weaken any incentive to search for non-subsidised jobs.

Figure 18. **Long-term unemployment and early transitional retirement beneficiaries**
Thousand persons, full-year equivalents



1. ETR = early transitional retirement.
Source: Statistics Denmark.

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Increasing labour supply

Weak performance for more than a decade

Policies to stimulate labour force participation were for many years given low priority because of other structural problems, notably high structural unemployment and low private savings. Although these were naturally perceived to be more urgent, they also became a pretext for doing nothing to enhance work incentives. It is only within the last three or four years that labour supply has become a high-priority policy concern. In 1997, the Danish government launched the medium-term structural policy programme D*enmark* 2005, which has subsequently been revised and followed up annually with the presentation of the fiscal bill (Regeringen, 1997*b*, 1998 and 1999). The current objective is to increase the labour force by 80 000 people between 1998 and 2005, while keeping annual average working hours constant.[18] The sum represents almost 3 per cent of the work force in 1998. Over the same period, demographic developments mean that the work force will decline by approximately 15 000 persons at present age- and gender-specific participation rates, making the target average annual labour-force growth rate (nearly ½ per cent) rather ambitious.

Labour-supply growth has been weak since the previous business cycle peak in 1987. Although the working-age population increased by 114 000, the labour force fell by 16 000, giving a participation rate that was 3 percentage points lower in 1999 than in 1987. Instead, the number of transfer recipients has more than absorbed the increased population (see Box 5). In particular, the number of people on voluntary withdrawal schemes, that is early retirement benefits, transitional early withdrawal benefits and paid leave schemes, has increased sharply (Figure 19). With employment picking up in the latest recovery, the fall in participation up to 1996 can be largely attributed to people shifting from unemployment to the above mentioned schemes which effectively withdrew them from the work force. Although the leave schemes have been scaled back and the early transitional retirement scheme has been abolished, the total number of persons on voluntary withdrawal schemes was still 100 000 higher in 1999 than in 1993.

Absence from the labour market due to sickness has contributed to a reduction in effective labour supply during the upswing. Starting from 36 000 persons in 1993 (full-year equivalents) the number of recipients of sickness benefit rose to around 50 000 in 1997, and significant budget overruns were recorded, especially from 1995 to 1997. More recently, the number of recipients seems to have stabilised around this higher level. Since the introduction of the tax financed sickness benefit scheme in 1973, the statutory rules, as well as the administrative practice, have been changed frequently. While the benefit level has followed that of unemployment benefit, such features as the employees' waiting period, the period during which benefits are covered by the employer, the maximum duration

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Box 5. Early retirement schemes

Three early retirement schemes exist in Denmark:

Early retirement (disability) pension is basically a disability pension scheme, although a benefit can also be granted for social reasons under certain conditions. The scheme applies to adults aged 18 to 66. Admission is dependent on discretionary assessment by the municipalities of applicants' working capacity, which must be reduced by at least 50 per cent. The number of recipients has stabilised at approximately 270 000 (9½ per cent of the work force) in recent years after a rather sharp increase during the 1980s and early 1990s. A framework for reform of this pension scheme was agreed in 1999 to be implemented as from 2003 (see below).

Early retirement benefit is a labour-market-related scheme currently available to wage earners aged 60 to 66; as from 2004, it will only apply to wage earners aged 60 to 64. It was introduced in 1979. Entitlement is dependent on a number of criteria, of which the most important are: 1) membership of an unemployment insurance fund and payment of contributions for 25 of the 30 years prior to entrance into the scheme; and 2) applicants must be entitled to unemployment benefit at the time of entrance. During the 1980s and early 1990s inflow to the scheme was cyclical, with high inflow in periods of increasing unemployment and low inflow in times of falling unemployment. However, during the latest upswing the inflow rate has kept increasing, taking the total number of recipients to more than 150 000 (5¼ per cent of the work force). Against this background a reform of the early retirement benefit scheme was adopted in 1998 (see below).

Early transitional retirement benefit was introduced in 1992 and extended in 1994. After a sharp increase in 1995, entrance to the scheme was closed as from early 1996. At that time there were almost 50 000 recipients, a number that has now fallen to 35 000. The early transitional retirement benefit applied to long-term unemployed (12 months or more) aged 50 to 59. The benefit level was 82 per cent of the unemployment benefit. Recipients are not subject to the availability criteria which applies to the unemployed.

of the benefit and the rules regarding reimbursement by the municipalities (who administer the scheme) have been changed on several occasions. Absence due to sickness has shown a somewhat cyclical pattern, and the experience from the one-day waiting period for employees between 1983 and 1986 indicates a quite strong response could be expected if such a feature was revived. The rise in the number of recipients since 1993 has primarily resulted from the increase in average duration of sickness spells and extensions that have watered down the importance of the formal 12 months limit. In part, this can be explained by changes in the early retirement pension (disability) scheme. As from 1996, sickness benefit recipients applying for such a pension remain eligible for sickness benefit while the application is being considered, even if the formal 12 months limit has run out.

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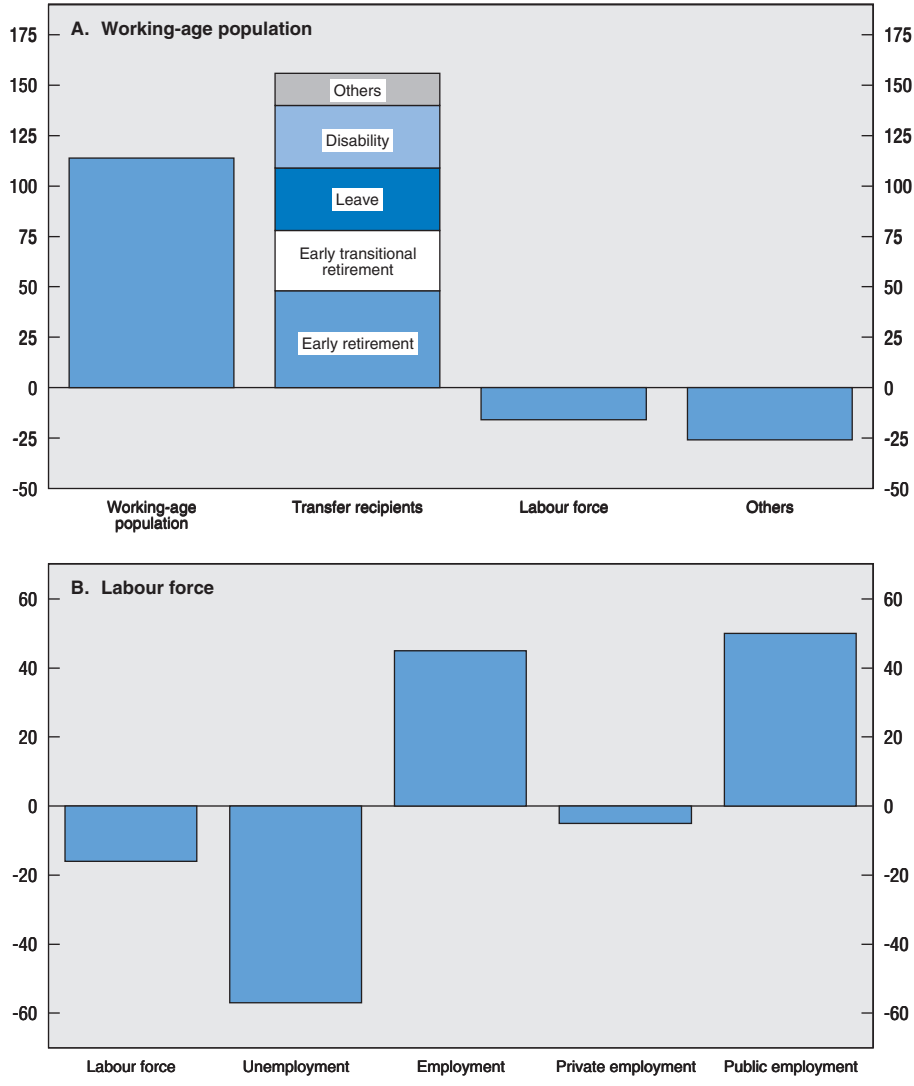
Figure 19. **Change in working-age population, labour supply and transfer recipients**

Thousand persons, 1988 to 1999



Note: The labour supply is calculated as the sum of registered unemployment and employment according to the National Accounts.

Source: Statistics Denmark and Ministry of Finance.

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Early retirement benefit reform has helped a little

As a part of the agreement on the fiscal bill for 1999, a long-debated reform of the early retirement benefit scheme was eventually adopted by a broad majority in the Parliament (Table 12). Since raising the entitlement age (60 years) was not politically feasible, the design of the reform is rather complicated, but it aims to raise the average age of entry through a mix of carrot- and stick-type instruments, namely:

- The reductions of the benefit level for income from private pension schemes for entry before 62 years of age have been increased.
- A tax discount is given to wage earners who stay in work beyond 62 years of age. The tax discount increases gradually with the time worked beyond 62 years of age up to a maximum of DKr 100 000. It is given as a one-off payment at 65 years, when the person becomes entitled to an old-age pension.
- The annual employee-paid contributions to the system have been approximately doubled, while the qualifying period has been raised to 25 years within the last 30 years – up from 20 years within the last 25 years.

It is too early to assess the impact of the reform on the average retirement age and therefore on labour supply. Based on current inflow rates, demographic developments will add another 50 000 recipients up to 2005 and approximately 100 000 up to 2010, and the underlying increase will be even bigger if the inflow rate to early retirement benefit keeps trending up. The government estimates that the reforms themselves will reduce the number of early retirement benefit recipients by 10 000 to 15 000 by 2005.[19] However, in the *Denmark* 2005 programme, only 25 000 additional benefit recipients are assumed (*i.e.* a greater number will remain in the labour force). This suggests that further initiatives will be needed to meet the targets.

Further policy changes required to meet the objectives

Even with the reform of the early retirement benefit scheme, the target increase in labour supply has been widely questioned, partly because the target is ambitious. For example, the Economic Council suggests that, in an unchanged-policy baseline scenario, labour supply will most likely edge down towards 2005 (Det Økonomiske Råd, 1999). Even if the assumptions underlying *Denmark* 2005 are applied, there are likely to be at least 50-60 000 labour force participants less than the target level. The shorter annual working hours contained in the recent collective agreements will also put a drain on total labour supply, not envisaged in the *Denmark* 2005 plan. Indeed, the Council predicts that the positive contributions from the phase-out of the early transitional retirement benefit scheme and the early retirement benefit reform will only just offset the reductions due to demographics and reduced agreed working hours (Figure 20).

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Table 12. **Reform of the early retirement benefit system**[1]

	Old scheme	New scheme
Old age pension	Entry age 67.	Entry age 65.
	Means testing: wage income above threshold reduced by a 60 per cent rate in basic state pension.	Means testing: wage income above threshold reduced by a 30 per cent rate in basic state pension.
Conditions for part-time work	Distinct transfer scheme for eligible wage earners working part time.	Unified scheme. Continuous reduction in benefit for the number of hours worked.
	200 hours work annually allowed in the main scheme without benefit reduction.	
Benefit level	100 per cent unemployment benefit of the first 2½ years. 82 per cent unemployment benefit of the last 4½ years. (Flat 100 per cent of unemployment benefit rate if withdrawal was postponed to 63 years).	Flat rate of 91 per cent of unemployment benefit. Flat rate of 100 per cent of unemployment benefit if withdrawal is postponed to 62 years.
Reduction of benefit for income from private pension schemes	Income from labour market pension schemes with continuous payments reduces benefits by a 60 per cent rate. No reduction if payments were voluntarily postponed to after age 67. No reductions for income from pension schemes with capital payments.	*Withdrawal before 62 years*: Equal treatment of all pension schemes 60 per cent of actuarial pension income, calculated from pension wealth at 60 years, is subtracted from benefits.
		Withdrawal after 62 years: Old rules apply.
Tax discount for postponement of withdrawal	No tax discount.	Tax discount if retirement is postponed 2 years or more. The discount equals 35 per cent of the early retirement benefit that could alternatively have been received. The maximum discount is a little above DKr 100 000.
Contributions and eligibility	Unemployment insurance contributions of 7¼ times maximum daily unemployment benefit rate – approximately DKr 4 000 per year before tax and DKr 2 400 after tax.	Unemployment insurance contributions are reduced to 6 times maximum daily unemployment benefit rate. New early retirement benefit contribution, at 7 times maximum daily unemployment benefit rate, is introduced.
	Eligibility requirement: payment of unemployment insurance contributions in 20 out of last 25 years.	Eligibility requires payment of unemployment insurance and early retirement benefit contributions for 25 out of last 30 years.

1. Only the main principles of the schemes are specified in the table and transitional provisions are left out for ease of presentation. These include a three-year phase-in of the early retirement benefit contribution, a phased-out floor for the imputed actuarial pension income and easier terms for payment of ERB contributions for people who were above 35 years of age on 1 July, 1999.

Source: OECD.

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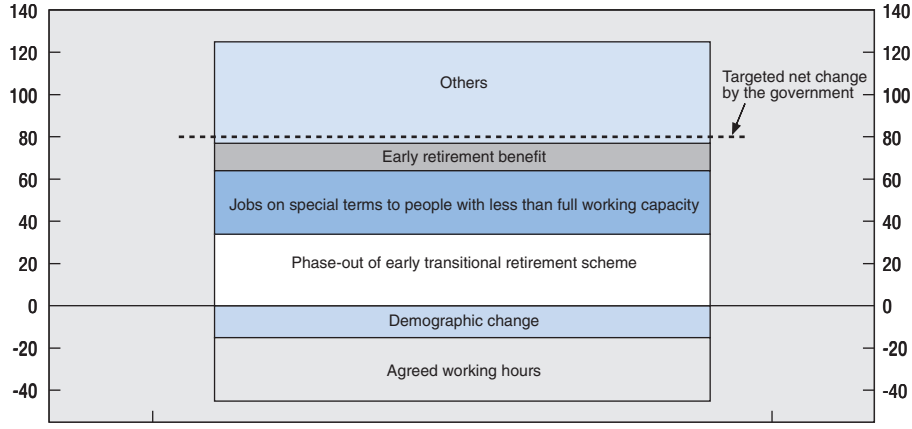
Figure 20. **Gross contributions to change in labour supply, 1998-2005**
Thousand persons, full-time equivalents



Source: Ministry of Finance; OECD.

The framework for a reform of the early retirement (disability) pension scheme[20] was adopted with the fiscal bill for 2000, setting out some main principles for a new system to come into force as from 2003. The benefit levels and structure, and thus the economic incentives of the future system, are still up in the air. It is unlikely, however, that the reform will add to the already assumed decline in the number of early retirement pensioners in the D*enmark* 2005 programme and the counterbalancing increase in the number of jobs for people with less than full working capacity (30 000 jobs). At best, it will be enough to achieve the objectives set out.

The bottom line is a discrepancy between the stated objectives and the policies in place, although the uncertainty surrounding the labour-supply projections is considerable. In light of this, additional policies to strengthen labour supply should be considered:

– While the overall participation rate is higher than in almost any other OECD country, labour-force participation among 55-64 year-olds is only around the average and some 15 percentage points lower than in Sweden and Norway (Figure 21). Although the early retirement benefit reform is likely to exclude further reforms, at least in the short term, policies to stimulate participation by older workers should be strengthened further.

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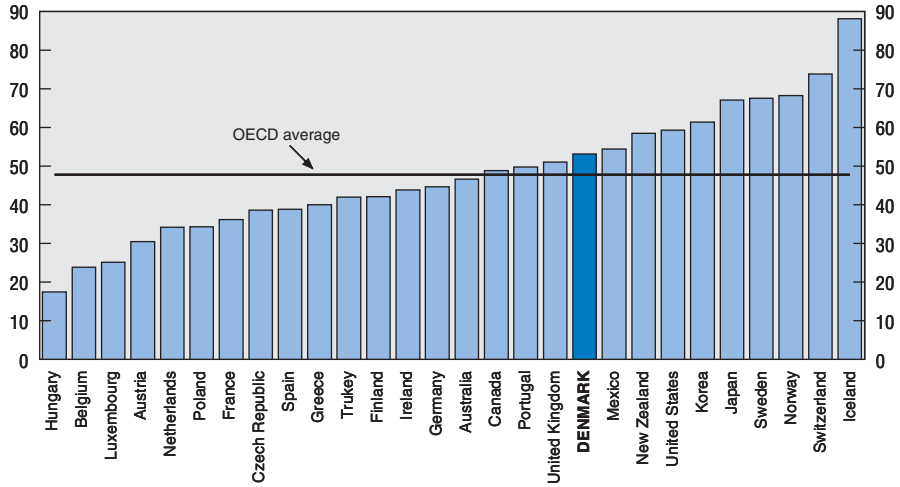
Figure 21. **Older workers'[1] participation rates across OECD countries**
Percentage of corresponding working-age population, 1998



1. 55-64 years old.
Source: OECD, LFS.

– Reducing marginal tax rates and aiming for earlier school completion could also boost labour supply (see below).

– The generosity of the sickness benefit scheme should be reconsidered. There is no waiting period for employees, and the replacement rate is high, especially given collective agreements to top up benefits. The formal limit to the duration of sickness benefit (12 months) is no longer effective.

– More concerted efforts could be made to increase participation among first-generation immigrants who have lower average participation rates than the working-age population as a whole. The positive effects would probably not be large, however.

Improving the efficiency of educational policies

The 1997 *Economic Survey* raised a number of concerns regarding educational policies:

– Despite allocating more resources to education than any other OECD country, outcomes were relatively poor. In primary school, international

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comparisons showed middle or below-average performance on basic skills in reading, writing and mathematics. In secondary education, a large share of students did not graduate, but instead dropped out of school.

– The low pupil/teacher ratio in formal education was not reflected in classroom contact time, because teachers were granted a high proportion of non-teaching (preparation) time. More generally, there was no link between performance and pay for teachers.

– The average age of those starting full-time employment was high due to excessive time spent in the education system and because of the prevalence of interruptions during secondary studies.

Some progress has been made on the recommendations of that *Survey*. Recent developments have been encouraging in some areas and, more generally, a shift in policy emphasis from quantity to quality can be observed. This is reflected in two recent comprehensive ministerial reports on the ordinary education system and adult education and training, respectively (Undervisningsministeriet, 1998; Arbejdsministeriet, 1999*b*). A much-needed sharper focus on cost effectiveness also appears to be underway, but it has not yet translated into improved performance.

Improving formal education

More attention has been directed towards acquiring basic skills in primary school, although the evidence of improvement is still anecdotal. In upper secondary school, 86 per cent of graduates obtained a vocational qualification in 1999, up 5 percentage points since 1994 and 9 percentage points since 1990 (*i.e.* the drop-out rate has fallen). However, this is still less than the government's target of 95 per cent and the best performing OECD countries (Sweden, United States and Germany.) As noted in the 1997 *Survey*, a number of initiatives taken in the mid-1990s, including a safety net for apprentices, have contributed positively. Probably more important, however, have been smaller cohorts and increased demand for apprentices during the upswing.

However, there has been a big increase in the average age of students graduating from secondary school, indicating that an increasing share of graduates are adults above age 25.[21] This indicates that no real progress has been made on recommendations to aim for a faster passage through the education system and a reduction in the corresponding high age of entry into tertiary education. These still need policy attention. Reforms should focus on the dual objectives of reducing unit costs and increasing labour supply. The amount of excessive time in the education system has been assessed to be equivalent to 80 000 to 100 000 extra workers, thus holding back a large potential labour force reserve (Regeringen, 1997*a*). Recommendations from an inter-ministerial committee to reduce this time,

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inter alia by strengthening economic incentives to complete education more quickly have not so far been transformed into concrete policy initiatives.

Progress has been made on education costs and performance. In the public-sector wage bargaining round in the spring of 1999, new work-time agreements were reached with the teachers in primary and secondary education. These agreements allow employers greater flexibility in organising teachers' working time and should ensure an increase in teaching hours. At the same time, the teachers were included in the new wage system in the public sector, introduced for other groups already in 1997. This system reduces the seniority dependency of pay increases, allows for greater flexibility and establishes a link between individual teachers' pay and performance. The agreements should thus help to create a new regime for education at the primary and secondary levels, but it requires careful and consistent implementation at the school level.

Adult education and training

Expenditures for adult education and training have increased strongly since the introduction of free uptake and "taxi-meter" grants in 1997 (*i.e.* appropriations increasing with the number of students completing courses). This change was implemented in order to strengthen demand signals in the allocation of resources across the various courses. As the 1997 S*urvey* recommended, to be effective the changes also needed user fees to play a major financing role. In fact the taxi-meter financing system failed because the user fees introduced for both employees and employers were negligible, and the intended gradual increase was never carried through. Instead, the user fees have been abolished altogether. Furthermore, the reform had perverse incentive implications for the public employment service (PES) as a large user of training courses. As noted above, ALMP measures were biased towards training courses that for the PES were a "free lunch", with payments going directly from the central government to the educational establishments. The free uptake and taxi-meter grants system was abolished in 1999 and expenditure ceilings restored, although at a much higher level. Moreover, as mentioned earlier, the PES must now pay all expenditures – benefit as well as operating costs – when supplying activation courses.

Initiatives to strengthen cost awareness and improve the incentive structure in the adult education system for employees were undertaken with the agreements on the fiscal bill for 2000. A financing reform will gradually reduce public involvement, as public expenditures are fixed nominally at their 2000 level. However, the financial responsibility of the social partners is expected to increase over time. Furthermore, a reassessment is intended of the list of courses to which public subsidies can be granted, the aim being to exclude those that provide only narrow, firm-specific competencies. The final objective is to improve the transparency of the system and create better coherence between the content of the individual

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courses.[22] The ambition is to make it easier for participants to build on earlier achieved competencies and, eventually, to obtain approved competence corresponding to upper secondary or skilled level.

Tax policies

The ongoing tax reform, described in more detail in the 1999 *Survey*, was primarily designed to enhance aggregate private savings and reduce differences in the tax treatment of returns on various types of savings. The most important measure reduces the tax value of the deductibility of negative capital income (which includes mortgage interest payments) from approximately 46 per cent in 1998 to 32 per cent in 2001 onwards.[23] Although that still leaves the tax value of mortgage deductions high by international comparison, it is partly counterbalanced by a municipal property tax. Hence, the distorting effects from the tax treatment of housing will clearly be smaller than in the past, although it is still too early to assess the exact quantitative long-term implication for private (financial) savings.

High marginal tax rates discourage labour supply

Since the mid-1980s Danish tax policy has followed the international trend of trying to reduce statutory formal tax rates while broadening tax bases. Major tax reforms were implemented in 1987, 1994 and 1999, in each case with a phase-in period. While these reforms have contributed significantly to improving national savings, their impact in reducing the distorting effects on work incentives are less impressive. The average effective marginal tax rate paid fell by 1.3 and 3.4 percentage points for all persons aged 25 to 59 years and those full-time employed, respectively, from 1986 to 1998. However, from 1983 to 1998 average marginal tax rates were virtually unchanged. The reduction in rates has been offset by a larger proportion of taxpayers in the top tax bracket (Table 13). The Danish authorities estimate that the average effective marginal rate will fall slightly up to 2002, while the proportion of taxpayers paying the top rate is projected to increase further. The proportion of taxpayers paying the lowest marginal rate is also estimated to increase, resulting from a 25 per cent rise in the threshold for the medium rate bracket.

The high proportion of top-rate taxpayers may be the most distinguishing feature of the Danish tax system. The income threshold for paying the highest marginal tax is only just above the average production worker level, much lower than in most other countries. With the compressed wage structure, however, most top-rate taxpayers pay the top rate on relatively little income. Therefore, while the top tax rate discourages additional work for a large proportion of the work force, it adds relatively little to total tax revenue. Put another way, significantly better work incentives could be achieved at a relatively small cost.[24]

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Table 13. **Statutory and effective marginal tax rates**

	1983	1986	1993	1998	2002[1]	Change 1983-1998
Statutory marginal tax rates						
Lowest rate	46.7	48.0	50.6	45.2	43.7	–1.5
Intermediate rate	61.1	62.4	58.2	50.7	49.1	–10.4
Top rate	71.9	73.2	68.7	62.0	62.8	–9.9
	Per cent					
Average marginal tax rates						
Persons aged 25-29	51.4	53.0	53.6	51.7	50.4	0.3
of which full-time employed	55.6	57.7	59.3	54.3	53.8	0.7
	Share of taxpayers, per cent					
Distribution of taxpayers (all persons aged 25-59)						
Lowest tax bracket	56.3	51.0	42.8	26.0	32.9	–30.3
Intermediate tax bracket	36.1	39.2	37.5	49.9	39.7	13.8
Highest tax bracket	7.6	9.8	19.7	24.1	27.4	16.5
	Share of taxpayers, per cent					
Distribution of taxpayers (full-time employed aged 25-59)						
Lowest tax bracket	40.3	35.4	22.8	9.1	11.9	–31.2
Intermediate tax bracket	48.5	51.0	46.9	54.2	46.5	5.7
Highest tax bracket	11.2	13.6	30.3	36.7	41.6	25.5

1. Estimated by the Danish authorities.
Source: Ministry of Finance (1999).

Relative to the OECD average, Denmark is characterised by a high participation rate but short average annual working time (and a large proportion of transfer recipients). Indeed, the distorting effects from the Danish tax system seem to work through short working time more than low participation rates. Whether or not the tax treatment of wage income depends on marital status may have implications for household work incentives. In contrast to Denmark, some countries – for example Belgium, France, Germany, Ireland and Switzerland – levy income taxes on households rather than individuals.[25] This is likely to impinge on household decisions as to whether only one or both spouses should be in employment. One-earner households may face very high marginal tax rates if tax allowances are reduced as the second spouse chooses to seek employment. A recent study (Callan *et al.*, 1999) estimates identical labour-supply functions for prime-aged women in Denmark, Ireland, Germany and the United Kingdom based on large samples of individual data. The design of the tax system is found to have a significant impact. Micro-simulations based on the estimation results suggest that women's participation rates would be reduced significantly in Denmark if, hypothetically, the design of either the UK or the Irish tax system were to be adopted (Table 14).

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Table 14. **Simulated effects of alternative tax systems**

	Tax Scheme[1]	Non-participation rate	Part-time rate	Full-time rate
Denmark	Own	0.107	0.293	0.600
	UK	0.303	0.245	0.453
	Irish	0.525	0.168	0.307
United Kingdom	Own	0.297	0.301	0.402
	Irish	0.642	0.195	0.163
West Germany	Own	0.443	0.267	0.289
	UK	0.340	0.258	0.402
	Irish	0.719	0.160	0.121
Ireland	Own	0.808	0.081	0.110
	UK	0.683	0.138	0.179

Note: The table concerns women in the age group 25-55.
1. Data and tax systems from: Denmark (1990), Ireland (1987), Germany (1991), United Kingdom (1991).
Source: Callan *et al.* (1999).

Capital income taxation has become less neutral

The distinction between (and separate taxation of) personal and capital income was introduced, with largely symmetric taxation of positive and negative capital income, in the 1987 tax reform. With the subsequent reforms negative capital income tax rates have been reduced significantly – from 2001 they will be approximately 32 per cent for all taxpayers. By contrast, the taxation of positive net capital income has become more progressive, as positive net capital income now enters the base for the top tax, implying a spread between the highest tax rate on positive and negative capital income of 25-30 percentage points. Furthermore, earnings from private share portfolios, share portfolios in private pension funds and other pension-fund portfolios are taxed at separate rates as low as 5 per cent (in the case of earnings from shares in pension-fund portfolios). Hence, the portfolio choice of persons with positive net capital income is distorted, as savings in ordinary interest-bearing assets are taxed at a higher rate than alternatives such as pension funds or housing.

In line with the reduction of the corporate tax rate to 32 per cent from 2000 onwards, the tax on individually owned companies has been reduced in parallel in order not to distort the choice of corporate structure. However, as the surplus from business activity is treated as capital income, the low tax rate on the current surplus implies the build-up of deferred taxes to be paid at the time of realisation of the assets in the firm. The deferred taxes cause lock-in effects, which impede business development, and the heavy capital income taxation should be assessed against such detrimental effects.[26] There is, on the other hand, a need to avoid incentives for personally owned businesses to relocate wage income to capital income, creating a new

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distortion. Deferred taxes also arise, for corporations as well as individually owned companies, as a result of favourable tax rules concerning depreciation: the typical 30 per cent depreciation rate in the tax account is higher than the average economic rate of depreciation (Skatteministeriet, 1999). Total tax expenditures stemming from business taxation are estimated by the Danish authorities to be approximately 0.7 per cent of GDP. The provisions regarding tax depreciation are to the benefit of capital-intensive industries and are therefore likely to preserve the present business structure rather than to stimulate the development of new industries, often in fast-growing sectors. Less capital-intensive firms would benefit more from a further reduction of the corporate tax rate financed by broadening the tax base.

As from 1 January 1999 Denmark changed the tax treatment of foreign and domestic holding companies' dividends. According to the new rules, holding companies are not taxed on such forms of income received from foreign subsidiaries. The change is seen by the Danish authorities as a simplification of previous rules, which aimed to tax the difference between the Danish corporate tax rate and that of the country hosting the subsidiary, but which in practice raised only very little revenue. However, a number of foreign companies have established a holding company in Denmark apparently as a consequence of the new tax code. The Danish rules, as well as those of other Member countries, are currently being examined and evaluated by the Committee on Fiscal Affairs' Forum on Harmful Tax Practices under the OECD Guidelines for Dealing with Harmful Preferential Tax Regimes in the Member countries (OECD, 1998).

Housing policies

The 1999 S*urvey* contained an in-depth analysis of policies for the housing market. That market operates in a complex way with several different segments: owner-occupied housing, private rented housing, rented housing provided by housing associations (known as social housing) and co-operative housing. The government has traditionally used housing policies to serve a number of different economic and social objectives. The S*urvey* stressed the importance of dismantling rent controls and extending property rights to improve mobility and property maintenance. It also noted the importance of continuing reform of tax and transfer systems to improve both resource allocation and equity, while modifying the tax system to reduce the contribution of the housing market to macroeconomic instability.

Rent controls prevail

No progress has been made on easing rent controls. Rent control sets a ceiling on prices in large parts of the rental sector, keeping rents well below the user costs in comparable owner-occupied housing. The largest discrepancies – up to 30 per cent – are between actual and market rents in the older part of the housing stock and therefore concentrated in privately owned rental dwellings. (Market

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rents can be charged for dwellings built after 1991.) Artificially depressing rents has discouraged landlords from undertaking expenditures for improvements or even doing maintenance, since the costs cannot be recouped through higher rents. In addition, the allocation process is heavily distorted; in fact the market for rental housing has been characterised as a lottery. The problem is well understood in Denmark, and many discussions have taken place over the years on how to eliminate or at least ease rent controls. But no consensus has been reached, and even a more gradual approach, discussed in the Report of the Rent Control Act Committee from 1997, of allowing market-based rent for new tenants seems unlikely to be taken further.

Distortions remain in the individual housing benefit scheme

A reform of the individual housing benefit scheme (which costs some ¾ per cent of GDP) was adopted in early 1999 and is being phased-in over a four-year period that started on 1 January 2000. In general, the reform implies a reduction of housing subsidies, as households with an annual income above DKr 100 000 must pay a higher proportion of their rent themselves. When fully implemented, the reform is expected to reduce total expenditures for the individual housing benefits scheme by around 10 per cent. One dominant and arbitrary feature of the scheme has been left unchanged however, namely the far more generous benefits to pensioners than to other households with comparable income – benefits to pensioners constitute approximately 75 per cent of all expenditures. Moreover, the interaction between the housing benefit scheme and the social welfare system remains complex and non-transparent, because discretionary supplementary cash benefits can be given to households with "large" net expenditures for rental housing. Thus, while the reform entails a modest fiscal tightening, it does very little to improve the allocation process or enhance transparency of the system. However, one additional – and potentially significant – change should be noted. The framework agreement for a reform of the early retirement pension scheme (see above) states that new pensioners from 2003 onwards will be entitled only to the basic allowance, instead of the far more generous scheme currently applying to them.

Urban renewal is still heavily subsidised

The financial terms for urban renewal of rental housing – most often privately owned – have been adjusted several times during the last decade. Fundamental problems remain, however, and no further policy initiatives have been undertaken recently. Thus, the recommendations in the last *Survey* still apply: cost control as well as the incentive structure are poor, and urgent policy attention is required. Since the regulation of rents leads to under-investment in maintenance and modernisation of the rent-controlled housing stock, it eventually results in an increased number of dwellings ripe for urban renewal support. The average

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square-metre cost of urban renewal programmes is higher than for new social housing – demolition and replacement would have been cheaper in at least some cases. The urban renewal subsidies cover some 70 per cent of total costs, often with full public financing at the margin.[27] This rather perverse incentive structure provides plenty of scope for improvement. A recent report (Finansministeriet *et al.*, 2000) provides a list of proposals to enhance cost awareness that could serve as a useful starting point for the design of policy changes.

Increased property rights needed in the social housing sector

Increased property rights in social and co-operative housing, as recommended in the previous *Survey*, would allow competition on more equal terms with owner-occupied housing. It would also help to make these segments of the housing market more attractive to a broader group of people. Furthermore, the Danish authorities have encouraged mixing social, owner-occupied and co-operative housing, arguing that it serves to improve the social environment in new building complexes by preventing "ghettoisation". In the existing social housing stock, however, no steps have been taken towards increasing property rights. As in other countries, granting increased property rights would be expected to stimulate the tenant's interest in the property and the immediate urban environment. However, in Denmark there is an additional dimension to the issue. The social housing stock is formally owned by self-governing companies or branches. Most of these still have mortgage debt to repay, but an increasing number will be debt-free in five or ten years from now. As the rents are set independent of the outstanding debt, the social housing sector will accumulate substantial amounts of capital. This build-up of wealth in self-governing housing branches seems inappropriate in itself, especially as it is not clear exactly who "owns" this wealth. It might also lead to future over-investment in housing, as wealthy social housing branches would probably re-inject the capital into the housing sector. Such "lock-in" effects of capital in the housing sector would be likely to distort overall resource allocation and should therefore be avoided. This issue needs policy attention, and one way to alleviate potential problems might be to increase property rights and clarify governance structures. However, to liquidate capital from the social housing sector in connection with increased property rights it would also be necessary to address the legally difficult question of the sector's ownership rights.

Competition and regulation

The importance of increasing the degree of product market competition has repeatedly been stressed in previous *Surveys*. While Denmark has made progress over the years, more still needs to be done. On the strictness of the overall regulation of product markets Denmark holds a middle position among OECD countries (Figure 22). Competition law has been reformed in line with EU principles,

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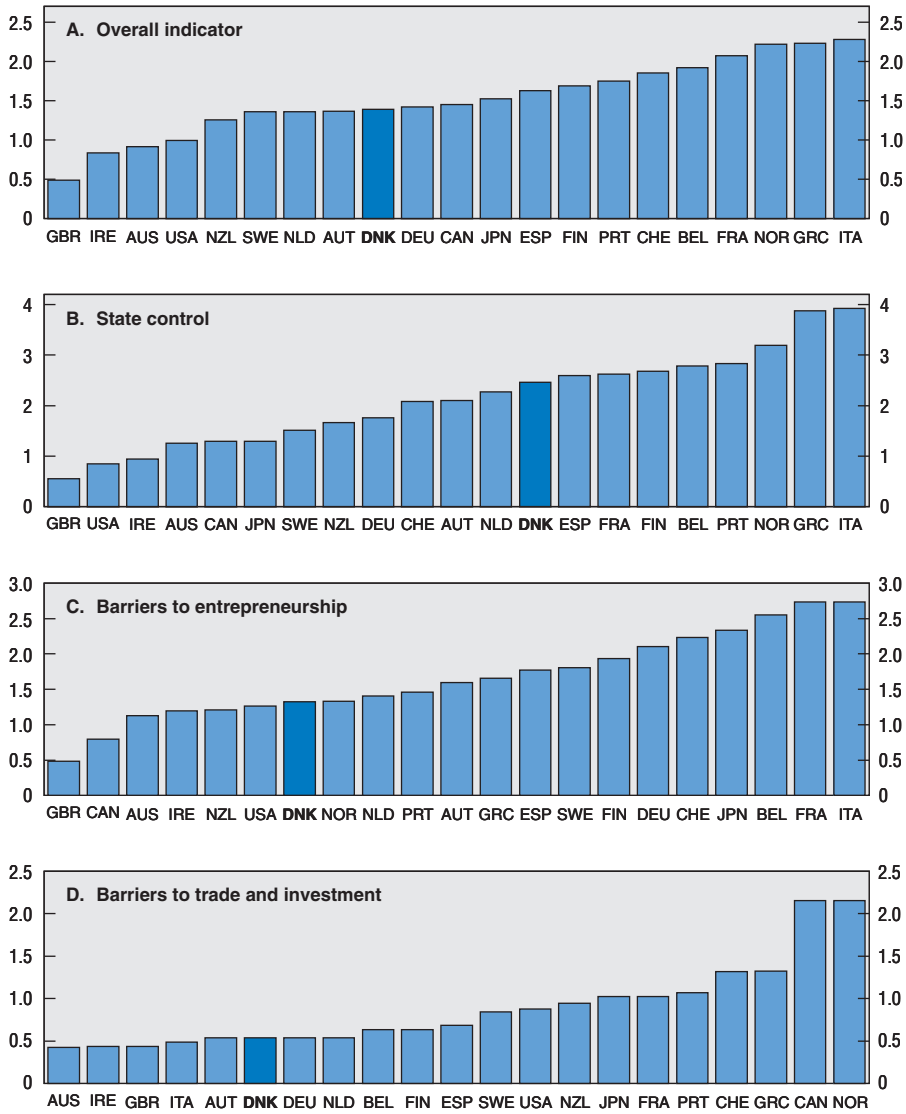
Figure 22. **Product market regulation**[1]
1997-98



1. Indicators of strictness of legislation. Scores can range from 0 to 6 with higher values representing stricter regulation.
Source: OECD.

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but the 1998 Act fell short in some areas, and some industries are subject to control only from sectoral bodies rather than from the Competition Authority. Effective competition still seems to be rather weak in some parts of manufacturing, while most utilities remain heavily regulated, as do a number of key service sectors, most notably transport. This suggests that there is considerable scope for efficiency gains to be realised from greater efforts to strengthen competitive forces in these areas.

Improving competition legislation

The framework for competition policy was strengthened with the 1998 Competition Act, which replaced the abuse principle with the prohibition principle. Denmark was one of the last EU countries to make this change. The new framework has helped raise awareness of competition policy issues, as has the more pro-active stance taken by the Competition Authority. The new instruments under the changed policy regime, "dawn raids" for example, have been used on several occasions. Extensive media coverage was given to a large-scale nationwide cartel among electricians and to the Competition Authority's decision to ban the widespread use of "report offices" effectively functioning as price-setting mechanisms. Currently the Competition Authority is reassessing all permissions to restrict competition granted under the old Competition Act and hopes to have completed the review by the end of 2001. Overall, the competition policy framework is now more in line with international standards, and some policy momentum seems to be underway to address outstanding issues. Nevertheless, the 1998 Competition Act was still vague with respect to merger control. The merger in 1998 between the two largest dairy product suppliers – the merged entity now holds total dominance in the sector with a market share of close to 100 per cent – is a recent example of a merger that was not assessed by the Authority as a consequence of the absence of merger control remedies under the current Act. Moreover fines for violating the competition rules have been insufficient. The potential size of such fines is smaller than in most other countries, and first-time violators cannot be fined. Against this background, a revision of the 1998 Act, addressing the above shortcomings has been agreed upon in May 2000.

Establishing a more up-to-date policy framework cannot resolve all competition problems overnight. The Competition Authority estimates that average consumer prices in Denmark are some 5 per cent above the level in comparable countries, which is consistent with earlier studies (OECD, 1993; Finansministeriet, 1997). Such cross-country comparisons are subject to large measurement problems, but using a broader range of indicators, the Competition Authority has identified 52 industries with signs of weak competition (Table 15). The list includes not only an array of manufacturing and service industries that have traditionally not been subject to public regulation but also some highly regulated sectors (whether

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Table 15. **Industries showing signs of insufficient competition**

	Concen-tration	Earnings	Prices
Mining and quarrying			
Oil and gas	x	n.a.	n.a.
Salt	x	n.a.	n.a.
Food, beverages, tobaccos	x	x	x
Bacon factories	x	x	x
Slaughterhouses	x	x	(x)
Fish meal plants	x	x	
Dairying	x	(x)	
Milk condensation	(x)	x	x
Bread plants	x		(x)
Sugar plants	x	x	x
Breweries	x	x	x
Mineral water plants	x	x	x
Tobacco			
Paper and graphics			
Publishing, books	(x)		x
Publishing, dailys	x		x
Publishing, weeklys	x	x	x
Chemicals			
Industrial gases	x	x	
Pharmaceuticals	(x)	x	x
Plastics	x	x	
Fertilisers, pesticides	x		
Building Materials			
Cement plants	x		(x)
Mineral wool plants	x	x	(x)
Tile works	(x)	x	(x)
Concrete goods plants	(x)	x	(x)
Mixed concrete	(x)	x	(x)
Mortar mills	(x)	x	(x)
Asphalt	x	x	
Electronics			
Switches, relays	x	x	
Loudspeakers	x	x	x
Antennas	x	x	x
Medical equipment			
Needles, syringes	x	x	(x)
Hearing aids	x	x	(x)
Spectacle lenses	(x)	x	(x)
Energy			
Gas supply	x	n.a.	x
Electricity supply[a]			
Retail, wholesale			
Super markets	x		x
Department stores	x		x
Building materials[a]			
Building and Construction[a]		x	x
Transportation			
Ferry service	x	n.a.	x
Air service	x	n.a.	x
Airports	x	n.a.	x
Railways	x	n.a.	x
Taxis[a]			
Financial sector			
Banks	x	n.a.	
Mortgage institutions	x	n.a.	
Payment services[a]	x		
Other service industries			
Private health care[a]			
Real estate agencies[a]			
Telecommunications[a]			
Postal services[a]	x		
Radio/TV production[a]	x		
Car repairs[a]	x		

Note: (x) indicates that the criterion applied is almost met; and n.a. indicates that earnings information are not available.
a) Indicates that competition is restricted for other reasons.
Source: Konkurrencestyrelsen (1999).

through legislation or intra-industry agreements). The Competition Authority is taking a broad approach to competition issues and raising awareness of the lack of competition, even where it is unable to act directly to improve the situation. This

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is the case in the financial sector, for real estate agents and car repairs, where competition problems partly relate to informational asymmetries inherent in these services. The rather large group of industries listed by the Competition Authority points to the need to pursue regulatory reform to encourage greater competition, as policy change in this direction so far has been slow and hesitant (see OECD, 2000*b*).

In manufacturing, competition problems are most widespread within the food, beverages and tobacco industry and in the production of building materials. These sectors exhibit high concentration, high prices and high earnings. In particular, industries like bacon factories and dairy products are very highly concentrated and sheltered from competition. High prices and concentrated market structures are also prevalent in some parts of the retail sector, most notably in everyday commodity sales, where the three largest vertically integrated chains have a 70 per cent market share.

The causes of weak competition are often specific to each industry in question and may have a long history. That is, for example, the case for pharmacies and for publishing and sale of books (see Box 6). In these over-regulated sectors performance is well below average by international standards, and reforms would have much scope for cost and price reduction (Figure 23). As regards pharmaceuticals, political agreement on the broad principles of changes in the subsidy scheme and the regulation of pharmacies was reached as part of the Budget for 2000.[28] Some groups of pharmaceuticals (generic substitutes) will henceforth be subject to competitive tenders, and only the prevailing brand will be entitled to subsidies. This change is aimed at strengthening competition among producers and reducing public expenditures through lower prices. The risk of cross-subsidisation is reduced by general regulation of the market for pharmaceutical products; in particular, purchasers of any product whose price in Denmark is set higher than the average price in other Northern European countries will not be eligible for reimbursement. It remains a shortcoming, however, that the Competition Authority does not have the power to override the sectoral regulation.

A liberalisation of the retail sale of pharmaceuticals is being planned, abolishing the long-standing monopoly in this area, although the details have not yet been laid out. To strengthen competition in the retailing of medicine effectively, the changes must entail a liberalisation of the present licensing system as well as of the existing imposition of equal prices throughout the country. Finally, the present prohibition against pharmacies negotiating rebates on their purchases will be abolished. The present difference of 30 per cent between pharmacies' and hospitals' purchase prices points to considerable scope for price reductions from this change, although the current differences may also be a symptom of cross-subsidisation.

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Box 6. Restrictions on competition in book sales

For more than a century the provisions regulating book sales have been laid down in collective agreements between publishers and booksellers. Only 75 of the largest publishers are included (out of some 1 400 altogether), but the agreements determine the conditions for the whole publishing industry. According to the agreements:

- Publishers are allowed to supply books only to booksellers complying with the trading regulations. Publishers set binding retail sales prices. These are typically effective for two years after release but can be extended in case a new impression is printed. Books below a certain threshold price (currently DKr 155) are not affected by these restrictions.
- Booksellers are allowed to sell only books supplied by a publisher complying with the trading regulations and must sell at the price set by the publisher. Thus, publishers who might want to break the agreements are effectively prevented from doing so. Requirements on the booksellers to have "*professional insight*" restrict the establishing of new bookstores.

While a number of other countries allow retail price maintenance in this section, Denmark stands out with respect to these exclusiveness rights:

- In Denmark, prices are fixed through agreements, and publishers are obliged to set the retail price. In addition, publishers are guaranteed exclusivity, as are booksellers.
- In Austria, Italy, Netherlands, Norway and Germany, prices are fixed through agreements and publishers are allowed or obliged to set the retail price.
- In France, Portugal and Spain, prices are fixed by legislation, although small deviations are allowed.
- In Belgium, Finland, Ireland, Sweden, the United Kingdom and the United States, there are no restrictions of any type.

The stricter regulation in Denmark has severely impeded competition and allowed the less efficient booksellers to stay in business as high costs have been passed on to retail prices; estimates by the Competition Authority suggest a 15-20 per cent surcharge on books compared to other EU countries. Moreover, Danish booksellers enjoy a much higher market share than in other countries. Ninety per cent of all books are sold through booksellers, while it is no more than 40 per cent in such countries as Sweden, Belgium, Italy, France and Portugal.

The regulations agreed between the main trade associations have until now been allowed as an exception from the general competition rules; this may have been an alternative to legislative regulation, as the current settings have held broad political support for cultural reasons. However, the cultural-political concerns can be dealt with efficiently in ways that are less detrimental to competition – and are already partly addressed in the extensively developed public libraries. Also, the experience of the United Kingdom and Sweden suggests that reducing regulation increases the number of bookstores and the total number of books sold, since the price elasticity on books is likely to be greater than one. Against

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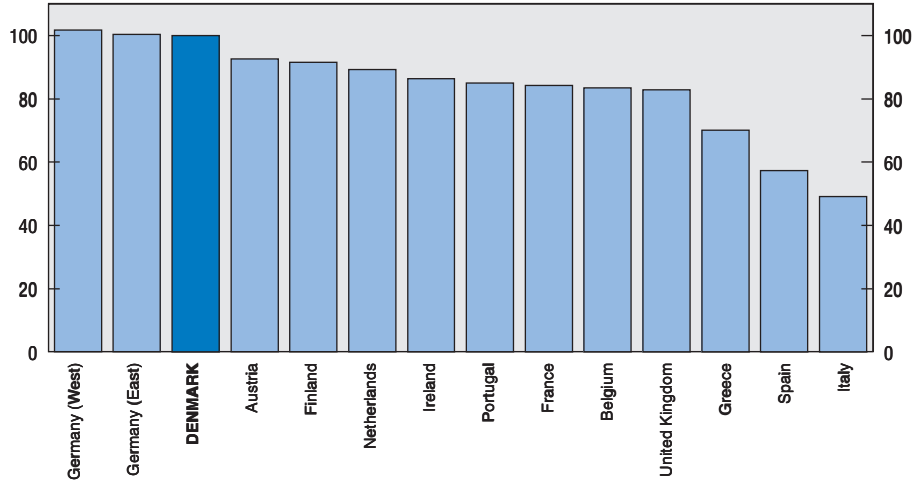
Box 6. **Restrictions on competition in book sales** (*cont.*)

the background of the new competition law from 1998 the Competition Authority has recently decided on a partial liberalisation. As from 1 January 2001 the exclusivity rights of booksellers and publishers will no longer be allowed and publishers will not be obliged to set fixed retail prices, although they will still be able to chose to do so in most cases. Indeed, these modifications are warranted.

Figure 23. **Pharmaceutical prices in EU countries, 1996**
Denmark = 100[1]



1. The comparison concerns the 100 generic groups of pharmaceuticals with the largest turnover in Denmark. Within each of these groups the pharmaceutical with the largest turnover is used for the comparison.
Source: Sundhedsministeriet (1999).

A *more competitive framework is slow to emerge in the utilities industries*

More competitive structures in the utilities industries are developing only slowly. The sole exception is telecommunications where TeleDanmark, the former public monopoly, was privatised in 1998. Competition has developed particularly quickly in the mobile phone segment and on long-distance calls, while TeleDanmark

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still holds almost total dominance over local services and cable TV and is the only supplier of all-in-one subscriptions. This reflects the fact that competitors' access to TeleDanmark's network hitherto has been too expensive.[29] Postal services remain a monopoly (PostDanmark) with full public ownership. The potential rationalisation gains there are likely to be substantial, but a transition to a more competitive structure is complicated by the presence of a large proportion of employees with public-servant status.

As in many other countries, the energy sector is in the middle of a transition process. Partial liberalisation is underway in the natural gas sector, following the EU requirement to open up that market from 1 August 2000. The electricity reform adopted in 1999 will gradually introduce competition in that sector from 2000 onwards (for the largest users) (see OECD, 2000). Businesses with an annual consumption of 10 GWh or more will have free choice of electricity supplier from 2001 and households from 2003. However, important elements of the reform are still being prepared, and the planned implementation of some elements of the reform have been postponed. Nevertheless, it has been decided that the production of, and trade in, electricity will be subject to competition, while transmission and network access and rates will be handled by the two regional system operators.

The electricity sector is characterised by considerable excess capacity, and the potential efficiency gains from full competition, free choice of supplier for all consumers and full vertical disintegration, have been estimated at 10 per cent of the total costs in the energy sector as a whole (Konkurrencestyrelsen, 1998). A thorough implementation of competitive structures in the electricity sector would entail closing down the least efficient producers and distribution companies, unless they could sell electricity abroad at competitive prices. The lengthy transition period in the electricity reform for wind turbines, including old and ineffective vintages, indicates that this reality has not yet been fully accepted, other than as a long-term goal.

The ownership structure in the energy sector (including the electricity sector) is dominated by municipalities and consumer owned co-operatives, even though networks and production facilities have been financed largely by central government. Originally this ownership structure reflected the perception that consumers' interests were best looked after under this structure. However, experience suggests that consumers have had very limited influence in practice. As in other sectors, the prevalence of co-operatives complicates the introduction of competitive structures. In addition, the dominance of municipalities tends to make warranted changes a political issue at the local level, and in any case the revenue that could be raised from selling municipal shares in electricity companies could have repercussions on fiscal policy in the short term. The latter complication has been dealt with

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by offsetting any realised revenue in each municipality's block grant, which, in turn, obviously removes any incentive for them to engage in such sales.

Notwithstanding these complications, it is a shortcoming of the electricity reform that it does virtually nothing to deal with them. The reform requires a formal separation of production, distribution and network companies (in legal terms) but not a separation of the owners behind these companies. Despite the electricity reform, therefore, the municipalities will remain the dominant owners in all parts of the energy sector and, as a consequence, vertical integration and lack of transparency will persist and the risk of cross-subsidisation will remain.

Competition problems are also quite widespread in public transportation, with city bus services being the exception. These have largely been contracted out to private companies, and average costs have been reduced by some 30 per cent. Other parts of public transportation, including taxi services, remain heavily regulated, however. Private long-distance bus services, for example, have generally been prohibited, and consequently there has been almost no competition to the monopoly railroad services in long-distance travel. Recently, the long-distance bus market has been opened marginally by a call for tenders on five specific routes, but the conditions have been widely criticised as being too restrictive.[30] The hesitation to allow long-distance bus services has been explained by the Danish authorities as a necessary consideration for the profitability of the investments made in the railroad sector. However, this should be dealt with in a different way, for example by reducing the present problems of cross-subsidisation within the railroad sector and making the subsidies for railroad services outside the main routes more transparent. This could create a more competitive framework in some parts of public transportation.

Assessment and scope for further action

The Danish economy has made considerable progress in recent years partly due to the structural reform process, an overview of which is provided in Table 16. In particular, the reorientation of labour market policy along the lines set out in the OECD Jobs Strategy has been noteworthy, but the good performance has also been underpinned by synergies between structural and macroeconomic policies. Notwithstanding the achievements so far, Danish labour market policy may be fragile in some respects. While the course of policy adopted fits well into an economy with high growth and rising employment, it may be vulnerable in case of more adverse economic developments as the required expansion of capacity could be considerable. Safeguarding and extending the results achieved so far by adding stronger economic incentives to the present policy of control and activation should be given high priority.

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Table 16. **Structural policy recommendations**

The table follows up on proposals from the 1999 *Survey* (Box 8) and summarises action taken since then. New recommendations, resulting from the analysis in this *Survey* are included as well. The follow-up on educational policies covers the period since the 1997 *Survey*, which contained an in-depth analysis of that sector

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Maintaining sound public finances (chapter II)		
Improve expenditure control (new).	Local government budgets have repeatedly surpassed the expenditure levels agreed with the central government. In addition, supplementary expenditures have resulted from objectives and high statutory standards set by the central government. Recent performance has improved as the outcome for 1999 and budgets for 2000 have been in line with the previous years' agreements.	Clarify the responsibilities between central and local governments as the tendency for the central government to impose standards and objectives on local governments may conflict with municipal autonomy. Adjust the current agreement framework with a view to strengthening expenditure control; as a concrete step, consider financial penalties on municipal tax increases.
Increase productivity in the public sector (new).	New public management tools, for example performance contracts with institutions and directors and company-style accounting, have been implemented more vigorously in recent years.	Combine the use of instruments enhancing incentives at the work-place level with across-the-board demands for productivity increases.
	Efforts to reap economies of scale by establishing more efficient administrative structures have proved to be difficult.	Strengthen efforts to develop instruments encouraging inter-municipal co-operation and voluntary mergers.
Increase the application of outsourcing and competitive tenders (new).	Outsourcing and tendering have spread only slowly and remain insufficiently used – around 7 per cent for municipalities and 10 per cent for counties.	Allow outsourcing to be applied in a broader range of services. Develop instruments to encourage more extensive use of outsourcing where this is allowed but not applied.
Extend and increase user charges (new).	Recourse to user charges in social services has been unchanged or declined during the past decade.	Reassess the statutory provisions regarding user charges for individual services. Local governments should be allowed to decide user charges more freely for a broader range of services.
Reducing structural unemployment		
Reduce benefit generosity.	No action.	Consider a reduction of the replacement rate and implement the annual income model for benefit calculation, allowing benefits to gradually decline with the length of the spell.

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Table 16. **Structural policy recommendations** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Extend the waiting period for benefits.	No action.	Proposal should be followed up. As a first step, extend the number of employer-paid days (currently two days).
Reduce UI benefit duration.	Initiatives from the 1999 labour market reform now being implemented (restricting the benefit period to four years and imposing longer qualification periods), but no further action has been taken.	Shorten duration further and abolish benefits for non-core purposes such as holidays.
Advancement of activation measures.	Full-time activation is now compulsory after six months for youth below 25 and will be so after 12 months for adults as from 2001.	Extend to unskilled in the age group 25-29 years. Adhere to a strict operational definition of what constitutes full-time activation.
Improve the efficiency of ALMPs.	A more consistent incentive structure has been established as from 2000, encouraging the PES to choose specific measures based on the assessed balance between price and quality.	Action taken should help to improve efficiency. Monitor closely the performance of the PES. Reduce the scope of early targeted activation within the first year of unemployment.
Strengthen policies towards older workers (new).	The Service Jobs scheme has been introduced, giving a DKr 100 000 annual wage subsidy (without time limit) for hiring people aged 48 or above who have been unemployed at least 18 months.	Carefully monitor the effects of the Service Jobs scheme and maintain it only if it leads to lower unemployment relative to other age groups.
	Extended benefit rights for unemployed aged 50 to 54 years were abolished with the 1999 labour market reform.	Follow up by removing the extended benefit rights for unemployed aged 55 or above as well. In addition, abolish unemployment benefit entitlement as a condition for early retirement benefit and impose the same activation requirements as for other age groups.
Increasing labour supply		
Reduce scope for early withdrawal from the labour force.	Early retirement benefit reform adopted and implemented.	The reform is a step in the right direction, but careful evaluation of the effects is necessary, and further initiatives should be considered if outcomes fall short of expectations.
	A framework for reform of the early retirement (disability) pension scheme has been agreed. The aim is to implement it in 2003.	It is too early to assess the framework agreement. Reform should strengthen work incentives and allow for regular reassessments.

Table 16. **Structural policy recommendations** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Integrate paid-leave schemes into wage settlements.	No action. Sabbatical leave expired in March 1999, following a decision from 1995.	Abolish the paid-leave schemes and shift the financing of childcare and educational leave to the social partners.
Adjust the sickness benefit scheme (new).	The statutory 12-month limit to sickness benefit has gradually been eroded by exceptions.	Introduce a waiting period and enforce the current 12-months time limit.
Lower marginal tax rates.	The tax reform currently under implementation has only small effects on average marginal tax rates paid.	As a first step, increase the income threshold for paying the top tax rate.
Education policies		
Secure basic skills, in particular at compulsory level.	New agreement with the teachers should increase teaching hours and introduce performance-related pay on a small scale.	Carefully implement the agreement at the local level.
	The new Danish Evaluation Institute is responsible for assessing the quality of education and teaching. A new act on teacher training for the Folkeskole (primary schools) aims at strengthening teachers' competence levels.	Ensure that these worthy initiatives pay off in terms of measurable improvements in performance.
Reduce drop-out rates from upper secondary education.	Drop-out rates have declined, due to small cohorts, increased demand for apprentices and better guidance processes.	Recent results have been encouraging. Efforts should continue to reach a completion rate of 95 per cent.
Reduce excessive time spent in the education system.	There are indications that long breaks between upper secondary and tertiary education have been reduced. The proportion of quota 2 (admission on the combined basis of marks at lower levels and other qualifications, in particular occupation) entrants to tertiary education is being reduced from 32 per cent of entrants in 1998 to 25 per cent in 2001.	Reduce the proportion of quota 2 entrants to tertiary education further. Continue efforts to stimulate direct transition from the ninth year of primary education to upper secondary education as the tenth year has proved superfluous for most purposes. Reduce the period for which students can receive educational grants, and set stricter time limits for completion of higher education.
Strengthen quality and cost-effectiveness in adult education and training.	"Taxi-meter" financing (*i.e.* where appropriations automatically increase with the number of completing students) and free uptake were abolished as from 1999 as total expenditures had ballooned.	Action taken was highly necessary as expenditures were out of control.

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Table 16. **Structural policy recommendations** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
	Nominal ceiling has been set on public expenditures for adult education and training. Additional expenditures are to be financed through employer-contributions to a newly established central fund (Arbejdsmarkedets Uddannelses Refusion).	The nominal ceiling is a step in the right direction, even if it has been set at a high level. But increased financing through employer contributions will not strengthen cost awareness at the enterprise level. Therefore, meaningful user charges for enterprises and/or participants should be consistently applied.
	Subsidies to courses with a general rather than a labour-market related content have been reduced somewhat in scope.	Try to narrow further the range of courses receiving subsidies by eliminating those with a narrow firm-specific content.
	A reform, aiming to create better coherence between the content of individual courses, has been adopted but not yet implemented.	The objective of the reform is appropriate and important. Careful implementation is needed.
Tax policies		
Reduce marginal personal income tax rates.	The 1999 tax reform reduces statutory tax rates but has a small effect only on average marginal rates paid.	Increase the income threshold for paying the top tax rate. Consider the scope for further broadening of the tax base to allow rate cuts. For example, the tax-value of deductible interest outlays will remain high by international standards. Extend the application of user charges to create room for cuts in tax rates.
Remove asymmetry in capital income taxation.	The spread between tax rates on capital income and interest outlays has increased with the 1999 tax reform.	Align more closely the tax treatment of capital income with that of interest outlays.
Reduce the corporate tax rate.	Corporate tax rate reduced to 32 per cent in 2000.	Reduce tax expenditures further to make room for additional cuts.
Housing policies		
Dismantle rent controls and extend property rights in social housing.	No action.	Remove the rent controls applying to private rental housing. Rent regulation in social housing should also be modified to eliminate unwarranted rent differentials within and across vintages.

Table 16. **Structural policy recommendations** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Clarify governance structures in the social housing sector (new).		A strategy to deal with the expected wealth accumulation in the sector should be developed in order to avoid lock-in effects of capital and over-investment in housing. Increasing property rights might be one element.
Improve incentives for cost reductions in urban renewal.	No concrete action. A recent report has pointed to a number of initiatives that could work in the right direction (Finansministeriet *et al.*, 2000).	Reconsider the current rules so as to avoid the perverse incentives in the present system. As a first step, implement policies that would increase cost awareness.
Reconsider the housing benefit scheme.	Reform of the housing benefit scheme is being phased in from 2000 to 2004.	There will remain distortions in the scheme. For example, benefits available to pensioners should be aligned with those for other groups.
Competition policies		
Strengthen the 1998 Competition Act.	Following a government proposal, political agreement has been reached to introduce merger control and fines for first-time violations.	The inclusion of merger control and penalties for "first-time offences" is a clear step in the right direction. Extend the coverage of the Competition Act by including for example the utilities sectors, tobacco products, pharmaceuticals and book sales. Tighten the law exempting practices that are "a necessary or direct consequence of public regulation".
Complete deregulation of public utilities.	A new electricity reform has been adopted, and limited deregulation of the natural gas sector is underway.	Monitor and ensure that electricity deregulation delivers effective competition. Increase further vertical separation in the sector where municipalities remain the primary owners in production, network and distribution companies. Go beyond the minimum deregulation of the natural gas sector set by EU requirements.
	City bus services have been subject to competitive tendering resulting in increased competition and lower costs. Long-distance bus services remain highly regulated.	Utilise competition to improve resource use in road and rail transport more consistently. In particular, long-distance bus services should be deregulated, the current call for tendering on five specific routes being an insignificant first step only.

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Table 16. **Structural policy recommendations** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
	In the telecommunications sector, competition has increased significantly in long-distance calls and in the mobile segment. But TeleDanmark, the former monopolist, is still dominant in local services and is the only profitable company in the sector. Moreover, it is still leading in cable TV, preventing competition between networks. Competitors access prices to TeleDanmark's network has recently been reduced significantly.	Carefully monitor that competition strengthens further and that cheaper access to the network becomes effective.

Source: OECD.

While the overall effects of labour-market policy have been very positive, the same is less true for the age group above 50. This is undoubtedly linked to weak work incentives. The unemployed over 50 have enjoyed special privileges in the unemployment insurance system that in practice has implied an indefinite benefit period. The elimination of the transitional early withdrawal scheme in early 1996 and of the special privileges of 50-54 year-olds in the unemployment insurance system in 1999 were steps in the right direction, although it remains to be seen if they will prove adequate. Further initiatives should include elimination of the special privileges accorded to 55-59 year-olds and reconsideration of the current requirement that early retirement benefit entrants be eligible for unemployment benefit, as the latter in practice prevents labour market policies from working efficiently for this age group.

Increasing the participation rate in the age group above 50 is crucial if the objective stated by the government – to increase labour supply by 5 per cent by 2005 – is to be achieved. However, the analysis in this chapter points to a shortfall of at least 3 per cent given the policy initiatives already undertaken. Thus, further action will be required to stimulate labour force participation and increase average effective working time. Besides a reconsideration of the generosity of the transfers system, this should include a reduction of marginal income tax rates, in particular in the income brackets above the level of unemployment benefits in order to strengthen work incentives further.

In educational policies, several of the recommendations contained in 1997 S*urvey* have been partially implemented, which should support structural progress in this area in the future. The general shift in focus from quantitative to qualitative objectives, noticeable in analytical work as well as policies, is most welcome. The working-time agreement with the teachers, a continuation of the fall in drop-out rates in upper secondary education and the establishment of consistent incentives for the PES to aim for value for money constitute the most significant changes. More needs to be done, however. The main policy concern remains the advanced age at which working lifetimes begin, which causes a double strain on public finances from heavy expenditures as well as the revenue impact of reduced labour supply. A significant policy response to reduce the excessive time spent in the education system should include a reassessment of the period for which students can receive educational grants and stricter time limits for the completion of tertiary education.

The last three tax reforms have helped to strengthen private savings considerably. By contrast, average effective marginal tax rates paid have not fallen much, and remain at levels restricting the supply potential of the economy, in particular by putting pressure on average working time. Therefore, genuine cuts should be considered, for example by increasing the income threshold for the top tax bracket. Moreover, the asymmetric taxation of positive and negative capital

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income, reinforced by the terms of the Whitsun package, should be reconsidered, as it causes unwarranted lock-in effects in individually owned private companies. Finally, corporate taxation would benefit from a reduction in the tax rate financed by a broadening of the base.

In housing policy, rent control in the private rental sector and the design of the urban renewal scheme remain the primary areas of concern. Rent controls should be dismantled and the urban renewal policy reformed to provide stronger incentives for doing maintenance and controlling costs.

The 1998 Competition Act was an important step in the right direction, and the recent decision to introduce merger control and fines for first-time violations bode well for further improvements of the competitive framework. The number of industries with signs of insufficient competition, as identified by the Competition Authority, is quite extensive and is one indication that significant scope for further improvement still exists. In particular, where competition problems are caused by excessive public regulation, policies should be reviewed; retail sales of books and pharmaceuticals and provisions regarding taxis and private long-distance bus services are obvious examples. Progress in the utilities industries has generally been slow. The recent electricity reform was helpful, although it is unfortunate that it does not ensure vertical separation within the sector.

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IV. Encouraging environmentally sustainable growth

Introduction

Environmental issues have been high on the policy agenda in Denmark for some time. Although the Environmental Protection Act, the principal current piece of legislation, was introduced only in 1973, a Nature Conservation Act had already been introduced in 1917, and the Danish Society for the Conservation of Nature, a non-governmental organisation with an important role in implementation of environmental policy, dates from 1911. Even though Denmark is relatively highly urbanised – 80 per cent of the population live in urban areas – there are few very heavy concentrations of population, and there have never been significant concentrations of heavily polluting manufacturing or processing industry. Natural-resource-dependent activities are few but important: Denmark is currently broadly self-sufficient in oil and gas products from the North Sea; its fishing fleet is one of the world's largest; agriculture is perceived as one of the most important industries, though providing only some 3 per cent of GDP directly, it is also the base of an important food-processing industry. Nature-based leisure activities are also important to many Danes.

Although Danish emissions of certain air pollutants are relatively high – per capita emissions of NO_x and CO_2 ranked sixth and seventh in the OECD, respectively, in 1995 – air quality, insofar as it can be compared across countries, ranks better than this.[31] Most air pollution is in fact "imported" – over 80 per cent of acid deposition in Denmark of SO_x and NO_x comes from outside the country, while Denmark is a net exporter of such pollution – around 90 per cent of Danish emissions travel elsewhere.[32] Surface water quality, affected both by deposition from air pollution and direct discharges, is much poorer relative to other OECD countries, however.

The Economic Council has experimented with ways to represent the condition of the environment and natural resources in the national accounts. This is not done systematically, but an exercise in calculating real net saving showed that whereas the national accounts showed net saving over the period 1986-96 at around 5 per cent of net national product, a "green adjustment" reduced this by

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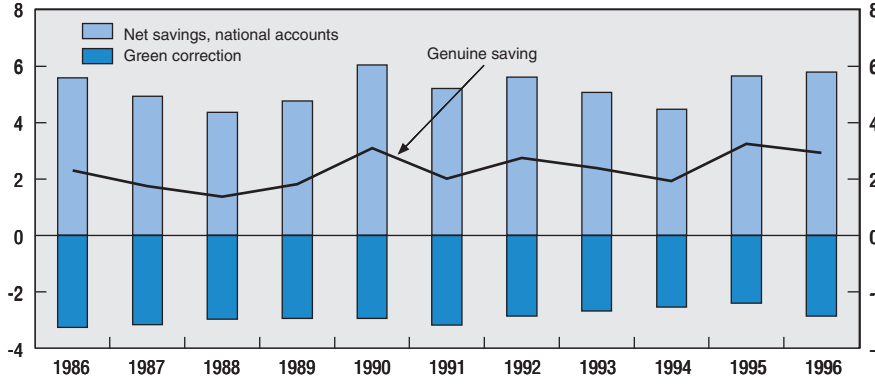
half (Figure 24). The green adjustment included corrections for depletion of oil and gas reserves, for emissions of greenhouse gases and other polluting gases,[33] but did not attempt to take account of water quality or even more difficult to value factors such as nature conservation or biodiversity. On this measure, environmental damage has a noticeable impact on welfare, with no clear trend over the period.

An active environmental policy is pursued, and in some cases, notably in respect of greenhouse gas abatement, ambitious targets have been set. A variety of different approaches to achieving the targets is used, varying from "command-and-control" regulation to taxes and charges (and, occasionally, both at the same time). This chapter looks at a selection of issues: water, waste, energy and transport policies with their links to air pollution and greenhouse gas emissions; these touch on most (though not all) of the principal environmental issues in Denmark, and are the areas where the analysis presented here suggests that improvements could be made to policies. The focus is particularly on elements of policy which appear to be excessively costly, either through inappropriate instruments or where unequal treatment of different activities encourages inefficient responses. The chapter opens with a brief outline of the policymaking institutions in the area of the environment; subsequent sections cover the policy targets and issues; a final section discusses the Danish approach to environmental targets and instruments, particularly the

Figure 24. **"Green" adjustment to net national saving, 1986-1996**
Per cent of NNP



Note: The "green correction" is the sum of the value of extraction of oil and gas, and estimated environmental cost of emissions of carbon dioxide, methane, nitrogen oxides, sulphur dioxide and volatile organic compounds (see footnote 32).
Source: Danish Economic Council.

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use of economic instruments, and to issues of balancing costs and benefits. Some conclusions and recommendations conclude the chapter.

Institutions

Denmark is unusual in having a combined Ministry for the Environment and Energy.[34] It is argued that this allows better co-ordination between environmental and energy policies, which might otherwise be antagonistic. Co-ordination is clearly important, since so many environmental issues are linked with the energy industry. There is, however, a fine line between co-ordination and subordination, and care is needed to ensure that conflicts of interest are resolved as far as possible through systematic analysis.

Environmental targets and policies are decided centrally, but, in common with education and many social services, implementation of many aspects of environmental policies is delegated to municipalities or regional ("county") level.[35] This includes in particular most issuance of operating or discharge permits, which are handled at county or municipality level depending on the size of the installation involved. Counties are also responsible for local and regional bus services, whose relation to environmental issues is discussed later. Municipalities are responsible for water supply and treatment and waste disposal, with a tendency for them to group together to provide these services as the necessary scale increases.

With a lot of responsibilities devolved to the local or regional level, the potential for democratic participation in many decisions is high,[36] but problems of lack of information or analytical capacity could emerge. The *Amtsrådsforeningen i Danmark* (Association of County Councils in Denmark) and the Kommunernes Landsforening (National Association of Local Authorities) provide a means for pooling such resources, as well as for joint negotiations with central government. In waste disposal there is less variation in approach than this decentralised structure might suggest; a national system of taxation is in place, allowing for internalisation of environmental costs in decentralised decision making, but centrally determined targets – not obligatory on local authorities – for different types of waste disposal appear to dominate local policy in practice.

The formal structure for integration of economic and environmental decision-making is well developed, although not unified, and its operation seems still to be evolving. Environmental Impact Assessments (EIAs) are made (under the 1992 Planning Act) in respect of various projects; responsibility for producing the EIAs lies in general with the county. Legislative proposals are subject to a requirement to produce a Strategic Environmental Assessment (SEA), which is based around a checklist of possible effects on the environment and which is the responsibility of the ministry sponsoring the legislation. Under a prime-ministerial decree, introduced in 1993 and revised in 1995 and 1998, the impact of environ-

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mental and other legislation on the economy must be assessed when it is presented to parliament. Although these arrangements provide a good framework for analysis of policy, the process is sometimes treated as a mere formality, with little independent checking of the accuracy of either the economic or environmental impact analysis – although the Finance Ministry will check any assessment of budgetary effects, it does not verify any wider impact analysis, nor does the Environment Ministry typically check the assessments of environmental consequences.

Since 1997 a report ("An environmental assessment of the finance bill") has been prepared annually by the Finance Ministry. Criticised by some for being an economic assessment of environmental policy rather than the other way round, the reports are useful sources of information and explanation of many aspects of policy including, in some cases, information on costs and benefits.

A recent report into the efficacy of environmental assessments in the legislative process concluded, at least in respect of the two case studies undertaken for the report, that environmental impact assessments' influence on Parliament's decision to adopt legislation was "murky and vague" (Elling and Nielsen, 1998). The same report said that claims in the bills that they would bring beneficial environmental effects were not substantiated or explained.

Local or county administrative decisions can be contested on appeal to the Danish Environmental Protection Agency and subsequently, in certain cases or on matters of important principle, to an Environmental Appeals Board. A number of non-governmental organisations, notably the *Danmarks Naturfredningsforening* (Danish Society for the Conservation of Nature), act as watchdogs for the environment, and in certain cases have the right to make complaints. Appeals in relation to EIAs or land-use planning decisions are made to another body, the Nature Protection Board of Appeals.

Although the judicial system is rarely used, this array of centralised appeal bodies potentially provides a basis on which the consistency of the results of decentralised decision-making across the country is maintained. Information on whether such consistency is actually achieved is not readily available.

Finally, international agreements play an important role in Danish environmental policy. Two major sets of policy issues – emissions of greenhouse gas and of acid rain precursors – are almost entirely transboundary concerns. Steps that Denmark takes on its own to curb carbon or SO_x and NO_x emissions will give it very little direct environmental benefit, certainly rather little in relation to the costs incurred. But Denmark has every interest in playing a leading role in implementing such agreements, if this helps to encourage other "upwind" countries to cut their own emissions, thereby benefiting Denmark.

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Policy targets and policy issues

The 1999 OECD Environmental Performance Review provides a comprehensive review of Danish environmental policies and targets. The aim of this chapter is to assess a selection of these and their role in enhancing growth with environmental sustainability. Two specific environmental issues are taken up below: water pollution, which became an important issue during the 1980s when growing contamination by nutrients threatened Denmark's traditional reliance on largely untreated groundwater for its water supplies; and waste management, where central government policy aims to reduce use of landfills to very low levels, employing a range of reasonably sensible economic instruments but supplemented by inappropriate nation-wide centrally-determined targets. A number of different environmental issues are then treated under the headings of energy and transport policy, where the interaction of taxation, regulation and deregulation, industrial policy, international agreements and constraints, makes for a very complicated picture.

Water

Total water resources[37] per capita in Denmark, though adequate, are among the lowest in the OECD. Intensity of water use is nevertheless similar to a number of other OECD countries, because abstractions per head are relatively low (Figure 25); low water usage is partly due to the relative lack of heavy industry, and the use of brackish or seawater for cooling in many power stations. Easy access to and previously high quality of groundwater supplies mean that Denmark is very unusual in that nearly all of its public water supply is from groundwater, in the past with very little treatment beyond simple filtering. However, surface and ground water quality had deteriorated substantially by the 1980s – in particular in terms of nutrient (nitrate and phosphate) concentrations (nitrates in ground and surface water, phosphates in surface water). As a result of this and the direct effects of surface water pollution, an ambitious plan for cleaning up surface water bodies was begun in 1987 and revised in 1998 as the Action Plan for the Aquatic Environment No. 2. Targets were set for different kinds of discharge and for different sectors – municipal water treatment plants, and industrial and agricultural discharges.

It is interesting that while municipal and industrial discharge targets have been met, at the cost of considerable investment and much higher water charges (see below), nitrate discharges from farms are well above target, even after the timetable for meeting it was relaxed. It is not clear whether the difference in response is due to much higher costs of reducing discharges on farms, which is a risk inherent in the kind of sectoral target quantity reduction approach often adopted in Denmark (the GHG reduction plans are another example of this), or

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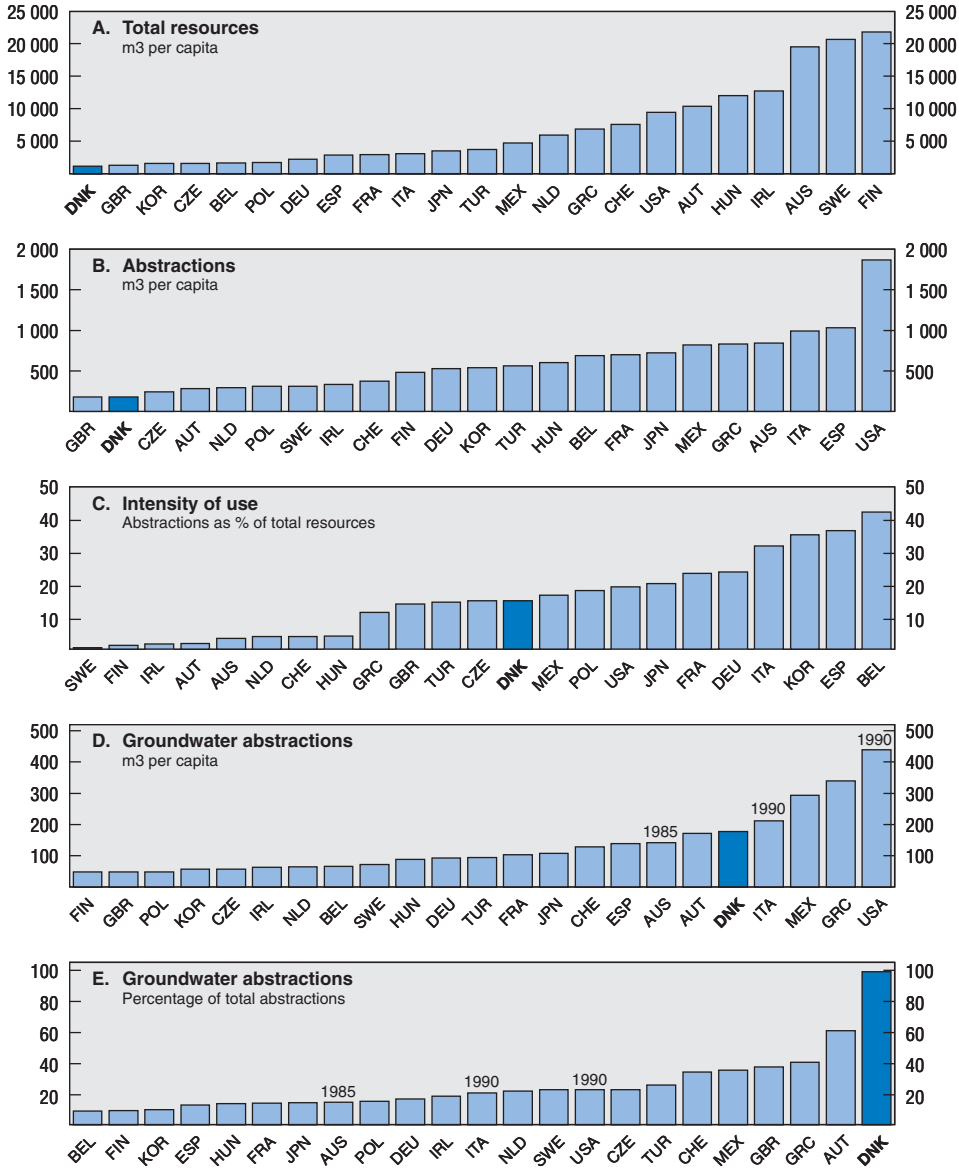
Figure 25. **Water resources in Denmark: an international comparison**
1997



Source: OECD Environmental Data Compendium, 1999.

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whether it is that procedures intended to enforce reductions in fertiliser use in agriculture were not strong enough.

Water and agriculture

A logical response to continuing water quality problems due to discharges from agriculture would be to subject discharges from agriculture to the same rates that are applicable in the general tax on waste water, a tax levied on the nitrate, phosphate and organic content of waste water. There are a number of special conditions and exemptions to this tax, and agriculture is probably the most important single exemption.[38] A fertiliser tax was introduced, but it applies only to non-agricultural uses (of fertiliser).

A tax on the nutrient content of all fertiliser (and animal feed) could be very unpopular among farmers. Instead they are forbidden to use excessive amounts of nitrogen. They are subject to a fine per kilogram of "excess" fertiliser use, the fine being from two to four times higher than the tax. Excess use is defined farm by farm as a function of land use, a maximum allowed use of nitrogenous fertiliser and the actual use thereof. Farms are obliged to keep[39] fertilisation accounts. This is a system with rather poor incentive properties – farmers are penalised for excess use, but not rewarded for economising on fertiliser use below their assigned levels as they would be if a tax on all fertiliser were applied. Thus the incentive to develop innovative means of economising on fertiliser use is high – perhaps too high – when farms are in excess and too low where the legal limits are being respected. It is true that methods and timing of fertiliser application, as well as the use of catch crops, have an enormous impact on the amount of nutrients that escape into water bodies for any given amount of fertiliser use, so that even a fertiliser/feed tax would not be ideal. But a tax could be more efficient than the current approach, lowering overall compliance costs, for any given targets, by giving incentives for those with lower abatement costs to reduce discharges by more than those with higher costs. Two important objections to such a tax that contribute to its unpopularity – the possibility of a heavy tax burden on farmers, and the rather loose relationship between fertiliser application and nutrient discharges already referred to – can be largely overcome (see Box 7).

Of course, beyond a certain point, further reductions in nutrient inputs to farmland imply degradation of fertility – it might seem bizarre to give tax incentives for this. However, if the environmental problem is sufficiently serious, this is the logical approach, just as governments are willing to accept the social costs of relocating or closing a sufficiently heavily polluting factory.

As in all European Union countries, and most OECD countries, it is ironic that agriculture is at the same time one of the most highly subsidised industries and one of the most highly polluting industries. Furthermore, the main subsidy, through high guaranteed prices, act on marginal production and thus are an incentive to

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Box 7. Taxing fertiliser?

There are two objections to a nitrogenous fertiliser tax: that it would be a heavy burden on farm finances, and, secondly, that it may be an ineffective means of reducing nitrogen leaching from farmland. The first is simple to deal with, the second more difficult.

Certainly, a tax paid on all fertiliser use would imply a large tax bill. But all farms could be given a tax credit equivalent to the tax payable on, say, their current fertiliser use target. If they use exactly their target amount they will face no net tax bill, as at present; if they use less they would be able to keep some of the tax credit. By creating more consistent incentives to reduce fertiliser use this would be a more efficient and effective method than the present one, even if the tax were set at a lower rate than the current fines. An alternative way of implementing a system with similar incentives would be to issue fertiliser-use permits and make them tradable. Whether a tax or tradable permit system were used, it could be progressively tightened if necessary, by increasing the rate of tax or reducing the total permit issue. Some revenue could be raised, under either system, by setting the tax credit equivalent to somewhat less than likely average fertiliser use or, in the case of a tradable permit scheme, by charging for all or part of the initial permit allocations.

The problem with the fertiliser tax as an efficient instrument is that while the tax can be calibrated on the amount of nitrogen applied to land, the target is the amount of nitrogen that leaches from the land into surface and ground water. But the nitrogen cycle on a farm is complicated, depending on the crops grown, use of animal manure and so on. Under some circumstances, an increase in the cost of fertiliser may cause shifts towards the production of crops that need less nitrogen in the soil, but which may also retain less nitrogen, so that even as nitrogen application is reduced, leaching could *increase*.

The only reliable way round this problem is to measure flows of nitrogen on and off farms (nitrogen can leave embodied in crops, animals or animal manure), levying a tax on the difference between the two, which is what must be left in the soil and liable to end up in surface water; see Hansen (1999), for a detailed description of a possible scheme. This might imply a heavy administrative and data-gathering burden and the costs of this might not be worth the gain in efficiency compared with a simple tax on the nitrogen content of fertiliser. However, in the Danish case, all the necessary information is already collected in the farm-by-farm nitrogen fertilisation accounts that cover all farms. The incremental cost of maintaining the reliability of the data (the incentive to misreport might be increased under a general tax) would probably not be very high. A tradable permit scheme could be applied here.

Any of these systems could be adapted to recognise that nitrate discharges are more damaging in some areas than in others, by making the tax payable, or number of permits required, per kilogram of nitrogen, a function of the estimated marginal damage done in that locality.

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more fertiliser use. While set-aside programmes reduce the amount of cultivated land and therefore the amount of land to which fertiliser is applied, they retain the incentive to intensive cultivation of the remaining land. From this perspective, a fertiliser tax on agriculture can be seen as part of moves towards a more sustainable agricultural policy.

Agriculture has had an enormous effect on the Danish landscape as well. Some programmes are underway to reverse some of the consequences of this influence. Watercourses that had previously been straightened to improve drainage are being restored to their "natural" beds, forest cover is being targeted to double over the next 80 to 100 years; some animals (beaver, native salmon) are being reintroduced into the wild too. These programmes are financed out of general taxation.

Water charges and taxation

The pricing structure for water supplies follows recommended "economic instrument" approaches, with full cost recovery and charges based on consumption (see OECD, 1999*f*). Up to the mid-1980s, water prices were among the lowest in the OECD. Subsequently, the heavy investment required to try to meet the targets of the aquatic action plans increased charges considerably and by 1995 the tax-inclusive price of water to households in Denmark was the highest among OECD countries for which data were available (OECD, 1999*f*). Since the water tax has risen substantially further, this is probably even more the case now. (As far as households are concerned, wastewater treatment and the wastewater tax described above is paid by local water companies and passed on in water charges.)

A rise in prices would be expected to finance interest and amortisation on the investment necessary to implement the targets of the aquatic action plans. It seems that the intention was in fact to avoid borrowing completely and to finance all the investment out of cash flow from increased charges, but some borrowing was nevertheless necessary despite the large price increases. With the addition of the water tax and waste-water tax, total revenue from water charges and taxes probably more than finances the ongoing costs of supply and treatment facilities, although it may also be the case that the relatively small scale of most water supply facilities keeps costs relatively high.

Industry is exempt from the water tax, which is paid only by households. Although the water tax is only one of four volume-related payments,[40] industry is in a sense cross-subsidised at the margin by households. The authorities argue that the industry exemption compensates for water suppliers' inappropriate charging structure – there is no fixed element and all costs are covered through a uniform volumetric tariff, which thus exceeds the marginal cost of supplying water. This might still involve some cross-subsidisation and leaves no-one paying the

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appropriate marginal cost (unless the external costs of water use are higher than the taxes); a restructuring of water charges would be appropriate. Consumption of water has fallen since the charges and taxes were increased, though the implied price elasticity of demand is low, a useful property from the point of view of raising revenue.

With water in Denmark apparently relatively scarce, surprising as this may seem, charging a resource rent – revenue from water charges and taxation exceeding the actual costs of supplying it – makes sense. While also being a convenient source of general government revenue, this has had the effect of reducing consumption, which fell some 20 per cent between 1980 and 1997. Water supply and treatment is mostly provided by municipalities, though one-third is provided by private water works. Municipalities are free to join with their neighbours to operate larger scale plants to benefit from economies of scale. The regional variation of charges appears mostly to reflect variations in local conditions, but the magnitude of such variation (the Confederation of Danish Industry reports that charges per cubic metre vary from DKr 5 to DKr 45) suggests that there may be efficiency gains available in some places and that local administrations do not have sufficient incentive to search for them – given their statutory right to charge whatever is necessary to cover costs and highly price-inelastic demand. The slow progress in reducing groundwater nutrient contamination increases the importance of incentives to search for cost-efficient treatment methods, as well as reinforcing the need for what is emerging as a significant cost-externality from agriculture to be addressed in that industry.

Waste disposal

Denmark makes considerable use of economic instruments in managing non-hazardous waste. A general tax on waste disposal was introduced in 1987 at a low level, rising during the 1990s and from 1993 onwards being increasingly differentiated according to the method of disposal (Table 17). With specific duties on a

Table 17. **The Danish waste tax**[1]

	DKr per tonne ($1= DKr 7 approx.)				
	1987	1990	1993	1997	1999
Delivered to incineration with electricity cogeneration	40	130	160	210	280
Delivered for other incineration	40	130	160	260	330
Delivered to landfill	40	130	195	335	375

1. Waste disposal charges also vary according to the method of disposal, averaging DKK 245, 325 and 285, for incineration with cogeneration, other incineration and landfill, respectively, in 1999.

Source: Ministry of Finance (1999).

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number of individual products (for example, plastic bags, tyres, disposable tableware, rechargeable NiCd batteries, electric bulbs) or materials, in addition to the general waste disposal tax, Denmark probably has higher taxation on waste than any other OECD Member country.[41]

Many of these taxes were first imposed, or increased, in the early 1990s as part of the shift towards "green" taxation which financed a necessary reduction in marginal taxes on income and labour. They are not typically calibrated from an explicit calculation of environmental damage, and the need to raise revenue may sometimes have taken precedence over structuring taxes so as to provide the desired incentive effects. Such a structure is not easy to achieve, especially for municipal waste. Collection and disposal of municipal waste are the responsibility of municipalities, which therefore pay the waste tax in the first instance. Local government is required to charge so as to cover costs including taxes. Thus, while a few municipalities charge individual households by the weight of their waste, the vast majority simply set flat rate charges to cover tax-inclusive costs. In the latter case there is no incentive for households to reduce their production of waste; there is some evidence[42] that waste production falls markedly if weight-based fees are imposed at the household level. The feasibility of such a system will vary according to the municipality – where there are many multi-occupied dwellings it would be much more difficult to pass on the by-weight charge to individual households. Only fourteen municipalities currently use such charging systems, and there are concerns that they lead to increases in illegal dumping.

In practice, waste policy is strongly influenced by central targets, whose emphasis is on recycling. For each of a number of different types and sources of waste, a certain percentage of recycling, incineration and landfill disposal is targeted. The overall target for the year 2000 is for a high level of recycling (54 per cent) and for maxima of 25 per cent and 21 per cent for incineration and landfill respectively. The recycling target share was already exceeded in 1996 (OECD 1999*e*), and there is now a target of 60 per cent recycling for 2004. Although the tax system described above provides some incentive for incineration, particularly if burnt in electricity generating plants, it was strongly reinforced by a 1997 administrative directive that requires that anything that can be incinerated, must be.

One interesting area in the recycling debate is glass bottles used in the beer and soft drinks industry. The use of metal cans is banned, and a mandatory deposit/refund scheme is in operation to encourage bottles to be returned to the point of sale. Domestic brewers can re-use multiple-use bottles once collected and thus avoid the DKr 0.5 tax on new bottles, but foreign brewers would face high transport costs in doing this and therefore in practice have to recycle (rather than re-use) the glass. Some claim that the combined effect of the ban on cans and the deposit-refund system is protectionist; the European Court is to rule on this later this year in an action brought by the European Commission. Certainly, if non-

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re-used bottles and cans paid only a tax similar to the landfill tax, the cost of supplying the Danish market from abroad would be much lower; however, this cost may not take into consideration other factors that may contribute to the marginal social cost of non-re-used bottles.[43]

The priority given to re-use and recycling reflects the "waste-hierarchy" approach increasingly used by many European countries. This approach can work well provided a comparison of benefits and costs – both economic and environmental – is allowed to influence the choice of waste-disposal method. The introduction of differentiated waste taxes has moved Danish policy further than most countries in this direction, but the retention of quantitative targets and outright bans (such as that on landfill disposal of any waste that can be incinerated) reflects a reluctance to rely completely on economic instruments. To a considerable extent this may be explained by the difficulty of agreeing explicit social valuations of environmental benefits and costs. While the command and control approach involves implicit and quite possibly inconsistent valuations, this approach may well nevertheless be more socially acceptable, especially since it can serve to implement European directives, such as the goals of the Landfill Directive.

Energy

The energy sector is at the heart of many aspects of Danish environmental policy, concerning both policy targets and instruments to achieve them. These include: international agreements on cross-border pollution;[44] an emphasis on renewable energy (that predates concern with global warming); ambitious targets for reducing greenhouse gas emissions; a mixture of central planning and economic instruments; and a structure of taxation influenced by sectoral and fiscal considerations as well as by environmental interests. The picture is complicated by the fact that the electricity sector is in the midst of being reformed, with the intention being to move towards a more competitive market system, though this process has only just begun, and it is not yet clear what the final outcome will be. This section begins by focusing on the complicated structure of energy taxation, which reflects many of the main environmental issues.

Denmark already had an energy tax in the late 1970s, which was generally levied on the energy content of different fuels; the rate was increased in 1986 and natural gas exempted. During the 1990s the tax structure has evolved, and there are now three different taxes: the energy tax, the CO_2 tax and the SO_2 tax. Table 18 shows the main elements. The number of different tax rates and the range of variation are striking.

Introduced in 1996, the SO_2 emissions tax rate, at DKr 10 per kilogramme, corresponds to approximately $1 200 per tonne, compared with a current market price for emission permits of around $200 per tonne in the US SO_2 trading scheme

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Table 18. **Tax rates on energy, 2000**

		Energy tax DKr per 10^9J	CO_2 tax DKr per tonne	SO_2 tax DKr per kilogramme
Fuel used for electricity production		0	0	10
Electricity		48[1]	100[2]	
Fuel used for all residential and commercial purposes, and industrial space heating	Coal	47	100	10
	Oil	48	100	10
	Gas[3]	48	100	10
Light industry	Without VA[4]	0	90	10
	With VA[4]	0	68	10
Heavy industry	Without VA[4]	0	25	10
	With VA[4]	0	3	10

1. The energy tax is levied on electricity supply at a rate equivalent to DKr 48 per GJ of coal energy, assuming that all such electricity is generated from coal at average efficiency rates for Danish power stations. In addition, two special duties are applied.
2. The "CO_2" tax on electricity is levied – except on that from renewable sources – at DKr 0.1 per KWh. This rate is equivalent to DKr 100 per tonne of CO_2 from coal, assuming that all such electricity is generated from coal at average efficiency rates for Danish power stations.
3. Gas is priced as a function of the tax-inclusive price of competing fuels. Gas suppliers pay DKr 40 per GJ tax, retaining DKr 8 "shadow tax".
4. VA: voluntary agreement.

Source: OECD, Danish Ministry of Taxation.

(and the $2 000 per tonne penalty for entities that exceed their permitted emissions in that scheme). The Swedish tax is somewhat higher; the environmental costs of such emissions vary a lot according to location, however. According to the Ministry of Taxation, the rate was chosen to correspond to current estimates of the marginal environmental cost of SO_2 emissions, although it is somewhat lower than the figure used in the green saving adjustment mentioned earlier, and much lower than that used in the Energy and Environment Ministry to assess the costs and benefits of the wind turbine programme (see below).[45]

The "energy tax" is payable only on household and non-commercial uses and industrial space heating. Exempting fuel used in electric power generation allows both domestically-generated and imported electricity used in Denmark to be taxed at the same rate, although there is no environmental reason why this should be necessary. The energy tax does not appear to have a specific environmental justification, in fact, although it is often – and sometimes misleadingly – recalculated as an implicit tax on CO_2.[46] There are a number of environmental externalities that arise from use of energy, so taxing it has environmental benefits. But since the externalities do not generally depend on energy content itself, it is not a precisely targeted tax, especially since industrial energy use is untaxed.

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The carbon tax also appears irrational at first sight. If the purpose of the tax is to internalise an externality due to CO_2 emissions, then the structure only makes sense if this externality is at least four[47] times higher in the case of emissions for space-heating purposes than for certain energy-intensive industrial processes, whereas in fact the external costs of CO_2 emissions are the same whatever the source. The exemption of SO_2 emissions from electricity generation also made little sense, though this exemption was removed at the beginning of 2000.

Two parallel, and difficult to distinguish, arguments are used to justify this deviation from a textbook uniform tax. One argument is that such a tax would damage the competitiveness of energy-intensive activities, leading to a loss of jobs and output; taken on its own, this argument is not very convincing, since the overall costs of reaching a given domestic emissions target are increased if heavy emitters are given special treatment (see Box 8). The second argument takes the first a step further: not only (it is argued) would output and employment in Denmark fall as, for example, cement production were curtailed, but the lost production capacity would move elsewhere, to countries where taxation is lower or absent, leaving global CO_2 emissions broadly as before. In these circumstances, given the Danish authorities' approach of targeting global emissions (see below), the optimal tax structure would not be uniform but would vary with the ease with which emissions intensity can be reduced. Once major foreign competitors pay the same rate as Danish producers, there will be no need for reduced rates for certain sectors. Under the Kyoto Protocol, such "leakage" is unlikely to be a major problem, since most of Denmark's trading partners will also be trying to reduce their emissions, and such leakage as occurs will be to developing countries.

Denmark is committed to the targets of the Kyoto Protocol, but is also maintaining its commitment to the target of the 1988 Toronto Agreement, to limit CO_2 emissions in 2005 to 80 per cent of their 1988 level. Although the wording of the Toronto Agreement does not specify that such reductions should be achieved without leakage, the Danish authorities interpret the target in this way, because they hope that the resulting demonstration effect will increase the chances of a global agreement being reached.[48] Once it is clear that the Kyoto Protocol will come into force, the authorities have stated that they will remove the differential provisions of the energy and CO_2 taxes designed to protect vulnerable industries and minimise leakage.

To counter the fear that setting low tax rates for heavy energy users gives no incentive to reduce emissions, the lowest rates are conditional on an enterprise reaching a voluntary agreement[49] (VA) on action to reduce energy consumption in the enterprise. This is an interesting experiment, which can be thought of as a way of implementing some properties of the tax-credit approach mentioned in Box 8, but which prevents firms closing down and keeping the tax credit. To be taxed at DKr 3 per tonne of CO_2 a firm has to undergo an energy audit, and undertake any energy efficiency improvements which would be profitable (defined as having

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Box 8. Competitiveness

Other recent Economic Surveys have discussed the influence of worries about competitiveness (in particular, those of Finland and Germany). In Denmark, too, systems of environmental taxation or regulation are distorted by a desire to reduce the extra tax burden facing certain industries compared with their competitors abroad, presumed to have less onerous regulatory of tax burdens. They are subject to less stringent regulation, exempted from taxation or allowed reduced tax rates, compared with other activities.

It is of course true that some industries will be particularly hard hit, forced to contract and perhaps to close down altogether, if they face the full burden of taxation or regulation while facing competition from overseas such that demand for their output is particularly price-elastic. But these will necessarily be the most polluting industries. From an environmental point of view they *should* contract if competitors in other locations can produce with less pollution, or if a lower price is attached to the relevant environmental damage in those locations.* The loss of employment has to be dealt with in the same manner as with any structural change in the economy. Similarly with regional problems – many of the *realpolitik* issues are actually to do with regional concentrations of particular activities.

Exemptions or reduced rates for heavy polluters are costly:

- Other domestic industries or activities are disadvantaged with respect to the protected industry;
- Overall costs of dealing with the environmental problem are increased.

To reach a given level of pollution abatement from Danish territory, exempting an activity means that the tax rate, or degree of regulation, on others has to be higher. Those activities then have to contract more than they otherwise would. Furthermore, since those activities will generally be less pollution-intensive than the exempted industries, the overall loss of output and possibly employment is likely to be higher than otherwise.

Exemptions are not the only way to soften the impact: compensation schemes (perhaps in the form of tax credits) or tradable permit schemes can reduce the competitive disadvantage without diminishing the environmental incentives.

A tax has an income effect and a substitution effect. In the long run it is the substitution effect that will change behaviour, but in the short run the income effect may threaten the survival of a firm. Compensation schemes using tax credits (as in Sweden for NO_x) or "grandfathered" tradable permits (as in the United States for SO_2 and NO_x) mean that the full tax rate can apply to all activities. The substitution effect is the same for all, allowing – in principle – cost-minimising adjustments overall. But some activities are sheltered from the income effects. Of course, since sheltered industries may delay their adjustment, income and substitution effects are not so easily separable in practice.

* The main text notes that Danish policy attaches importance to pollution in other countries as well as that in Denmark. Such an altruistic approach leads to a different interpretation of the competitiveness issue; but it can also be used by sectoral interests to argue for special treatment.

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Box 8. **Competitiveness** (*cont.*)

In Denmark, a particularly important example arises in the CO_2/energy tax. There is no "correct" level of tax on different forms of energy – a number of different environmental problems are linked to them and the resulting costs are not easily valued. But the inefficiencies of using such a wide range of tax rates for CO_2 are clear; even if the cost (and therefore the "correct" rate) is not known, there is a definite overall quantitative target, and efficiency requires that all emitters face the same marginal cost.

Heavy energy-using industries, such as steel and cement, successfully argued that they should not be subject to the full tax, since they compete with companies overseas not facing the same constraints. Regional concerns probably bolstered their case (they tend to be dominant employers in the regions where they are located). In order to be entitled to have lower rates of tax applied, firms must reach "voluntary" agreements (VAs), which amount to an obligation to undertake energy-saving investment (identified as profitable in an energy audit). In its effect on abatement incentives, the VA system has some similarities to the tax credit approach: trying to give firms an incentive to reduce emissions without burdening them with heavy tax payments on the bulk of their emissions. But the marginal costs of abatement will differ across firms, and a new set of incentives will need to be introduced in the future, since emissions will certainly grow too fast, or diminish too slowly, in the absence of the continuing incentive that a higher marginal tax rate would provide.

Thus, although it is true that international co-ordination of tax rates, regulations or permit schemes would allow more efficient solutions to emerge, reducing the marginal abatement incentives for the major offenders on competitiveness grounds is not efficient and is rarely the only means available to reduce adjustment costs.

a pay-back period of four years or less) if the DKr 25 tax rate were applied (to be fully equivalent to the tax credit system would require carrying out all investments profitable at the full DKr 100 per tonne rate). If the energy audits were reliable and continuously updated, and in the absence of transaction costs, this would be equivalent to applying the tax at DKr 25 with a tax credit roughly equal to DKr 22 per tonne of baseline emissions. Krarup (1999) casts some doubt on the efficacy of entering into making voluntary agreements, concluding that the undoubted energy efficiency improvements that have been seen would in many cases have occurred even without the voluntary agreements. However, the authorities have extended the scheme (albeit with some amendments) based on the results of an independent evaluation carried out for the government, which indicated that the scheme is effective. Although the scheme is ingenious, it probably entails addi-

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tional administrative costs, as well as, at best, preserving the distortion inherent in the difference between the DKr 100 tax rate applied to households and the DKr 25 rate for industry.

Taxing carbon emissions will eventually lead to a switch towards less carbon-intensive fuels. The Danish authorities have accelerated this trend over the last decade by strongly supporting renewable energy in general and wind turbines in particular.

Wind turbines

There has been a rapid expansion of wind-generated electricity in Denmark, now providing about 9 per cent of total supply, with some 5 300 wind turbines. Technology has made rapid progress over the past decade, and, as the rated capacity of new wind turbines has risen from 55 kW in the mid-1980s to the present 750 kW, average production costs for newly installed wind turbines (in a prime location) have fallen by around half, to around DKr 0.30 per kWh.[50] The move towards larger wind turbines is expected to continue with the projected sea-based wind turbines of 2 000 kW or more,[51] development of which is intended to provide the bulk of the planned doubling of renewable energy production. Danish wind turbine producers have profited from the rapid growth of their domestic market, and are thought to supply around 60 per cent of the world export market.[52]

The high cost of producing wind turbine electricity has required substantial public subsidies to enable the sector to expand, totalling almost DKr 4 billion over the past six years and reaching almost 0.1 per cent of GDP in 1998 (Table 19). These subsidies come in three main categories: production subsidies, tax subsidies for co-operatively-owned wind turbines, and guaranteed prices for wind-generated electricity.[53]

- The current production subsidy was introduced in May 1992 (replacing a tax-exemption introduced in 1980). The subsidy has two components: 0.10 DKr per kWh for both independent wind turbines and those owned by the main generators; 0.17 DKr per kWh paid only to independent wind turbines;
- Co-operatively-owned wind turbines are exempted from tax, leading to tax expenditures in the order of DKr 70 to 90 million per year;
- The guaranteed price was determined as 85 per cent of the consumer price for electricity excluding taxes and subsidies. The effect is that electricity distributors (the purchasers of wind-generated electricity) pay 0.12 to 0.13 DKr per kWh more for wind turbine electricity than the estimated marginal feed-cost to the grid of 0.20 DKr per kWh.

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Table 19. **Wind turbine subsidies, 1992-1998**
DKr million

	1993	1994	1995	1996	1997	1998
Production subsidy						
Wind 0.10 DKr	100	118	136	122	193	216
Wind 0.17 DKr	134	153	165	155	262	390
Tax expenditures	67	67	69	76	83	88
Price subsidy	143	158	165	157	256	359
Total	444	496	535	510	794	1 053
Unit subsidy, DKr per KWh	0.43	0.44	0.46	0.42	0.41	0.38

Source: Ministry of Finance; Statistics Denmark; Ministry of Environment and Energy; Danish Wind Turbine Association; OECD calculations.

The size of the total subsidy relative to production costs has increased over time. Although the amount per kWh has declined by 11.5 per cent in real terms since 1990, wind turbine efficiency has increased four times faster. An average wind turbine installed in 1998 has production costs of around 0.32 DKr/kWh and sells at 0.60 DKr/kWh, a margin of nearly 100 per cent compared with a 5 per cent margin ten years earlier. As a result, the post-tax rate of return on investment in wind turbines (with low risk since output prices are guaranteed by the government) had risen to between 10 and 17 per cent[54] by 1999, compared with post-tax rates of return of around 4 per cent on government bonds.

In the future, the effective rate of subsidy is likely to decline as the system of "green certificates" is now under implementation and the subsidies described above will be phased out; it should provide a more cost-effective way of meeting what is itself a rather arbitrary target – satisfying 20 per cent of electricity demand from renewable sources by 2003. Green certificates will represent units of electricity generated from renewable sources; electricity consumers will be obliged to acquire certificates corresponding to a yearly quota of their total electricity consumption (20 per cent in 2003); renewable electricity generators will supply certificates according to how much they generate and supply. Hence the payment for wind electricity will consist of the market price plus the price of green certificates. Using certificates, rather than insisting that each distributor actually purchases 20 per cent of its electricity from renewable resources, provides useful flexibility, for example in avoiding unnecessary transmission costs.

The level of subsidy is uncertain in such a system. The government has set upper and lower limits on the price, of 0.27 DKr and 0.10 DKr per kWh respectively. The upper limit caps the subsidy (at the current level) even if not enough "green" electricity is generated – distributors will pay a "fine" of 0.27 DKr per kWh of the difference between the quota and the number of green-certificates they have purchased on the market; the lower limit guarantees a subsidy to wind turbine operators even if capacity installed exceeds the quota.

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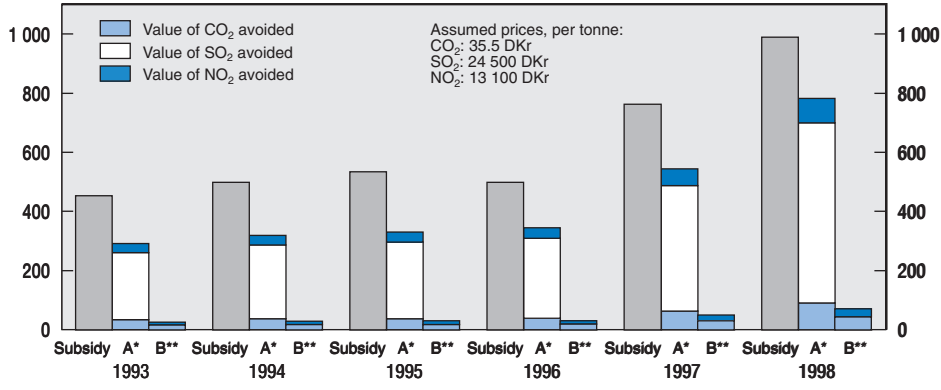
The aims of the wind turbine programme appear to be threefold: reducing greenhouse gas and other emissions, improving security of energy supplies and supporting technological development. The government has not published any kind of cost-benefit analysis of the programme, perhaps because valuation of some of the benefits is difficult. It is possible, however, to put approximate valuations on the emission reductions and compare the resulting figures for benefits with the expenditure on subsidies. The quantity of emissions saved depends on how the electricity would have been generated in the absence of wind turbines. In Denmark's case it is coal generating plants that have been run down, so coal emissions might be the obvious comparator, but they could have been replaced by gas generation rather than wind turbines. Figure 26 illustrates an approximate comparison, though the figures are based on a number of assumptions about technology and valuation (see Box 9) and does not consider, for example, the subsidies to the gas distribution network.

This is far from a full cost-benefit analysis, but some implications are clear, particularly since it makes more sense to assume that wind turbines displace gas-generated electricity than that they replace the "dirtier" coal. This is because

Figure 26. Costs and benefits of the wind turbines programme

DKr million



* A: value of emissions avoided if wind turbines replace coal.
** B: value of emissions avoided if wind turbines replace gas.

Note: The prices used here are illustrative, not intended as an explicit estimate of the environmental costs of emissions. They correspond to those used for the "green adjustment" shown in Figure 24.
For each year the second and third columns represent the value (see footnote 32) of the emissions that would have occured if, respectively, coal or gas fired power stations had been used to generate the electricity that was produced in that year by wind turbines.

Source: OECD.

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Box 9. Valuing the benefits of wind turbines

Figure 26 – which is for illustrative purposes and is not intended as a definitive statement on the net benefits of the wind turbine programme – compares estimates of the costs of the wind turbines programme with estimates of some of the benefits, specifically the reduction in emissions of carbon dioxide and oxides of sulphur and nitrogen. To do this, it is necessary to estimate both the quantity of emissions avoided and the environmental cost per unit of emission. The quantity avoided depends mainly on the kind of fuel assumed to be supplanted by wind, but also on some technology assumptions – combustion techniques, use of "scrubbers" for SO_x and NO_x, for example. Valuation depends on assumptions about environmental damage avoided and its social worth. In the present exercise both sets of assumptions are relatively crude but make use of easily available information.

Emissions avoided[1]

It is assumed that the coal (gas) stations replaced by wind turbines would have emitted CO_2 at the average rate, per kWh of electricity generated, of all Danish coal- (gas-) fired stations in 1997. In the absence of data on SO_x and NO_x emissions from Danish stations, it is further assumed that tonnes of SO_2 and NO_2 emissions in those stations would have borne the same proportion to CO_2 emissions as was the case in typical US stations in 1998. The CO_2 assumption is likely to be accurate; those for SO_2 and NO_2 could be out by a factor of 2, especially for coal stations, depending on whether emission control technology would have been in place or not, for example.

Valuation

Emissions were valued as in the "green adjustment" to national saving referred to earlier: CO_2, SO_2, and NO_2 valued at DKr 35.5, 24 500 and 13 100 per tonne, respectively. The valuation attached to SO_2 may be rather high (alternatively, the current tax on SO_2 emissions, which is only 40 per cent of the value used here, is rather low). The prices used in this analysis were used by the Danish Economic Council in 1997. Only one year earlier, in 1996, the Economic Council used much higher prices. Using these higher prices, the wind turbine subsidy per kWh would substantially exceed the environmental costs of coal-generated electricity but not that from gas. This indicates that the uncertainty in the valuation of environmental benefits is quite high, reinforcing the need both for caution in the use of cost-benefit analysis where valuation is uncertain and for consistency in the use of whatever information is available. The Ministry of Finance and the Ministry of Environment and Energy are at present working on a more thorough investigation of existing studies on the valuation of benefits in the environmental field.

The assumed valuation of CO_2 is at the low end of the likely range for trading under the Kyoto Protocol; unless trading under the Protocol involves a number of significant developing countries, not currently signatories, the CO_2 price in such trading is likely to be two or three times higher than that used here. See OECD (1999*g*).

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Box 9. **Valuing the benefits of wind turbines** (*cont.*)

With these assumptions, reduced CO_2 emissions account for about 12 per cent of the benefits in replacing coal-generated electricity, SO_2 and NO_2 emissions accounting for 78 per cent and 10 per cent respectively. For gas stations, the corresponding percentages are 62 per cent, 0 and 38 per cent.

1. CO_2 emissions data for Denmark are from the International Energy Agency. Data on emissions from US stations is taken from the "Emissions Scorecard" of the Acid Rain website of the U.S. Environmental Protection Agency, *http://www.epa.gov/acidrain/score98/es1998.htm*. Data used are from the Emissions Scorecard, "Appendix B: 1998 Data for SO_2, NO_x, CO_2, Heat Input, and Other Parameters" – *http://www.epa.gov/acidrain/score98/table_b2.zip*, as on 25 January 2000. Thus, per GWh of electricity generated, it is assumed that coal-fired stations would have emitted 920 tonnes of CO_2, 8.96 tonnes of SO_2 and 2.24 tonnes of NO_2; for gas, the corresponding assumptions are 452, 0 and 0.76 tonnes, respectively.

although the output of coal-fired stations has been reduced, they are generally far from the end of their economic life. Only the variable costs of running them would enter cost calculations, whereas the full capital cost of wind turbines and gas stations (gas's contribution to electricity generation is also expanding) is relevant. At the illustrative prices used, and if capital costs are not too dissimilar, the environmental benefits of using wind turbines instead of gas are far less than the subsidy to wind turbines.[55] If the choice were only between coal and wind turbines, the gap is narrower, but it would be better to replace coal with gas rather than wind unless either the omitted net benefits were substantial, or CO_2 emissions were valued at over DKr 250 per tonne; such a valuation is likely in the near term only if the Kyoto Protocol is implemented and enforced with quite severe restrictions on international trading of emissions permits, and if the ability of Denmark to cut back on non-CO_2 GHGs such as methane and NO_2 were small (whereas in agriculture there are probably significant low cost abatement options).

However, environmental objectives are not the only goals of the wind turbine programme. One cited by the authorities is technology stimulation.[56] The technology argument is bolstered by the fact that Danish wind turbine producers are among the world's most important. Valuing this is difficult, however. Although exports of wind turbines are substantial (reaching about DKr 11 billion in 1999), this means little unless the value added in their production exceeds the value that would have been produced in alternative uses of the resources. This means that Danish wind turbine producers have to be earning a rent, due either to their technology or to their market power. This may be true, but the market for wind turbine supply seems to be fairly competitive, as well as being quite heavily depen-

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dent on subsidies in other countries. For example, California was an important market until the degree of subsidisation there was cut substantially.[57]

It would be interesting to see the partial analysis used here more fully developed with better information from the authorities. A forward-looking analysis would also be appropriate: since the costs of the wind turbine subsidy are likely to fall if the green certificate system is successfully implemented, and the valuation of CO_2 is likely to rise, the results would almost certainly be more favourable to wind turbines.

Carbon trading in electricity generation

As noted earlier, the position of the electricity-generating sector is rather anomalous. A major emitter of NO_x and CO_2, it has been exempt from taxation on these (though a sulphur tax has been applied, as described earlier), and consumers pay an electricity tax unrelated to the fuel input (except that it is refunded to renewable sources). This may be explicable by the fact that electricity supply has been highly regulated and largely publicly-owned, so that it may have been felt that market signals would have little effect anyway. With the ongoing deregulation of this sector, economic instruments are being introduced. As from 2000 electricity generators will pay a CO_2 tax of DKr 40 per tonne (*i.e.* less than one-quarter of the value attributed to CO_2 abatement in the cost-benefit exercise summarised in Figure 26), but only if their emissions exceed a certain threshold (which is being reduced each year). (Table 19 takes no account of this tax.) The intended move to a tradable permit system – trading is planned to begin in January 2001 – will provide more effective incentives than the tax-threshold system because it provides a reward for reducing emissions below the threshold rather than only penalising the excess. The possibility that EU state aid rules might prevent the implementation of the scheme was removed when the European Commission approved it in April 2000. Having such a system in place will leave Denmark well prepared for the introduction of international CO_2 trading under the Kyoto protocol.

How efficiently all these taxes, subsidies and permits will achieve their environmental and other goals remains to be seen. The system is complicated and likely involve distortions that will, *a priori*, mean excessive abatement costs in some industries and unjustifiably low abatement in others. Most of the CO_2 tax could be replaced by merging with the cap-and-trade system currently planned only for electricity. Given the quantitative nature of the targets for SO_2 emissions, the SO_2 tax could also be replaced by a cap-and-trade system, though the authorities expect the SO_2 emissions to fall well below the target, on current policies. The major obstacles to such integration would be the carbon leakage or competitiveness problems discussed earlier, and also the possible loss of significant tax revenues. With SO_2 and CO_2 trading schemes in place, the green certificates would be unnecessary for wind turbines, provided either the permits were sold or auc-

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tioned by the government, or that an initial allocation were given to renewables producers.

One problem with both the green certificates and the CO_2 trading schemes may be how to deal with imported hydro-electricity from Sweden and Norway; the three countries already operate a common market in electricity, with Denmark typically importing hydro-electricity but exporting fossil-fuel-generated electricity in years when precipitation has been too low to leave a surplus for export from Sweden or Norway.[58] The current intention is to allow only Danish "green" electricity generators to participate in these markets. Such discrimination may be difficult under EU competition rules; the Danish authorities hope that these will be reconsidered in the near future with a view to allowing such restrictions.

Gas

The market for natural gas, in which Denmark is currently self-sufficient from North Sea supplies, is characterised by a very high degree of vertical integration[59] and an interesting approach to taxation. Up to 1996 taxation was in fact implicit rather than explicit, with prices being set (by regulation) as a function of the tax-inclusive prices of alternative fuel in any given market; this means that gas companies naturally charged the highest possible prices as opposed to the lowest possible prices to consumers (Det Økonomiske Råd, 1999). The resulting "shadow charge" generated implicit revenue of DKr 2¾ billion in 1996, equal to 11 per cent of all environmental and energy related taxes. If the alternative fuels were competing in a competitive market, with taxes set as a function of the environmental costs of those fuels, this might be quite a sensible way of extracting the resource rent associated with the gas, whose actual production cost is quite low and whose environmental externalities are below those of other non-renewable fuels. The implicit subsidy in the form of the special tax treatment of natural gas has made it possible for the distribution companies to service their debt obligations as a result of the rapid development of the natural gas supply network.

Since 1996 explicit tax charges on gas have been introduced, partly because the existing arrangements were incompatible with EU liberalisation, and the "shadow" tax revenue has more than halved. A more competitive natural gas market is planned as part of EU-wide liberalisation; as part of this, a move away from the policy of charging maximum possible prices could speed the substitution of gas for coal and oil, helping Denmark to reach its ambitious CO_2 targets.

Transport

A number of interesting questions arise in transport policy, both in the context of mass transit and automobiles. Public transport receives quite large direct subsidies and is not subject to certain environmental taxes, while private automobiles operate under a tax system which imposes very high fixed

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costs for car ownership, while the costs of car use are very similar to other western European countries.

Public transport

As in most European countries, train, inter-city and local bus services are subsidised, with subsidies generally covering a large part of total costs (Table 20). The aims of subsidisation are to keep down urban congestion, to ensure that a maximum number of people have access to public transport, and to help to reduce automobile use. In recent years there have been improvements in the efficiency with which public transport services are provided, with most services being provided by private contractors chosen by competitive tender and the level of services specified by the relevant county administration; as Table 20 shows, costs per passenger-kilometre travelled in deregulated buses are much lower than in trains, the latter mostly both state owned and operated. But full advantage has not been taken of these changes so as to allow market forces to increase economic efficiency and thereby improve environmental efficiency by internalising some of the environmental costs and allowing comparisons between costs and benefits

For example, the public transport sector in general is exempted from fuel taxes.[60] This means that efforts to improve fuel efficiency and thus reduce pollution depend on particular programmes to develop low-emission vehicles without giving operators any general incentive either to use such vehicles, or to reduce emissions by changing operating methods or the mix of vehicles used. Although a full bus produces lower emissions per passenger-kilometre than a typical car, when buses are far from fully loaded cars may have lower environmental costs – under some circumstances pollution could be reduced by switching from bus or rail to private cars, but there are no economic mechanisms to test for the advisability of such a switch.

It might be worthwhile accepting higher pollution levels from running partially empty buses if these costs were offset by other benefits. For example, the social benefits of late-night rural bus services are an important factor used to

Table 20. **Costs and subsidies in public transport, 1997**

Kroner

	Buses	Railways	
		State	Private[1]
Cost per passenger kilometre	1.61	2.03	0.39
of which financed by subsidy	0.83	1.06	0.16

1. So-called private rail lines are in fact jointly owned by the state (70 per cent) and municipalities (30 per cent). They tend to be small lines in rural areas.

Source: Konkurrenceredegørelse (Competition Authority Report) 1999.

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justify subsidies. But little information is available on how valuable these social benefits are. The counties typically have no information on passenger numbers on any particular local bus route and apparently do not take such considerations into account when deciding whether to provide services or not. They are more likely to measure the value of a service according to how much protest arises when it is removed. Such a measure is certainly likely to be related to the social benefit – which may be of the nature of an option value: even though they may rarely use such bus services, people benefit from the knowledge that they can do so. If these arguments are thought to be important, surveys of passenger-use, and of potential use, would provide additional objective information.

Private cars

Denmark has very high taxes on private car ownership. The vehicle registration tax is levied at 180 per cent of the VAT-inclusive price over DKr 53 000 (at 105 per cent up to that price);[61] VAT is payable at the standard rate applied to the price exclusive of vehicle registration tax. In addition, an annual "pollution tax" is payable, at rates according to the tested fuel consumption of the model.[62] The pollution tax rises linearly (at DKr 460 per 0.5 litre per 100 km) for consumption above 5 litres per 100 km. The minimum rate is DKr 460 with fuel consumption at 5 litres per 100 km, but the tax rises more than proportionately with fuel consumption, increasing by a factor of eleven as fuel consumption doubles from the 5 litres per 100 km threshold. The taxes on motor fuels, however, are similar to those in neighbouring Germany. This is a deliberate policy, lest motorists avoid them to buying their fuel in Germany. According to Danish government estimates, switching of fuel purchases just by normal cross-border travellers (*i.e.* not taking account of any additional cross-border traffic that might occur) as a result of a significant fuel price differential between Denmark and Germany, would make such action prohibitively expensive for the budget (though good for the German budget). The national cost of increasing the petrol tax above the German level would thus be higher than the environmental benefit. The petrol tax and the annual pollution tax are slightly regressive, while the registration tax is progressive, as a higher proportion of the income of well-off households is used to pay this tax than for low-income households.

With such high taxes on ownership of private cars, it is not surprising that car ownership is relatively low, particularly when account is taken of the fact that incomes are relatively high – car ownership generally increases with per capita income (Figure 27). Figure 27 also illustrates, however, that total distances travelled by car in Denmark[63] do not appear particularly low, given average incomes. The implication is that cars are on average used more intensively in Denmark than in other countries – since user costs are no higher than typical European levels,[64] distance travelled is less out of line than the stock of cars. It

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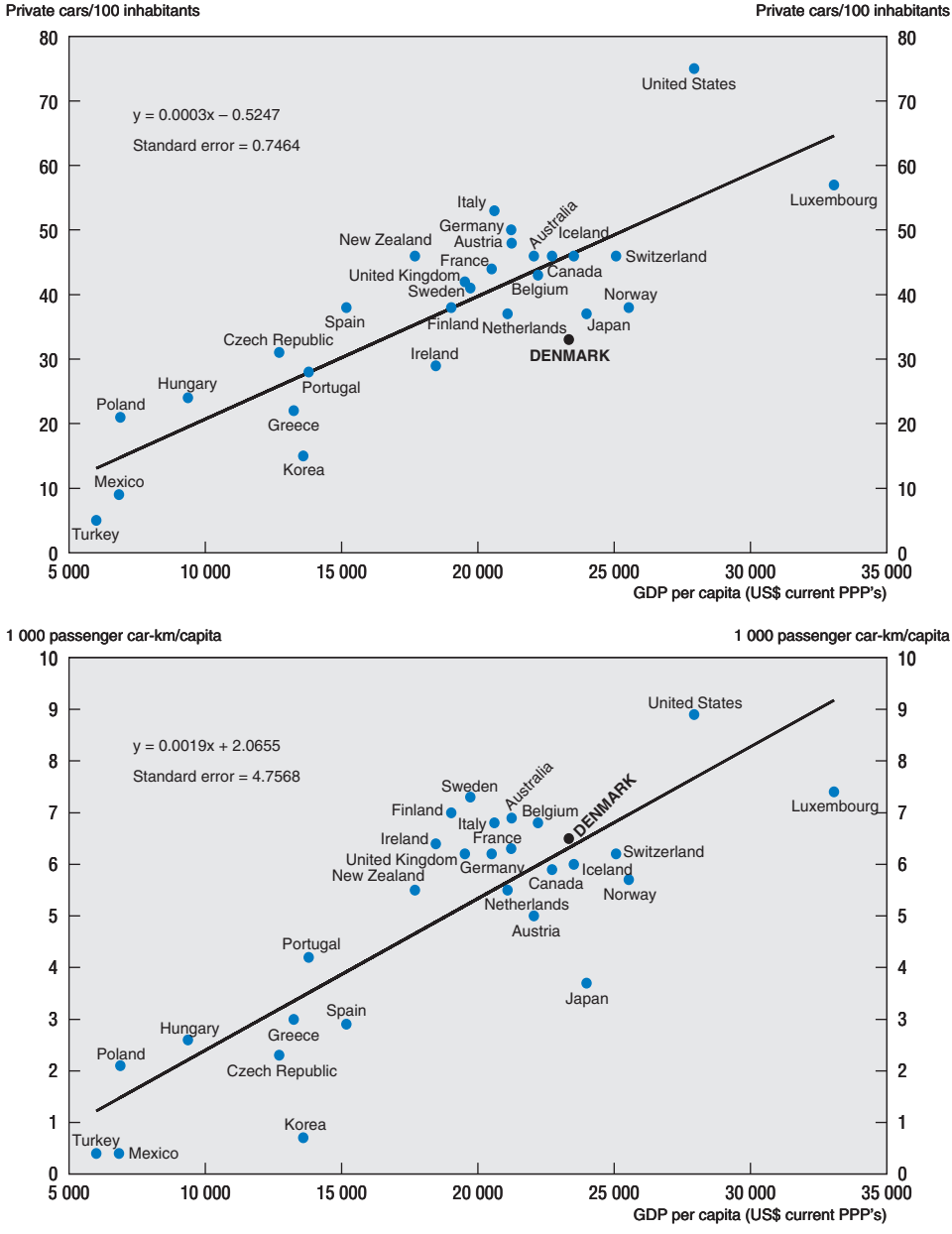
Figure 27. **Private car ownership and usage: an international comparison, 1996**



Source: OECD, *National Accounts* and *Environmental Data Compendium*.

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is also likely – although firm data are not available to support this hypothesis – that the high fixed costs of ownership mean that cars are kept longer before being scrapped. If true, this would slow the rate at which innovations, including those improving the environmental performance of cars, are diffused.

The registration tax serves both environmental and more traditional revenue-raising purposes – its proceeds represented about 1½ per cent of GDP in 1998. The resulting high fixed cost of car ownership does have economic and social ramifications costs, however, since Danish households benefit less from the flexibility provided by car ownership than they would otherwise, with the effect likely to be concentrated in poorer sections of society.[65] Furthermore, the registration tax is not strongly related to most of the externalities from car use (most are proportional to fuel consumption or distance travelled), even if it does provide a strong incentive to buy cheaper cars, which tend to be lighter and have smaller engines, thereby reducing fuel consumption. However, removing the tax entirely might leave private cars facing a lower tax burden than the value of total negative externalities. Thus, the Danish authorities are considering the possibility of changing the base of the tax so that it would be more closely aligned with the many different source of externalities and not just incentives to buy cars with better fuel economy. Nevertheless, increasing the annual "pollution" tax would serve both to make up for lost revenue and to align car taxation better with environmental externalities. Gradually removing the exemptions for vehicles with only two seats (mainly for business use), which have no environmental justification, could also make up for lost revenue.[66]

An integrated approach to transport and the environment

This section has only touched on two of the many issues involved in transport policy. But in both cases considerable costs are imposed, in one case on public finances and in the other on the household sector, by policies seeking to direct behaviour in certain directions because of social and environmental externalities. The two approaches to a large extent complement each other – making it costly to use private cars, and cheap and easy to use public transport – and thus may be thought of as part of an integrated transport policy. There is no real sign, however, of consistent integration of environmental externalities in the two sectors. Crudely, it seems as though the same quantity of pollution is treated as more costly when it is emitted by private cars than when emitted by public transport.

A better integrated approach need not imply, for example, eliminating "uneconomic" rural bus services. It may be perfectly rational to subsidise them, while taxing their use of fuel; the competitive tendering process would then have incentives to which to respond. (Indeed, if it is rational to impose a fuel-economy-related registration tax on private cars, it may well be rational to do the same for buses; as argued above, however, the rationale for this in the case of cars is not

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clear.) To achieve such integration effectively first requires identifying more explicitly both the positive and the negative externalities generated by different modes of private and public transport, and then quantifying them. Secondly, the structure of tax and subsidy needs to be aligned as closely as possible with these externalities. In the case of externalities generated by congestion, for example, neither a high registration tax nor a high fuel tax are necessarily the most efficient response; as technology permits it, road pricing may become a more important instrument.

Policy overview

It is often argued that as countries become richer, they tend to have cleaner environments, either because economic growth implies the use of more advanced, and therefore (so the argument goes) cleaner, technologies or because richer consumers demand a cleaner environment. It is certainly the case that Denmark, with one of the highest per capita incomes in the OECD, appears to have few acute environmental problems, though against a background of quite difficult-to-resolve air and especially water pollution problems. This relative absence of severe problems may be partly geographical and historical accident, with little heavily polluting industry – partly as a result of environmental policy – and few concentrations of population.

It is also true that Danish policy has recently adopted ambitious environmental goals, in some cases even when the environmental benefits will not directly accrue in Denmark. The most striking example is in the case of greenhouse gases, where the both the unilateral target for 2005 and the target adopted in the EU burden-sharing agreement under the Kyoto Protocol for 2008-12 require an adjustment effort significantly greater than in most countries. It is true that there are indirect benefits from having a "green" image in the world community. In addition, even a self-interested policy could involve apparently altruistic behaviour, if it is part of a process of building and maintaining confidence in multilateral agreements with other countries.

Denmark has made increasing use of economic instruments and has gone further than most OECD Member countries, as reflected, for example, in the green tax reform, which should reduce the economic costs of environmental policy. These instruments have so far been frequently applied as supplements to, rather than replacements for, command and control measures. One problem arises from the high degree of regulation and/or public ownership that remains in some key areas of the economy – notably in energy and transport. This means that while the economic instruments provide market signals, many of the agents in the market (public power and bus companies, for example, in the issues that have been discussed in this chapter) are not always particularly constrained to

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follow them. To achieve environmental goals it is thus often thought necessary to supplement the economic instruments with policy directives; hence, for example, in the case of waste, a system of taxes to encourage recycling has been introduced, but it is quantitative directives that seem to have the major impact on local waste management. Of course, even where this is the case, "green" taxes are useful sources of revenue, causing fewer unwanted distortions than the taxes they have replaced.

Furthermore, where the economic instruments seem likely to have a strong effect – that is where they may actually be costly to ignore – they are often overridden by special arrangements exempting particular industries or activities, on the grounds that they will impose heavy costs on them (especially those subject to international competition, as discussed earlier). Examples are the exemption of heavy energy users from taxes that should reduce carbon emissions, and of public transport from fuel taxes, as well as the very mild restrictions on nitrate emissions from agriculture. These policies result in highly unequal marginal incentives to abate pollution across different parts of the economy. Where the policy targets are relatively easily attainable, the costs of this inefficiency may be fairly low; this is less and less the case, however, as targets become more ambitious, and efficiency considerations need to be given greater weight. In general, worries over international competitiveness can be dealt with without distorting marginal incentives, although some of the solutions, such as "grandfathered" tax credits or emissions permits, may mean sacrifices in terms of cost effectiveness or give rise to domestic competition worries if they act as entry barriers. Concerns over "leakage" and competitiveness may be used by industry pressure groups to argue for exemptions that are not justified on environmental grounds. Nevertheless, there may be occasions where a differentiated approach is justified, especially where action is taken unilaterally in anticipation of concerted international action.

One result of the willingness to deal with worries about competitiveness by practically exempting some of the most polluting industries is that final consumers end up having to bear a larger burden of any abatement than they otherwise would. Nitrate and carbon abatement policies are two examples, the implicit tax rates faced by households are many times higher than those faced by agriculture and industry in the respective cases. It might be argued that it is anyway the consumer who pays the final costs, whatever policy is adopted. However, loading many of the economic incentives onto the final consumption stage means that opportunities for input substitution or technical changes in the production process may be ignored. Furthermore, if economic instruments can be used to establish uniform prices and incentives for innovation, the apparent need to recycle tax revenues in the form of subsidies for technologies that may in truth be sub-optimal would be diminished.

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Conclusions

Given the selective nature of this chapter, the following cannot be a balanced assessment over the whole range of Danish environmental and natural resource policy, but draws conclusions from the previous discussion of the issues of water, waste, energy and transport. It also discusses the potential role of a more extensive use of explicit analysis of costs and benefits, where the points made certainly have an application wider than the particular issues or sectors included here.

Water

In dealing with water quality, as with all externalities, all sectors should ideally face the same incentives to reduce polluting discharges; the government has appeared reluctant, or unable, to impose this. Given the diversity of sources of nitrate and phosphate discharges, the main source both of eutrophication problems and, it seems, of deteriorating groundwater quality, complete uniformity is not achievable – measuring household discharges, let alone taxing them, is infeasible, to take one example. However, the steep increase in water charges and taxes over the last decade is indicative of the considerable costs that cleaning water is imposing on the economy and therefore of the importance of dealing with it in a cost-effective way.

To this end, to the extent that water charges and taxes are imposed for environmental reasons, they should be set at similar rates on all users. Current rebates to industrial users or high rates for households do not seem to make sense, unless they are conditional on already removing pollutants that the taxes or charges are intended to deal with. It is better to correct the pricing structure directly rather than dealing imperfectly with it through differential taxes. Agriculture almost certainly has not been faced with appropriate incentives to reduce its nutrient discharges. The complicated biological and physical mechanisms at work make it difficult to assess what the marginal cost of reducing discharges would be, but the amount of information already available on farm-by-farm nitrogen balances, including inputs of both fertiliser and animal feed, would almost certainly support a scheme taxing estimated net discharges, or submitting them to a tradable quota scheme, with prices or quantities linked to the severity of the local water quality problem. If feasible, such a scheme should be less costly to farm output than the current plans simply to restrict fertiliser use to a certain proportion of the "optimum". A simpler though less well targeted approach would be to tax fertiliser (or, again, use tradable quotas). Ideally, similar conditions to those set for agriculture should apply to freshwater fish farming, whose exemption even from current restrictions is surprising.

Under a taxation scheme, the problem may arise that the costs to farmers (ultimately to landowners) may be very high. As in other cases, the appropriate response is not to start with low tax rates, but to offset some or all of the burden

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through tax credits, or through issuing "grandfathered" quotas without charge. Any such arrangements could reasonably be transitional, implementing the polluter-pays-principle by phasing out tax credits, or moving to auctioned quotas.

Waste

In the area of non-toxic waste disposal, Danish policy has developed quite a strong system of economic instruments, with differential tax incentives for recycling, incineration with or without co-generation of electricity and heat, or landfill, along with a cost recovery principle covering operational costs. If these taxes are thought to represent the social and environmental costs of the different methods of disposal, it should be possible to phase out the constraints of national targets for the proportion of waste recycled, which do not make sense if environmental externalities have been suitably internalised. This is especially so when set at a national level because such targets are unable to take into account collection, transportation and processing costs that may vary both by location and through time. Economic instruments allow such costs to be taken into account, and thus might result in levels of recycling higher in some areas, lower in others. But allowing waste taxes alone to regulate the treatment of (non-hazardous) waste requires an explicit assessment of the cost of associated externalities. Agreement on these may be more difficult to achieve in practice than agreement on quantitative targets. It is also the case (though this is not in itself an argument against recycling incentives) that regulations requiring recycling of glass bottles, with specific restrictions on the kind of bottles allowed, and banning metal cans as drinks containers, by some are claimed to be protectionist measures disguised as environmental ones.

Denmark also has a range of taxes on individual containers and packaging products, as a function of the product and material used. When set in line with environmental costs, such taxes are highly appropriate. Some proposed aspects of this system of individual taxes have been modified on the grounds of excessive administrative complexity. It is worth monitoring the costs and benefits of the operation of this system, in order to improve its working in Denmark but with the useful externality of providing information for other countries on their advantages and disadvantages.

Energy

Energy policy and taxation have been discussed extensively in this chapter. From the point of view of the targets for D*anish* emissions of greenhouse gases, energy taxation appears distorted, and the renewables programme, now largely based on wind turbines, seems to have incurred costs much higher than any environmental benefits achieved so far (the innovative green certificate system will substantially reduce the cost of the wind turbine subsidy for any given target,

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however). However, the two greenhouse gas targets – that self-imposed for 2005 as well that agreed for 2008-12 in the EU burden-sharing agreement under the Kyoto Protocol – are more ambitious than those for many countries, and were partly designed with the aim of encouraging other countries to make greater efforts.

The result is a set of policies that, while using economic instruments, does not minimise the costs of reaching the targets, except when those are defined to exclude reductions due to activities that may be displaced abroad. Activities where the loss of output in reducing GHG emissions is likely to be the lowest are taxed relatively lightly and receive subsidies to finance profitable investment, while other activities are heavily taxed. It is true that if all sectors paid the same tax rate, some industries might suffer cost increases sufficient to close them down, resulting in distributional outcomes that might be considered to be unfair. This would have three principal consequences: loss of capital, loss of jobs and "leakage" abroad of GHG-intensive activities.

For a given emissions-reduction target, both job destruction and loss of capital (through reduced profits) will be greater if a differentiated tax rate is introduced.[67] Concern over losses due to falls in the value of emission-generating capital is best dealt through a temporary compensatory scheme, which might take the form of tax credits or allocation of emission permits by "grandfathering".[68] Such schemes would probably not eliminate the need for job losses,[69] but by minimising overall output reductions they are likely to minimise any shrinkage in employment as well.

However, it should be acknowledged that "grandfathering" would not deal with concern over the environmental cost of emissions leaking abroad: since marginal incentives remain the same as for a uniform tax, firms' behaviour would be unchanged, at least in the long run. Hence, in the case of leakage lowering the tax rate does have some beneficial effect – emissions of GHG and other pollutants abroad are reduced. Furthermore, where leakage is likely to occur only in the short run, ending when other countries implement their own GHG-abatement programmes, such action may prevent an industry closing down unnecessarily and possibly permanently.

Whatever the conclusion on the latter, within the electricity generating industry itself there is room for more efficient policies than the current or imminent combination of wind turbine subsidies, obligations to purchase renewable energy and CO_2 trading. Since the environmental benefits of wind-generated electricity are largely reduced emissions of SO_x, NO_x and GHGs, consideration should be given to replacing the green certificate programme by integrating tax and trading arrangements for all these emissions across all generators. The resulting output of "renewable" electricity may fall below the 20 per cent target but the only loss in this case would be in security of supply. It would be useful for the govern-

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ment to explicitly identify the risks against which wind turbines provide such security, in order to assess the most cost-effective way of ensuring it.

Transport

This report has not covered the whole range of transport policy but has touched on idiosyncrasies in two areas: subsidies to public bus services and vehicle taxation. Transport is an interesting case because subsidies to public transport are sometimes justified on environmental grounds, yet badly designed subsidies can lead to excessive environmental costs when passenger loadings are too low. In such a situation, cost-benefit studies are necessary, but they are conspicuously lacking, perhaps because basic data are frequently not available. Allocation of these subsidies is the responsibility of county authorities and it would appear they have insufficient incentive, or perhaps insufficient information, to improve or optimise the use of the funds. Since both the costs and benefits are local, it would not be sensible for central government to impose a uniform subsidy policy on the counties, even though most finance for county and municipal policies comes from block grants or earmarked taxation. Regardless of whether the overall benefits of these subsidies exceed costs or not, encouraging, perhaps mandating, systematic cost-benefit analysis of them, and of other subsidies, would be helpful. Such analysis should include considering how best to deliver subsidies, in particular how to internalise environmental costs. Ending the exemption from fuel taxes would be one obvious step.

Just as environmental considerations (among others) may justify subsidies to public transport, so they may justify taxation of private transport. In Denmark, private cars are subject to a very high level of taxation. Since most of the relevant environmental externalities are related to car use rather than to possession, the very high vehicle registration tax is not very effective as an incentive to reduce externalities for those who possess cars, but appears to reduce car ownership significantly. The authorities accept this but argue that the alternative – higher fuel tax – is ruled out by the impossibility of raising fuel taxes in Denmark significantly above those in neighbouring countries.

Again, some cost-benefit analysis is required to clarify matters. The high fixed costs of car ownership produce a burden on those on car owners, who tend to be wealthier. However, for any given type of car, the burden on the poor is greater. This may in turn generate pressure for subsidies to public transport which in principle can be measured. The simple cross-country regression presented in this chapter at least suggests that although the high fixed costs result in relatively low levels of car ownership, distance travelled per person is not obviously lower than in other countries, even when income levels are taken into account. Potential costs from a lower turnover of the stock of cars, and hence a higher average age, which means

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that embodied technological improvements in environmental performance are incorporated later than elsewhere, should also be taken into account.

Assessing costs and benefits

Many aspects of Danish policy towards natural resources and the environment have advanced in the direction of improving cost efficiency during the 1990s. In particular, there has been increasing use of taxes and charges in a number of areas – SO_2 emissions, waste and water pollution, for example. Politically it was convenient that many of these changes began as part of general tax reform efforts to reduce marginal income tax rates. The efficiency improvements in adopting such approaches are largely independent of so-called double-dividend gains, however. Further extensions should be pursued and need not depend on this justification. Revenue-neutral schemes or even those that may reduce overall revenues may be appropriate in some areas: they should not be ruled out *a priori*.

Ideally environmental taxes should be set equal to, or at least as a consistent function of, the value of likely environmental damage arising from the taxed activity. Generally this has not been done. This is partly because it is not easy to get such information. However, the approach adopted has been to set taxes according to some estimate of the impact of taxes on the level of the tax base – targeting a particular reduction in the damaging activity; those targets are often set according to an international agreement or EU directives, with which Denmark has to comply. Measuring success according to whether this target is just met, or over- or under-achieved, then comes naturally but is partly mistaken, depending on whether the target made sense in the first place. It may well be that a targeted decline in a polluting activity is achieved, even if the tax is set beyond its optimal level; conversely, the tax may well have been optimal, even if the reduction in polluting activity had been less than anticipated.

To assess the overall efficiency of these targets, or to assess outcomes, one needs much the same kind of information on the value of damage as is needed to set the "optimal" tax in the first place. The Danish authorities should undertake more systematic research to quantify the costs of environmental damage: to enable taxes to be set more coherently, but, perhaps even more importantly, to set more efficient policies in the areas where a "command and control" approach is retained. This need not always imply doing fully original research each time – shadow prices and costs should not generally change from one context to another, and results from other countries can be useful. It may be difficult to establish definite social valuations for many things – a range of values may be the nearest it is possible to get. Using such information to establish bounds on benefits and costs is still better than having no such information at all.

Some things cannot easily be quantified or valued (for example, where what may be thought of as "rights" may be concerned). Consideration of these may

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sometimes reasonably overrule a quantified cost-benefit analysis: yet even in these cases, such analysis is useful, in particular because it is likely to increase transparency and consistency in decision-making. The Danish authorities, notably in the Finance Ministry and in the Ministry of Environment and Energy, have started programmes of work to assess costs and benefits of policy, notably in the areas of waste, waste water, air pollution and traffic. The results will no doubt help to implement further improvements in the integration of Danish environmental and economic policies.

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Notes

1. The "Whitsun" package made substantial changes in the taxation of both capital and labour income, to be phased in over a four-year period. These were described in detail in the previous S*urvey*.
2. However, even if Denmark were to adopt the euro, the 10-year government bond yield differential would be unlikely to disappear completely, because it is a smaller issuer of bonds than Germany and therefore slightly less attractive (Andersen, 2000). Dutch government bonds, for example, currently have a yield premium of around 15 basis points.
3. Statistics Denmark has revised the method used for calculating the CPI as from 1 January 2000. The new calculation uses geometric instead of arithmetic means, and the index weights have been updated. The main difference between the CPI and HICP since the January 2000 revision is the weight of the housing component, which is much larger in the CPI.
4. This is in line with earlier OECD estimates which indicated that the tax-to-GDP ratio would need to rise less than 4 percentage points by 2030 to keep net debt constant (Roseveare *et al.*, 1996).
5. See Chapter III for further description of this programme.
6. The average annual real growth rate in public consumption from 1993 to 1999 was 2.6 per cent. The – striking – difference in growth in public employment and real public consumption results from a surprisingly low increase in the public consumption deflator in the national accounts.
7. The main responsibilities of the counties are hospitals and most other health-care services, secondary education, road maintenance and social services supplied, for example to handicapped persons, drug abusers, criminal youth, etc. Most public services are provided by the municipalities, in particular primary schools, childcare and elderly care.
8. This overspending was financed through increased municipal income tax rates.
9. The tax auditing of corporate businesses was officially taken over by the central government in 1998, but the job is still partially delegated to municipalities.
10. This enumeration of municipal responsibilities is not complete. The role of local governments in the conduct of environmental policies is described in Chapter IV, and the extensive involvement of municipalities in the utilities industries, particularly the energy sector, is discussed in Chapter III.
11. Introduced in 1998, "Flexjobs" apply to people with less than full working capacity. The central government gives a wage subsidy in the range of one-third to two-thirds of the minimum wage depending on how much the working capacity is reduced. Flexjobs are considered subsidised work in the sense that they do not qualify for unemployment

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benefits. It is the responsibility of the municipalities to assess whether applicants' working capacity is sufficiently reduced to qualify for a Flexjob. Employment in Flexjobs can be in either the public or the private sector. The number of Flexjobs has increased quite rapidly, from 2 500 at the time of their introduction in mid-1998 to almost 7 000 at the beginning of 2000.

12. Primary schooling can be provided by the private sector through the "friskole" concept, according to which private, independent schools automatically receive funding per recruited pupil corresponding to 80 per cent of the average expenditure in public schools. This provides a useful competitive check on public-school performance.

13. The continental European countries have somewhat less generous unemployment benefit schemes than Denmark but stricter EPL. The United States, United Kingdom, Australia and New Zealand are the only countries combining easy EPL with comparatively low replacement rates.

14. Expenditures for daycare are considered, in this study, to be linked to having a job, and are not incurred during unemployment. This assumption may be questioned as the unemployed must, according to the rules, be available to the labour market and ready to start job or activation at short notice. Including expenditures for daycare in both cases lowers the mentioned figures by approximately 4-5 percentage points.

15. Cash benefit recipients are those unemployed not eligible for unemployment benefit. ALMPs for these people are administered by the municipalities instead of the regional labour-market authorities.

16. The youth unemployment rate has declined from 13-14 per cent early in the 1990s to approximately 5 per cent by the end of 1999 – which is a fall from above to below the average rate. This reflects the fact that earlier compulsory activation has been combined with much stronger economic incentives for unskilled youth: they are forced to take up education at a benefit level only half the normal level. The stronger incentives for youth have quickly and easily led to higher employment rates, helped by the healthy macroeconomic situation and by modest employment protection.

17. By virtue of these extended rights the effective unemployment benefit period may be as long as nine years: a person who becomes unemployed just after his/her 51st birthday will be just above 55 by the end of the unemployment period. This is just enough to be entitled to a prolonged benefit period up to 60 years. Up to 1998 the extended benefit rights applied to 50-54 year olds as well.

18. This was announced in August 1999. The original labour-supply objectives were revised downwards, partly because they were unrealistically high, but also because the corresponding policy changes were slow to evolve and because the expected slowdown in the inflow to early retirement benefit never emerged.

19. This assessment was embodied in the explanatory memorandum to the law proposal and was applied in the D*enmark* 2005 programme as well. In a more recent report, the very long-term impact has been estimated at 20-30 000 fewer beneficiaries (Økonomiministeriet, 2000).

20. Early retirement (disability) pension may be very generous, depending on the applicant's civil status and housing situation; altogether, benefits may be as high as 70 per cent of the average production worker's wage level, even at the lowest step. The scheme is administered by the municipalities, and there is some evidence that the system has been used for economic reasons, particularly in marginal areas with limited job opportunities. However, administration has been tightened in recent years. The inflow to the pension scheme has been on a downward trend for about a decade,

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although this is partly the counterpart to increased inflow to the early retirement benefit scheme. In addition, the inflow dropped sharply in 1999 in response to the increase in financial responsibility for the scheme given to municipalities. In addition, preventive policies, such as earlier follow-up in the sickness benefit scheme and more consistent rehabilitation efforts have been emphasised in recent years.

21. Many of these adults would be eligible for grants well above the ordinary level of student grants.

22. The structure of the system for adult education and training is very confused and far from generally-accepted principles of good regulation. The system contains several thousand approved courses, supplied by more than 600 educational establishments. The administration and regulation takes place with reference to seven different laws. In addition, three different remuneration schemes exist, each of which has its own legislative foundation (Arbejdsministeriet, 1999*b*).

23. The reform also reduces tax subsidies to private pension savings and continues the increase in environmental taxes from the 1994 reform.

24. Recent studies (Graversen, 1997; Graversen and Smith, 1998) indicate that the average compensated wage elasticity may be around –0.5. This is higher than found in earlier Danish studies (Smith, 1990 and 1995; Pedersen and Smith, 1996) but in line with results from other countries – see Leibfritz *et al.* (1997) for an overview. Assuming an elasticity of –0.5, an increase in labour supply of 1 to 2 per cent could be achieved by raising the threshold for the top tax bracket enough to reduce the proportion of taxpayers facing it, to the mid-1980s level, that is 8-10 per cent. Such an increase would reduce the average effective marginal tax rate by 3-4 percentage points.

25. The only element of joint taxation in the Danish tax system is the ability to transfer between spouses unexploited basic personal allowances and deductions in the middle and top tax brackets, the latter pair where taxable income is below the corresponding income threshold.

26. In general, deferred taxes cannot be passed on to a new owner when the company is sold; succession is allowed only if the sale is between spouses or in case of restructuring of the firm – that is where the deferred tax is not taken out of the firm.

27. Post-renewal rents can be raised in proportion to the landlord's share of the investment undertaken. Thus, neither the rent increase nor the increase in the market value of the renewed property reflects the quality improvement. This implication should be taken into consideration in case of a future easing of rent controls. Otherwise landlords would benefit twice: first from not having incurred the maintenance costs and subsequently from being allowed to keep the capital gain resulting from the largely state-financed quality improvement.

28. The agreements are currently being prepared for implementation.

29. In September 1999 the National Telecom Agency ruled that TeleDanmark had to lower their prices on various interconnection services by 15 to 35 per cent. This was implemented in February 2000. Then, an additional cut of 15 to 25 per cent was imposed as from May 2000.

30. For example, the statutory rebates granted to children and pensioners using DSB (the train operator) will not apply if using a private long-distance bus.

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31. By per capita SO_2 emissions in the mid-1990s Denmark ranked thirteenth. SO_2 emissions have fallen very rapidly over the last two decades, those of NO_2 much less so. Available cross-country data on air quality compares large cities only. See OECD (1999*d*).

32. See OECD (1999*c*), E*nvironmental Performance Reviews*: D*enmark*, Table 8.3, p.191.

33. The full list of gases included, with their marginal costs (DKr per tonne), is: CO_2 (35.5), methane (702), N_2O (12 000), NO_x (13 100), SO_2 and volatile organic compounds (both 24 500).

34. For more details of institutions, see OECD (1999*e*).

35. There are 14 counties, excluding Copenhagen and Frederiksberg (which serve both municipal and county functions), and 275 municipalities in Denmark.

36. The procedure for deciding on whether bus services are socially useful was described informally to the Secretariat as to withdraw the service and then see how great the ensuing protest was.

37. This is measured as precipitation less losses due to evaporation and transpiration from plants.

38. Wastewater from fish farms is not subject to the waste water tax. Businesses registered for VAT can obtain a reimbursement of 97 per cent of the tax exceeding DKr 20 000 a year if 80 per cent of the production is connected to the processing of fish or the production of cellulose or of sugar. A reimbursement of 70 per cent of the tax exceeding DKr 20 000 a year is possible if 80 per cent of the production is connected to production of organic pigments, of pectic substances, gelatine, etc. or of vitamins. Thus, some of the largest polluters are subject to an intricate set of exemptions, a pattern that occurs in energy taxation as well.

39. A similar approach is used for phosphates in the Netherlands.

40. The four are water and waste water charges, paid to the local water company, and the water and waste water taxes, which accrue to central government.

41. The current tax on retail containers distinguishes 22 kinds, with rates differentiated by material and size. Its complexity probably imposes substantial administrative costs. No analysis on whether environmental benefits justify these costs appears to have been undertaken, although the measures have been subject to a number of revisions, partly to try to reduce the administrative burden.

42. See Andersen (1998), quoting Statens Byggeforsknings Institut (1996). Andersen (1998) reports that in two communities that adopted weight-based fees in the mid-1980s residual (*i.e.* non-recycled) waste fell to about 100 kilo per capita, compared with 200-300 kg per capita in communities with recycling facilities but without a weight-based charge.

43. The deposit, equivalent to several thousand krone per tonne, is very high compared with the waste disposal tax – see Table 17 – but the appropriate comparison may be with the marginal social cost of bottles left in the street rather than of those arriving at a landfill site.

44. Specifically, the relevant accords are the EN-ECE Convention on Long Range Transboundary Air Pollution (for acid rain), under which a new "multi-effects" protocol was signed in December 1999, and the United Nations Framework Convention on Climate Change.

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45. Prior to 2000 electricity generation did not pay the SO_2 tax directly, since Denmark was limiting its emissions under an internationally-agreed quota scheme that adjusted for electricity trade. A supplementary tax on electricity consumption, calculated according to the implicit sulphur content of electricity used domestically, nevertheless allowed the revenue to be collected – largely for fiscal purposes. This device circumvented EU restrictions on differential taxation of traded electricity. From 2000 onwards the quotas under the agreement (following the Göteberg Protocol) no longer adjust for trade in electricity.
46. When so calculated, it is some DKr 600 to 700 per tonne of CO_2 – much higher than the CO_2 tax itself. The precise figures depends on the fuel, since the energy tax is calculated on the energy content, not the CO_2 content, which varies according to the fuel.
47. Comparing the highest and lowest nominal tax rates would give an implicit ratio (DKr 100 versus DKr 3 per tonne) much higher than four. The lowest tax rate, at DKr 3 per tonne, is applicable only to firms who have accepted the terms of energy-saving voluntary agreements, which, the Energy and Environment Ministry argues, has an effect equivalent to that of imposing the CO_2 tax at DKr 25 per tonne.
48. The Toronto Agreement was largely superseded by the 1992 UN Framework Convention on Climate Change and the Kyoto Protocol to that Convention. Very few countries are likely to meet the Toronto targets, though Denmark will get nearer than most. According to government analysis, current measures are sufficient to reduce emissions in 2005 to 15 per cent below the 1988 level (Finance Ministry, 1999).
49. This may be a misleading use of the term "voluntary", since there are significant penalties for not complying.
50. Renewable sources of energy, with special emphasis on wind energy. Committee on New and Renewable Sources of Energy and on Energy for Development. Report of the Secretary General. Prepared by R. Redlinger, P. Dannemand Andersen and P.E. Morthorst, Risø National Laboratory. The marginal cost of coal-generated electricity is around DKr 0.20 per kWh.
51. A. Larsen and J. Munksgård (1996) Samfundsmæssig værdi af vindkraft, AKF Hovedrapport, København.
52. The six largest Danish wind turbine producers, who each export more than 80 per cent of their production, are among the ten largest producers in the world.
53. Other indirect costs are associated with necessary investments in the electricity grid to accommodate new windmills, although since such investments are difficult to separate from other investment costs, and they are not included in the presented calculations.
54. According to the Economic Council.
55. Using the full amount of the subsidy as a cost may be an overstatement, since in recent years much of it has become a transfer payment rather than a resource cost. On the other hand, the distortions involved in raising revenue are usually taken to imply that the full economic cost of subsidies substantially exceeds the simple budgeted amount. No adjustment for either of these factors was made in the illustrative calculations presented here. Possible environmental costs such as visual impact are also ignored, though they may be important.
56. Energy security is also cited. Since coal is entirely imported, wind turbines may be a more "secure" source of supply in time of war (the earliest power-generating wind turbines were built during the second world war); for the moment, Denmark is self-sufficient in natural gas so this would be a secure source too.

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57. A further argument would be that, if these successes were to be taken as a justification for the wind turbine programme, it would be an example of a government successfully "picking winners"; in a cost-benefit analysis the costs of "losers" that have been picked would have to be set against these benefits

58. This left Denmark with a definitional problem following the Kyoto negotiations. In the base year 1990, Danish imports of hydro-electricity were particularly large, and hence its emissions of CO_2 from fossil fuel burning unusually low, which means that the targeted reduction in emissions is more stringent than if 1990 had been a "normal" year. Danish negotiators have argued that emissions from exported electricity should be treated as being emitted by the country importing the power, but this has not been accepted.

59. The government owns directly Danish Oil and Natural Gas (DONG), of which Danish Naturgas A/S is a daughter company. Until the market opening on 1 July 2000 DONG-Naturgas has the sole concession on buying natural gas for the Danish market as well as a monopoly on transport and management of inventories. GAS, together with four regional municipally-owned companies, is also responsible for the sale and distribution of natural gas.

60. According to the Competition Authority Report, of subsidies attributable explicitly to bus operators, approximately one-tenth were due to fuel tax exemptions (*Konkurrenceredegørelse*, 1999, p. 129).

61. The vehicle registration tax has been high in Denmark since the second world war, rather than as a result of concern in recent years about the environment.

62. This system was introduced for cars registered as from July 1997. The previous system, under which the tax was a function of the weight of the car, still applies for cars first registered before that date.

63. However, the Danish authorities argue that these data may be unreliable.

64. It could be argued that since depreciation of a car is partly a function of distance travelled, the high *ad valorem* registration tax acts to increase depreciation and hence the cost of car use. The strength of this effect depends on to what extent depreciation is actually perceived as one of the costs of car use, and how closely second-hand car prices are in fact related to distance travelled.

65. There are possible counter-arguments. The high cost of car purchase means that many fewer young people – including those who will later be well off – own cars than would otherwise be the case. Apart from higher use of (subsidised) public transport, bicycle use is also consequently quite high. It may be that this has resulted in a structure of endogenous "tastes" such that the demand curve for car use is shifted downwards. The reversal of such an effect, if it exists, by a significant reduction in the registration tax, might mean that the level of fuel taxes needed to keep traffic to its current levels might conceivably have to be so high as to have even "worse" distributional consequences than the present system. There appears to have been no empirical work in this area.

66. Another solution would be to increase fuel taxes in a graduated way, as a function of the distance from the German border, but administrative and political complexities probably rule this out.

67. This is the case, except to the extent that the burden on direct energy use by final domestic consumers is increased. In this case job losses will be fewer; but welfare would fall too as the incentive to reduce households' indirect consumption of energy is reduced.

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68. This is true provided that the distribution of tax credits or emission permits is not a function of actual or future emissions. In Sweden such a compensatory scheme is used in the NO_x charging system.
69. Unless the CHG-intensive companies are better able to generate jobs in other activities than the average company.

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Annex I

Changes in user charging

Information on the overall application of user charges is scarce. The only indicator at hand is a calculation of the sale of goods and services from public-sector entities in per cent of public consumption. This ratio rose from 8.7 per cent in 1977 to 13.2 per cent in 1997 (Finansministeriet, 1999*a*). Changes in the composition of services, for example for demographic reasons, could affect the calculation significantly, however. For example, the total spending on primary schooling (where there are no user charges) has been increasing less than average during the latest decade as the number of pupils have declined. On the other hand, the number of children in daycare facilities (with user charges even if at declining levels) has increased sharply. The net effect of these two examples would be an increase in the overall ratio of user charges, even though the only underlying policy change was a reduction of user charges in daycare facilities (see below). Therefore, cautious interpretation is necessary, and the developments in individual expenditure areas should be assessed separately. For most social services, user charges have been unchanged or reduced over the last two decades. The list below reports some of the most significant changes.

- *Childcare*. Municipalities have traditionally been free to set payments for childcare within a limit of 30 per cent of the costs, which in practice has been a binding constraint. In 1991, however, a brothers-and-sisters discount of 50 per cent was imposed. Moreover, the means-tested subsidy to daycare was broadened as part of the Whitsun package in 1998. Although the agreements with the municipalities for 1999 and 2000 have allowed for small increases in user charges for after-school facilities, the general trend during the past decade has been to reduce the proportion of costs paid by parents significantly.
- *Elderly care*. User charges for practical help (cleaning, etc.) for the elderly were abolished in 1989. The municipal agreement for 2000 included a reintroduction of means-tested payments for such services, but the government has backtracked from implementing it. Some cost savings have been realised in this area by municipalities where these services have been contracted out to private companies, but controlling expenditures remains notoriously difficult because rationing is the only instrument at hand. The coverage rate has increased by approximately 10 percentage points since 1989 (Finansministeriet, 1999).
- *Medicines*. In 1989 a threshold of DKr 800 per year was introduced, before which consumption was to be paid fully by the consumer. Consequently, the public expenditure on medicines fell by 20 per cent. The threshold was abolished in 1991, and public expenditures and the volumes consumed have risen sharply since then.
- *Adult and vocational training*. In 1997 expenditure ceilings were abolished and schools were allowed to accept all applicants and have the costs reimbursed by a "taxi-meter"

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principle, *i.e.* each student completing a course automatically increased central government reimbursements. The user charges were abolished in 1998 after strong pressure from the social partners, while the free uptake to schools remained in force. Public expenditures to adult and vocational training ballooned in 1998 as a result. Expenditure ceilings have now been reintroduced, although at a higher level. Price signals to regulate demand remain very weak, however, and expenditure control relies entirely on rationing.

– P*hysiotherapy.* Free physiotherapy for handicapped persons was introduced in 1989. Public expenditures have risen tenfold since then, while expenditures for ordinary physiotherapy have actually decreased slightly. One reason is that the administrative practice has changed. The group of patients being referred for free treatment by general practitioners has become still broader, indicating that abuse may be taking place. Against this background, it has been decided, as part of the agreement with the counties for 2000, to put a ceiling on the number of free treatments per year. Such quantitative regulation does not change incentives, however, and is clearly a second-best solution.

– D*ental care* is an exception to the general trend. Public subsidies have been reduced from 40 per cent of the costs in 1980 to approximately 25 per cent currently. Total public expenditures have consequently remained broadly unchanged, despite a relatively strong increase in the number of treatments.

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Annex II

Chronology of main economic events

1998

November

The central bank lowers the deposit rate to 4 per cent and the repo rate to 4.25 per cent.

Agreement is reached on the 1999 fiscal bill, incorporating the proposals for labour market and housing-policy reforms. Restrictions on investment in social housing and higher stamp duties on housing transactions are maintained. The early retirement scheme is modified: the entry age of 60 years is maintained, but tax stimuli are introduced to encourage workers to stay on in the labour force until the formal retirement age of 65 years (previously 67 years). Eligibility criteria are tightened and a separate insurance premium for early retirement is introduced for members of UI funds. Old-age pensioners are encouraged to go on working by lowering the offset for labour income in their basic (non-funded) pension.

December

The central bank lowers the deposit rate to 3.50 per cent and the repo rate to 3.95 per cent.

1999

February

The central bank lowers the deposit rate and the repo rate by ¼ percentage point to 3¼ and 3½ per cent respectively.

The wage round in agriculture, horticulture and the pork processing industries leads to two-year agreements with expected increases in total wage costs of 4 per cent for 1999 and 3½ per cent for 2000. As part of the agreements, three extra days of annual leave are introduced, two in the first year and one more in the second.

In the public sector, the wage round results in three-year agreements at all levels (state, counties and municipalities) covering the period from 1 April 1999 to 31 March 2002. For central-government workplaces the total increase in hourly wage costs during the period covered is 7½ per cent, of which 1 per cent is to finance the introduction of three extra days of leave. On top of these increases, the index arrangement is prolonged, compensating central government employees for 80 per cent of the difference between the wage increase in the state and the private sector. In counties and municipalities, the increase in wage costs over the three-year period is assessed to approximately 10 per cent, including three extra days of leave. The wage agreements extends the coverage of the new wage system introduced in

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1997; all employees in counties and municipalities are covered by the new wage system. A new working time agreement between the municipal employers and the schoolteachers is included.

March

The central bank lowers the repo rate by 10 basis points to 3.40 per cent.

The two-year wage settlement in the financial sector is estimated to increase wage costs by around 3½ per cent in 1999 and 3 per cent in 2000. Extra days of leave are not a feature of the wage agreements, reflecting that the sector already introduced five extra days in the preceding wage round.

The central bank lowers the deposit rate and the repo rate by ½ percentage point to 2¾ and 2.9 per cent respectively.

May

A reform of the electricity sector is adopted by parliament, thereby implementing the EU-directive on the internal market for electricity. The reform sets ceilings to the electricity-suppliers' emissions of CO_2. The direct subsidies to renewable energy are henceforth to be paid by the consumers via the electricity price and the reform aims to establish an efficient market for renewable energy from 1 January 2003 the latest – from that day all consumers are to be allowed to choose their own supplier.

June

The central bank lowers the repo rate marginally from 2.90 per cent to 2.85 per cent.

Agreements on the broad line of local government budgets in 2000 are reached with the central government and approved by the standing fiscal committee in parliament. For the municipalities the period is three years, while it is one year with the counties, although with a three-year sub-agreement for the health sector. The agreements allow an increase in local government consumption by 1.1 per cent in 2000 while tax rates should be kept unchanged.

August

The government presents the fiscal bill for 2000 in which a surplus on the central as well as the general government financial balance is estimated – DKr 7.9bn. and DKr 28.8bn. respectively. The budget proposal does not contain new major structural policy initiatives. It aims to ensure, together with the agreements on the local government's economy, an overall neutral fiscal stance for 2000.

October

Local-government budgets for 2000 are adopted and broadly in line with agreement with the central government.

November

The central bank raises the deposit rate by ¼ percentage point to 3.00 per cent and the repo rate by 45 basis points to 3.30 per cent.

Political agreements on the fiscal bill are reached. The outcome is slightly more expansionary than assumed in the original proposal. The fiscal effect is estimated by the Danish

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authorities at 0.1 per cent of GDP rather than neutral as in the proposal. Three structural policy initiatives are included:

- A reform of the financing of adult education and training is launched, setting an upper limit (in nominal terms) to public expenditures and introducing employer contributions to a new central fund (Arbejdsgivernes Uddannelses Refusion) which is to finance expenditures above the ceiling.

- A framework for reform of the early retirement (disability) pension scheme is agreed upon, although most of the crucial elements, including the benefit structure, are still undecided. An Act is to be presented during the first half of 2000, with the aim to implement the reform as from 1 January 2003.

- Initiatives to improve competition in pharmaceuticals are put in place. Most important, the principles of a deregulation of the retail sale of pharmaceuticals are agreed.

2000

January

Wage agreements covering most of the private sector are reached after a break-through in manufacturing. The agreements cover a four-year period beginning from 1 April 2000. The main elements are the extension of extra holidays from the previous wage round to cover a full sixth week, and an increase in pension contributions from employers and employees. In most industries total pension contributions will be at approximately 9 per cent by the end of the four-year period, which is close to the end-contributions originally aimed for when the build-up of pension contributions in blue-collar industries started in the early 1990s. In general the agreements also increase the minimum wages during the period, leaving the negotiations on increases in actual wages to take place at the enterprises level (only a small fraction of industries set wages paid at the central level). Assuming full carry-over from increases in minimum wages to average wages the agreements suggest increases in wage-costs in the range 3½ to 4 per cent per year during the period covered. The actual costs depend on the enterprise-level wage bargaining.

February

The central bank raises the deposit rate by ¼ percentage point to 3.25 per cent and the repo rate by 30 basis points to 3.60 per cent.

March

The central bank raises the deposit rate and the repo by ¼ percentage point to 3.50 and 2.85 per cent respectively.

The Prime Minister announces a referendum on Denmark's participation in EMU to be held on 28 September 2000.

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BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

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4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

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